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(54) **MEDIA DATA DELIVERY**

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H04W 88/08 (2009.01)

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CPC **H04W 28/06** (2013.01); **H04W 88/08** (2013.01)

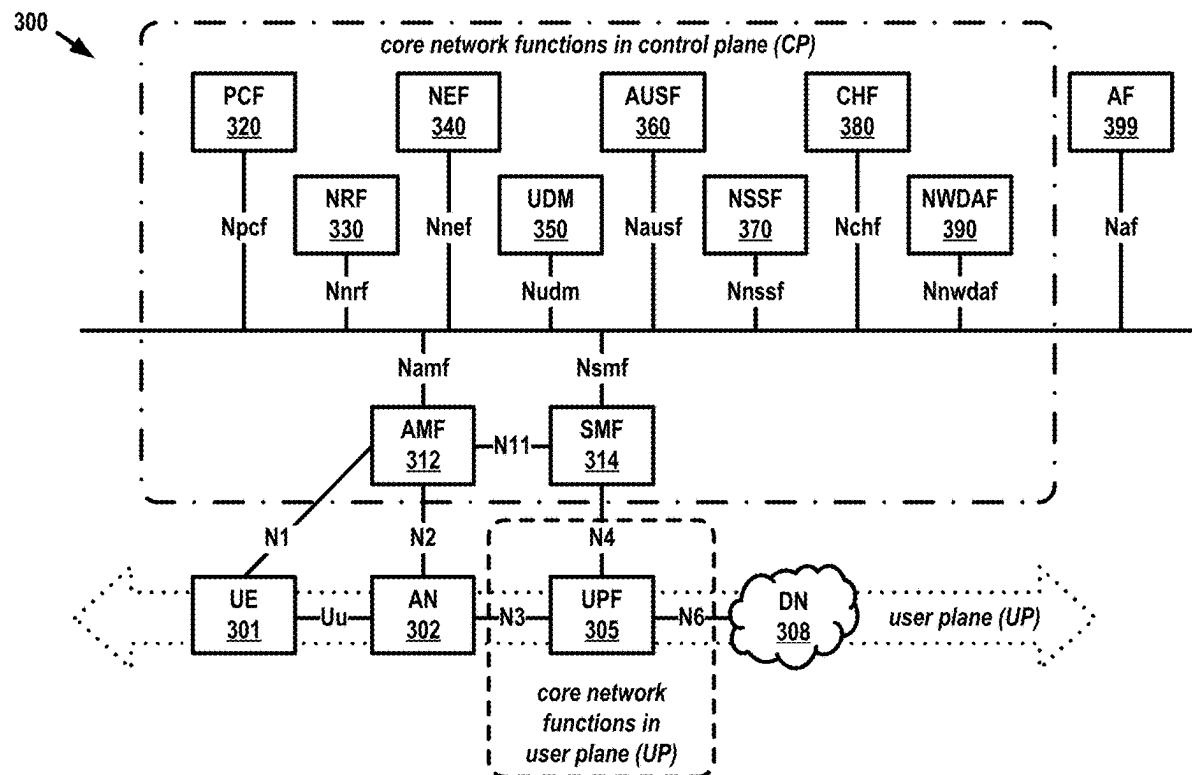
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(21) Appl. No.: **19/189,981**

(57) **ABSTRACT**

A first base station (BS) sends, to a second BS, a message comprising an information of a service data unit (SDU) of a protocol data unit (PDU) set.

(22) Filed: **Apr. 25, 2025**



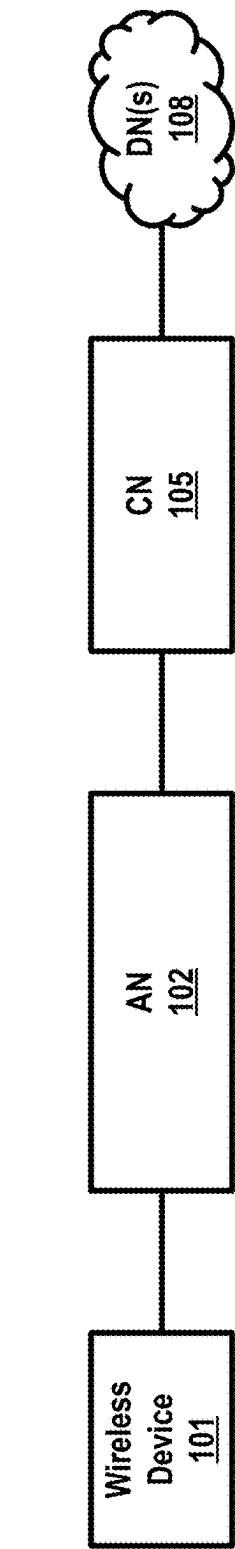


FIG. 1A

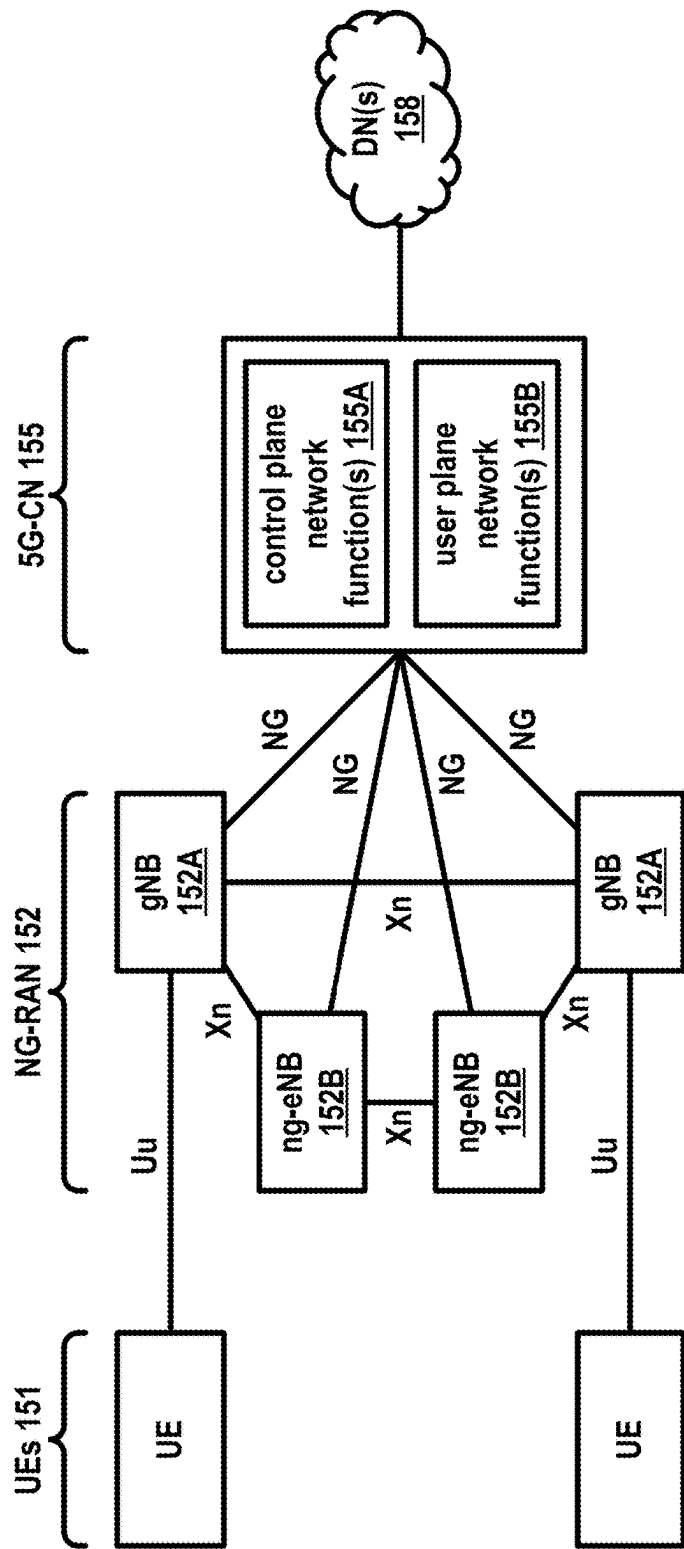


FIG. 1B

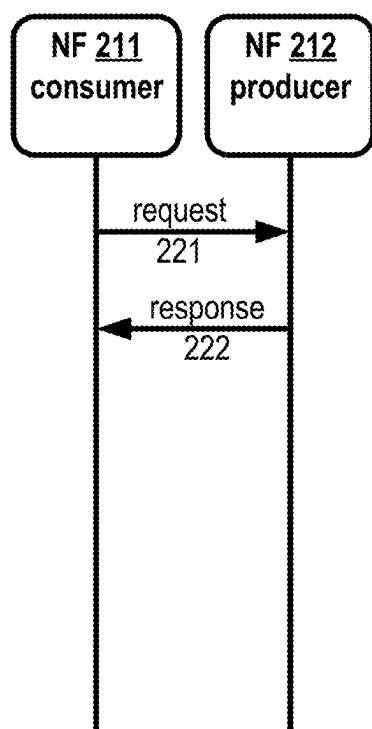


FIG. 2A

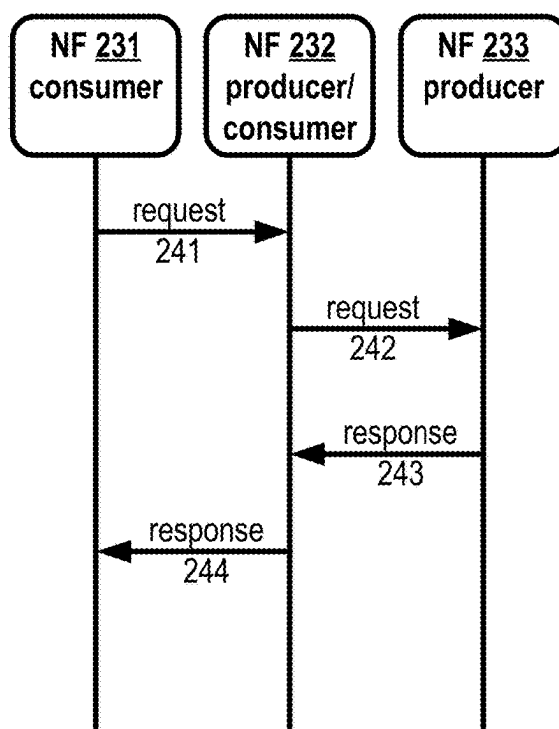


FIG. 2B

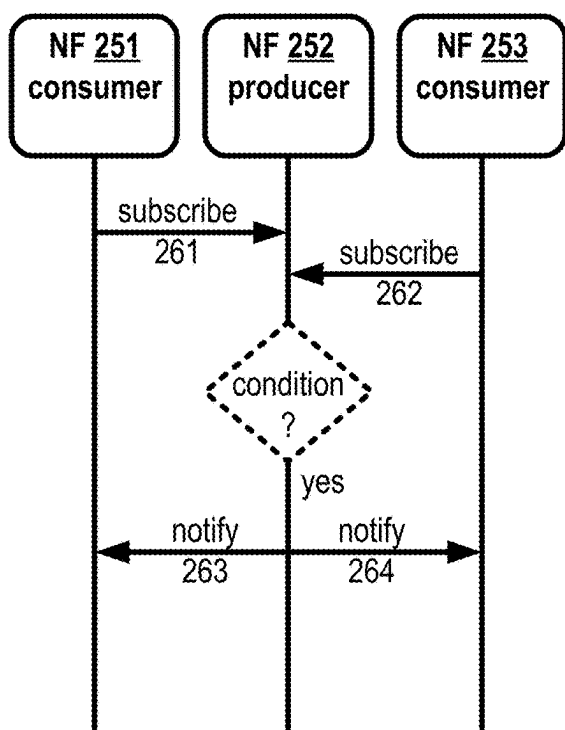


FIG. 2C

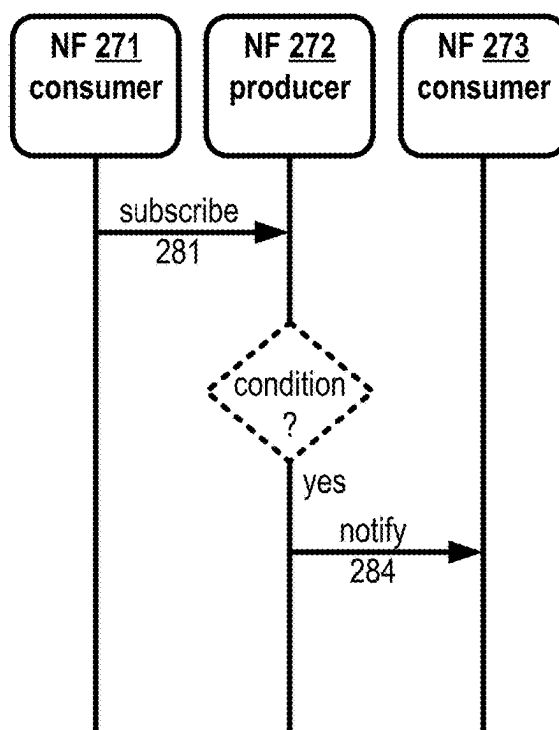


FIG. 2D

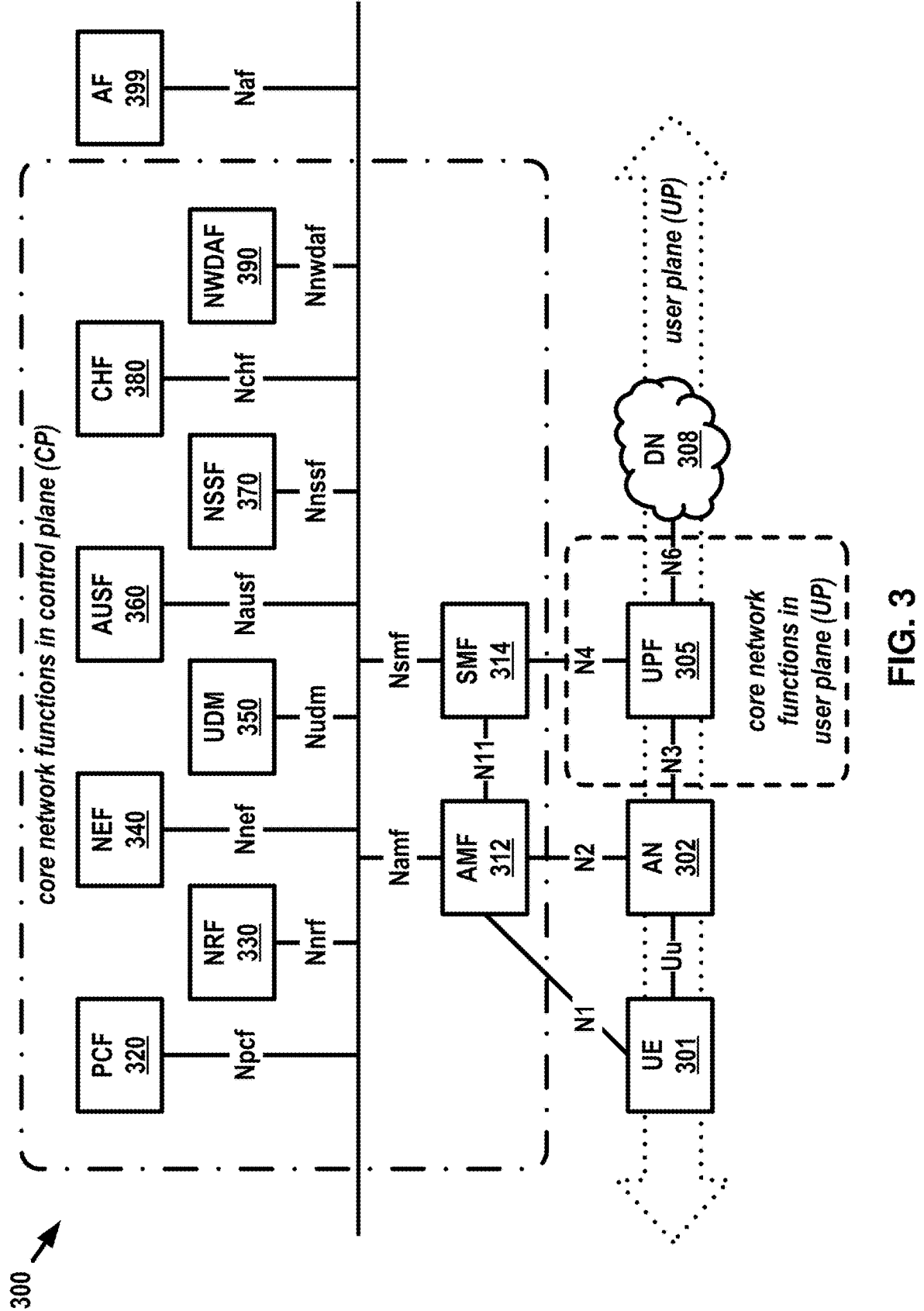


FIG. 3

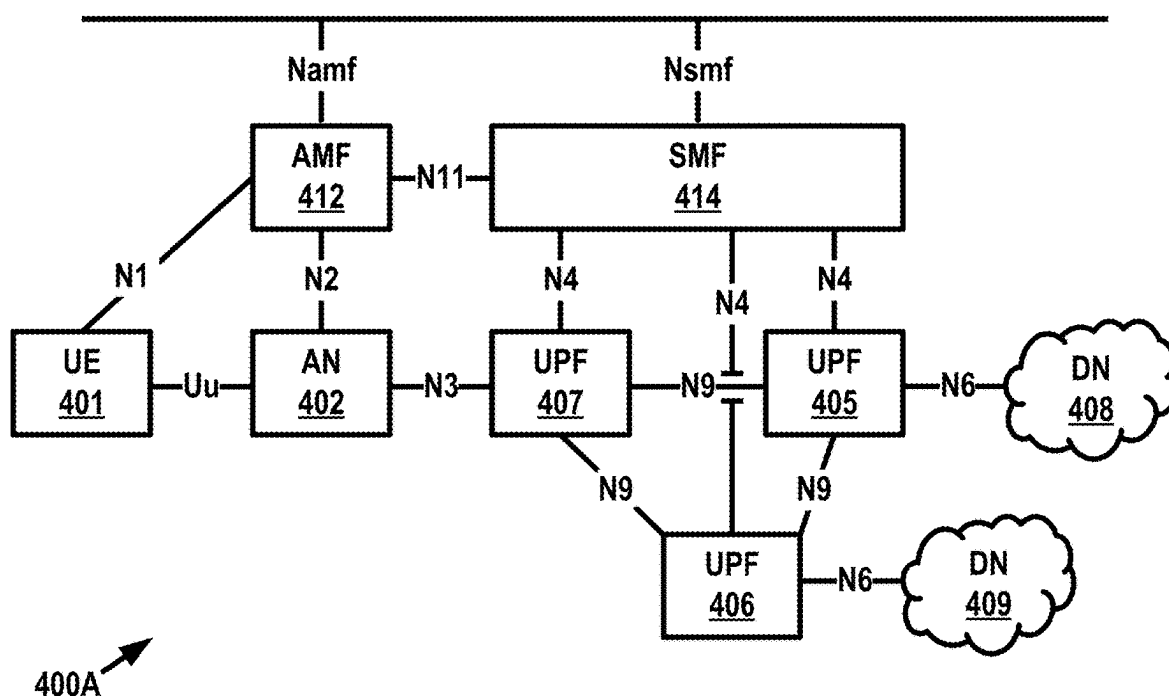


FIG. 4A

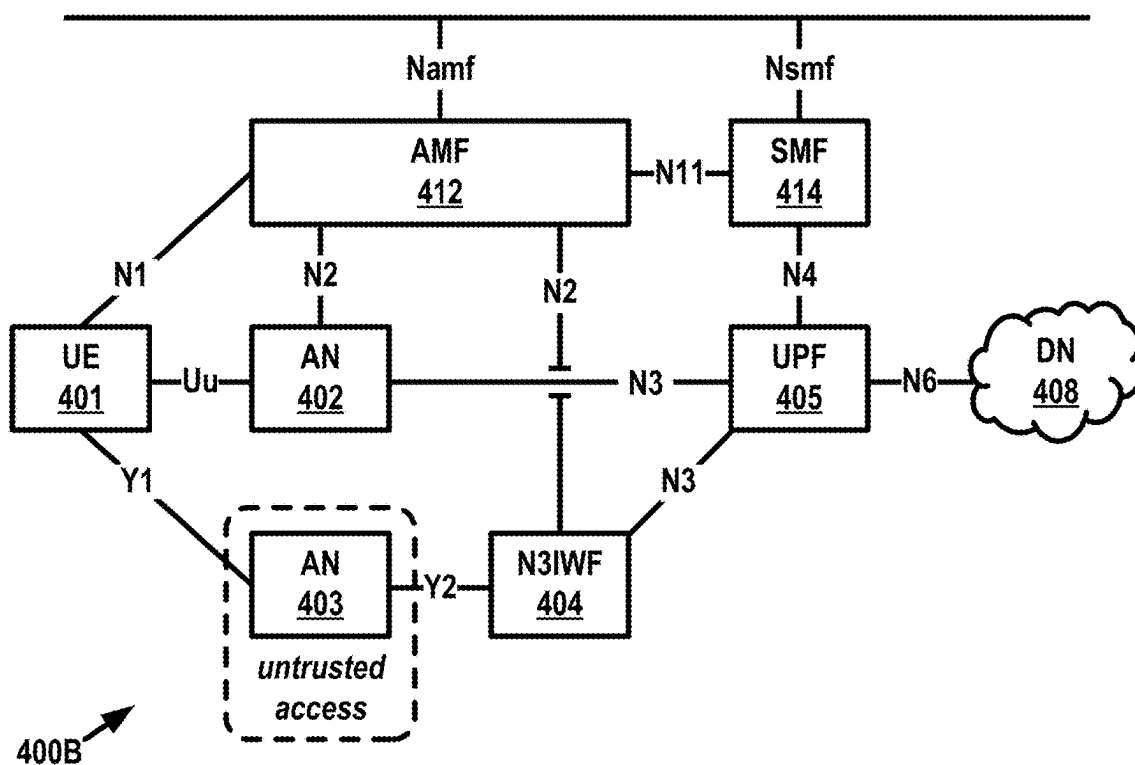


FIG. 4B

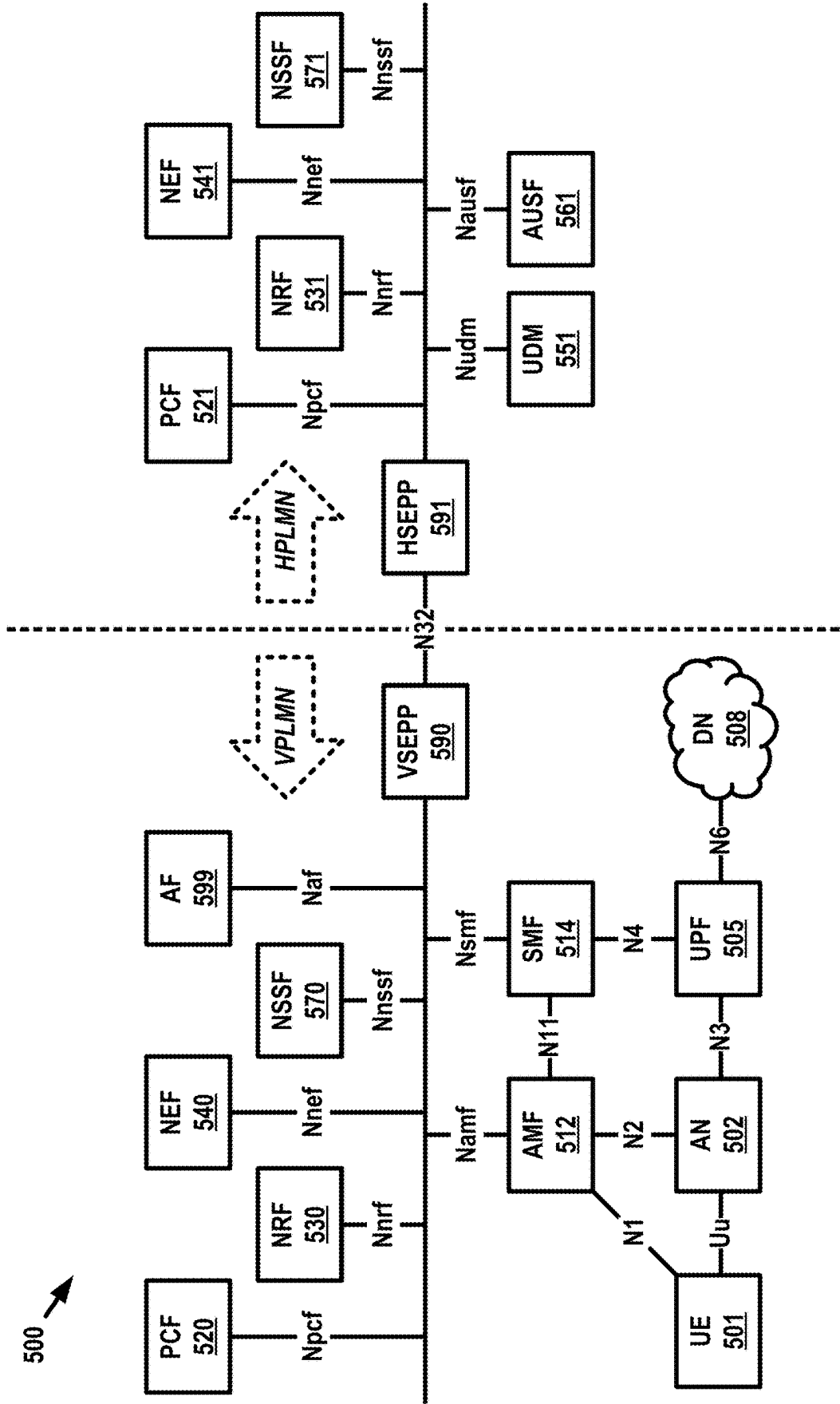


FIG. 5

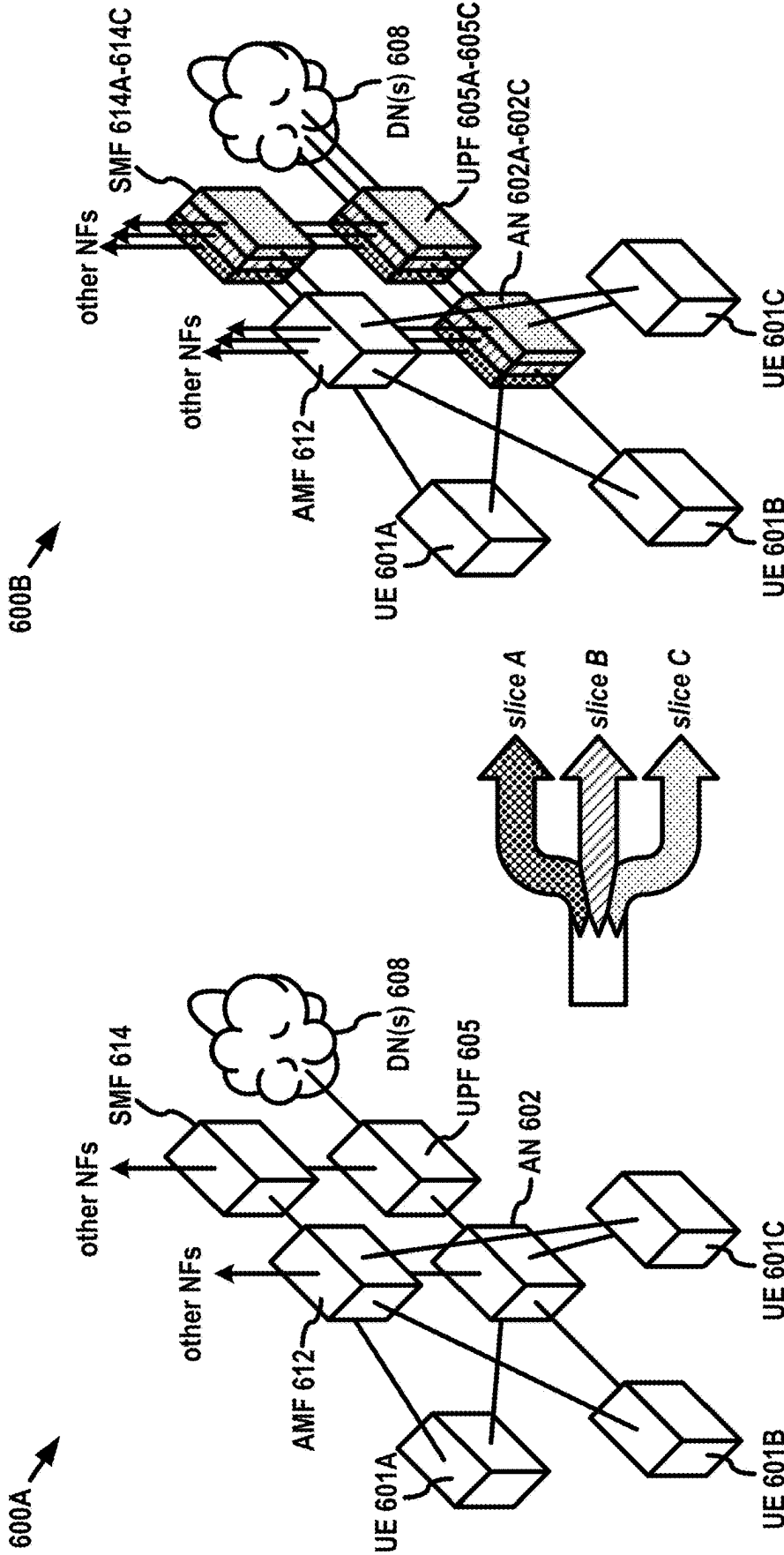


FIG. 6

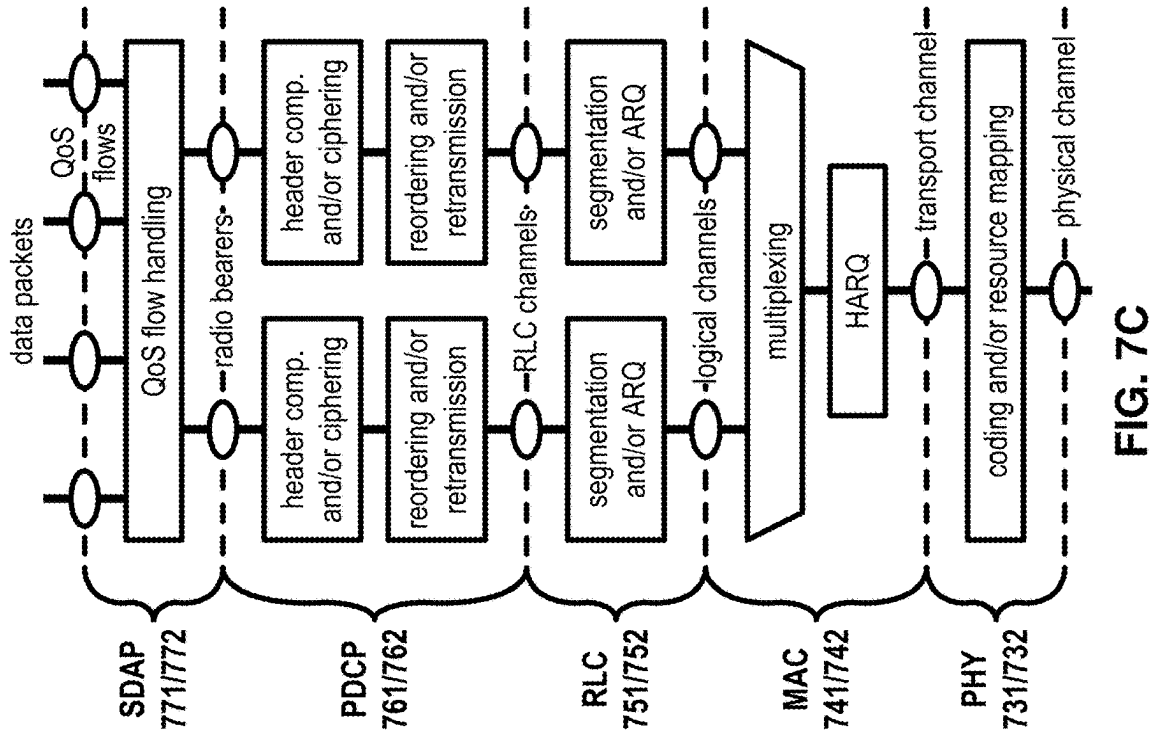


FIG. 7C

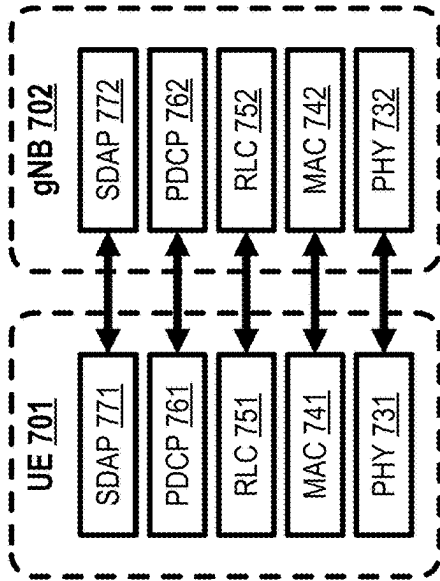


FIG. 7A

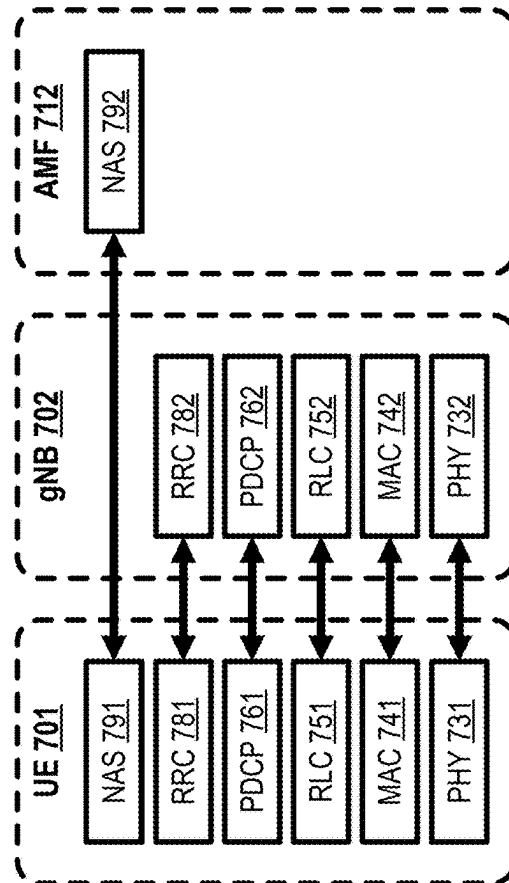


FIG. 7B

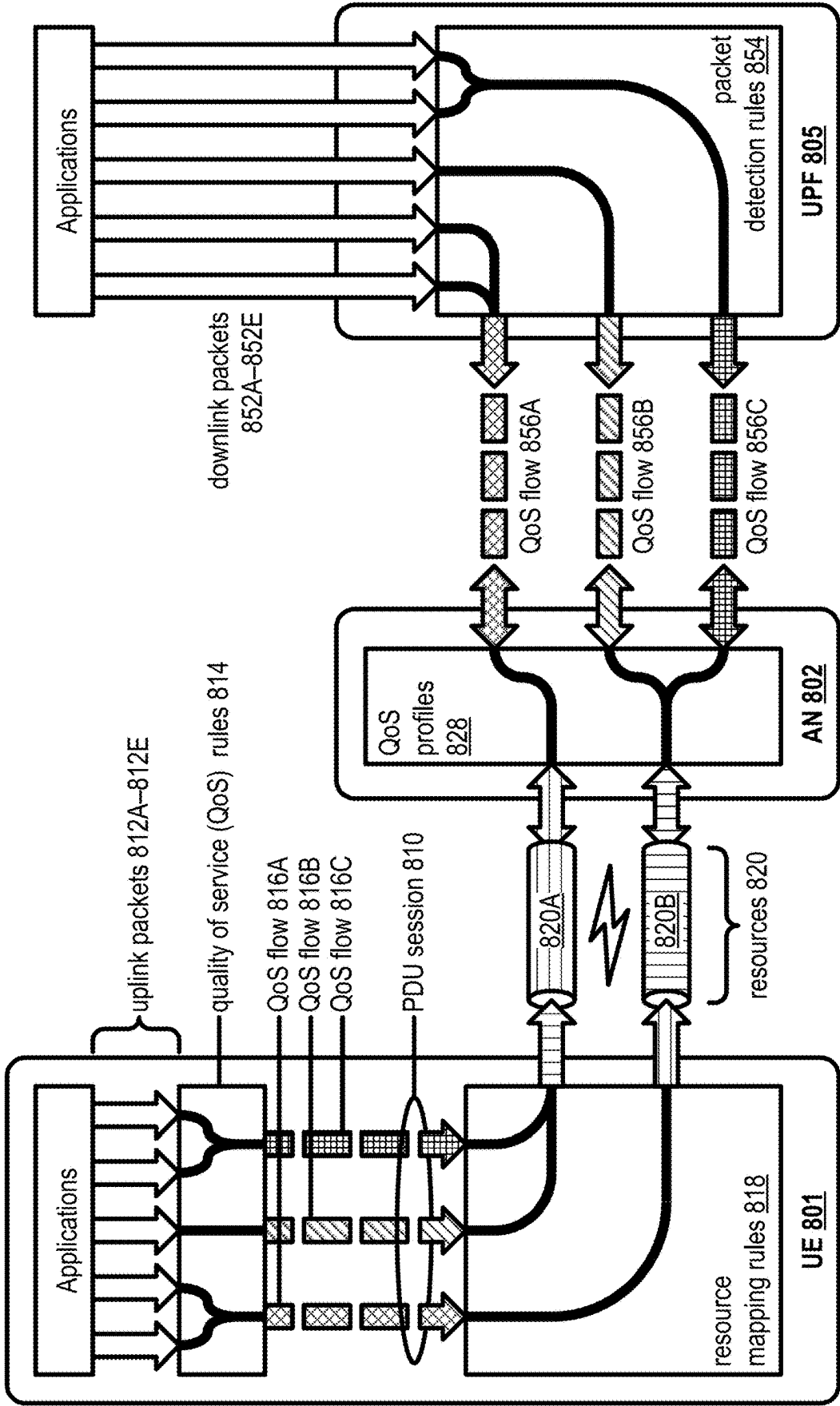


FIG. 8

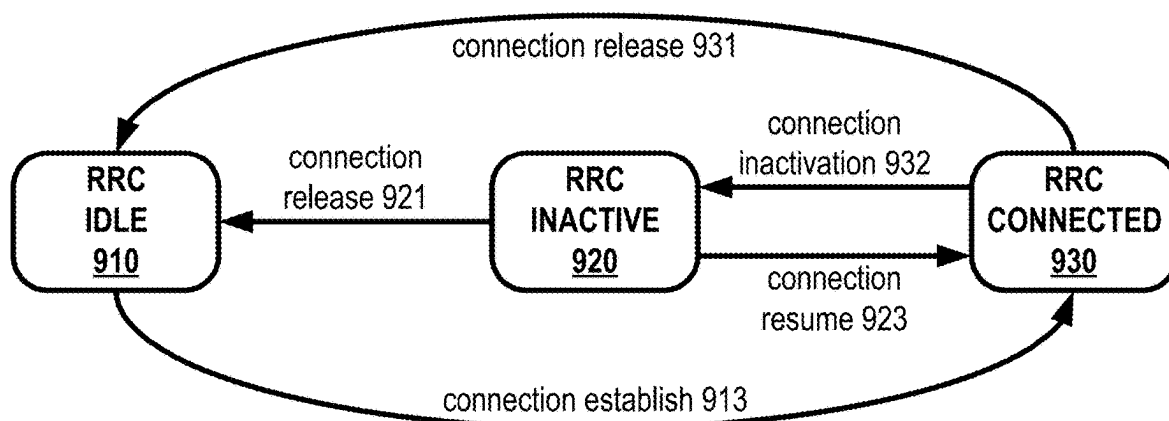


FIG. 9A

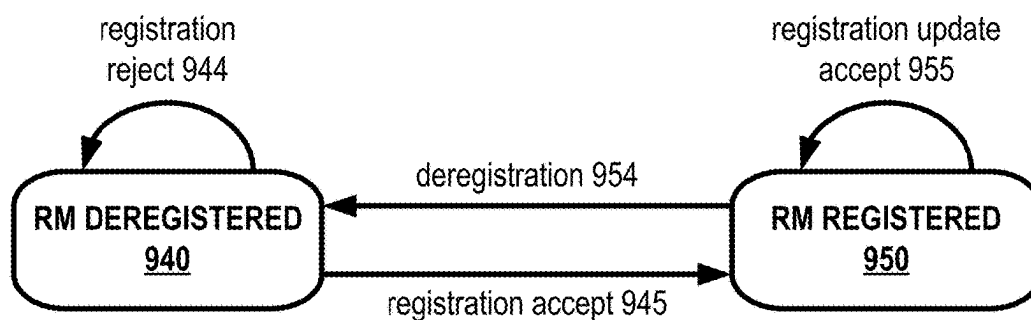


FIG. 9B

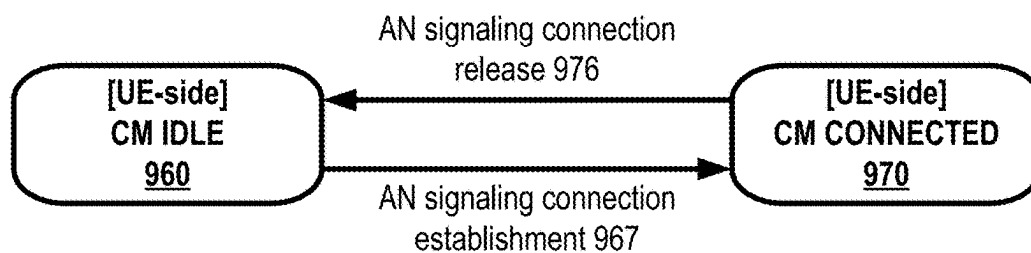


FIG. 9C

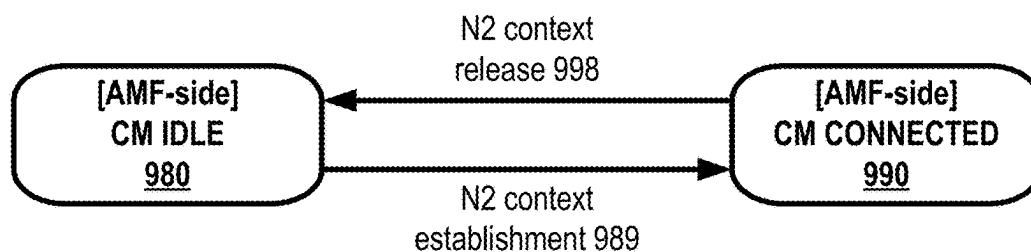


FIG. 9D

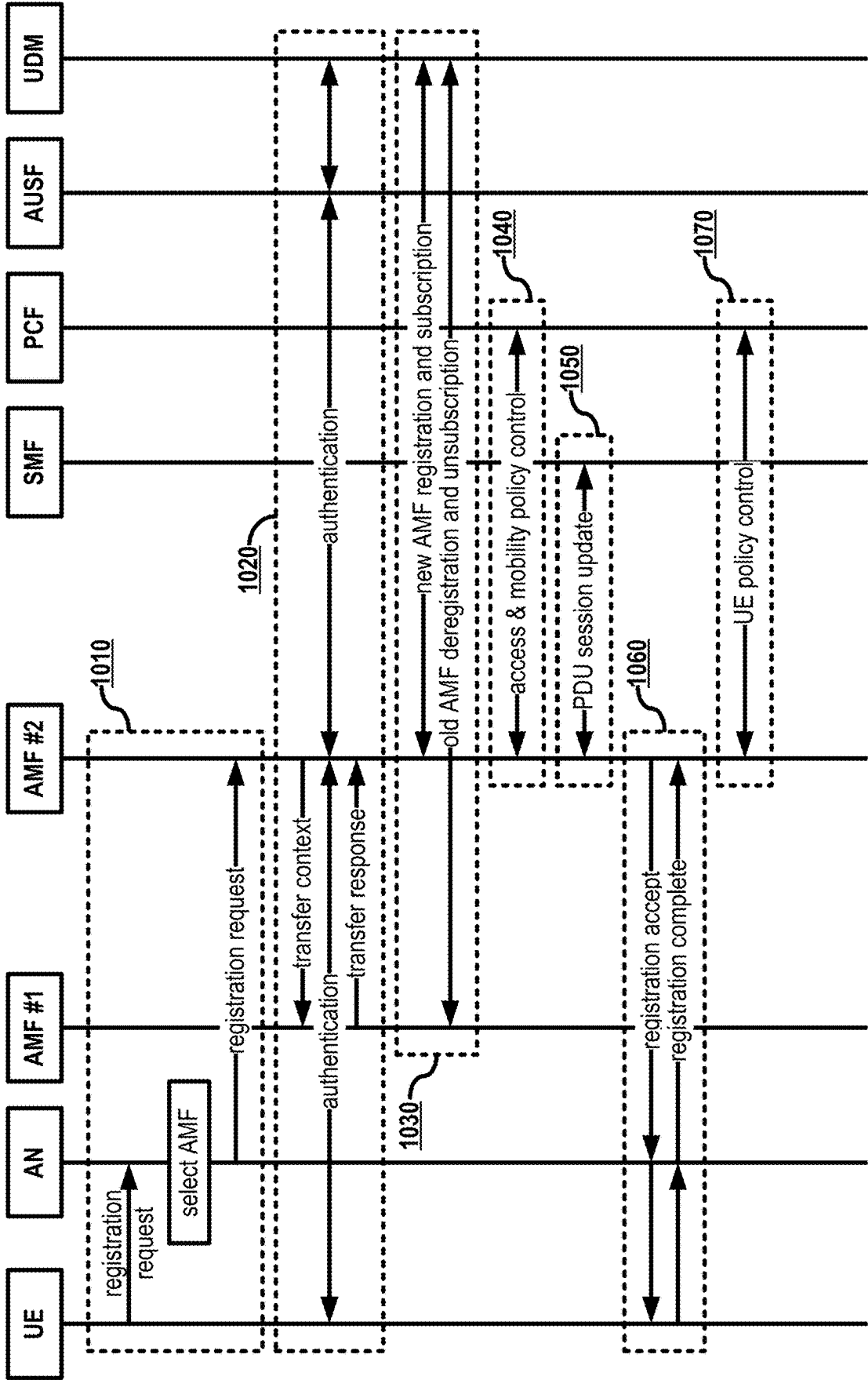


FIG. 10

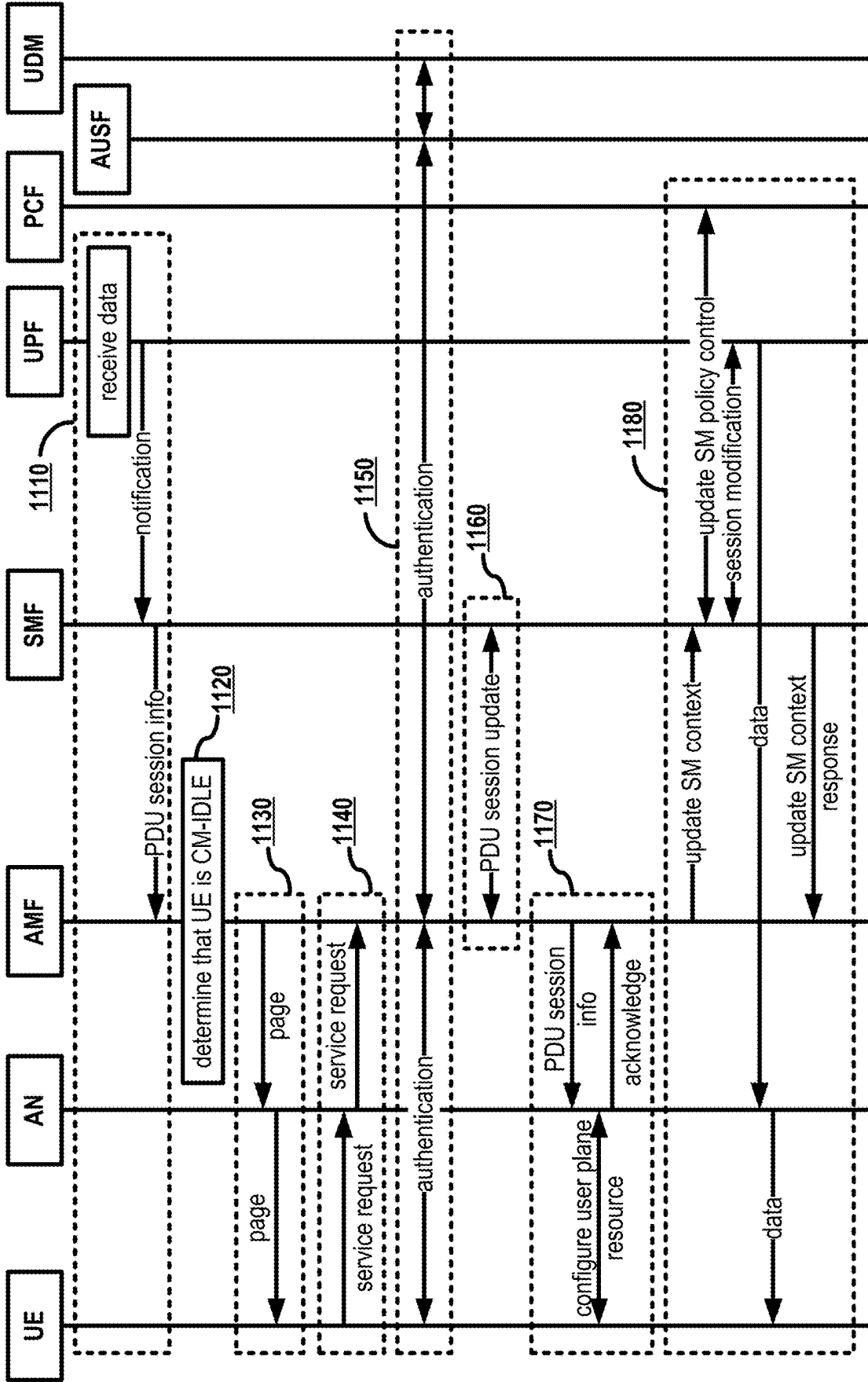


FIG. 11

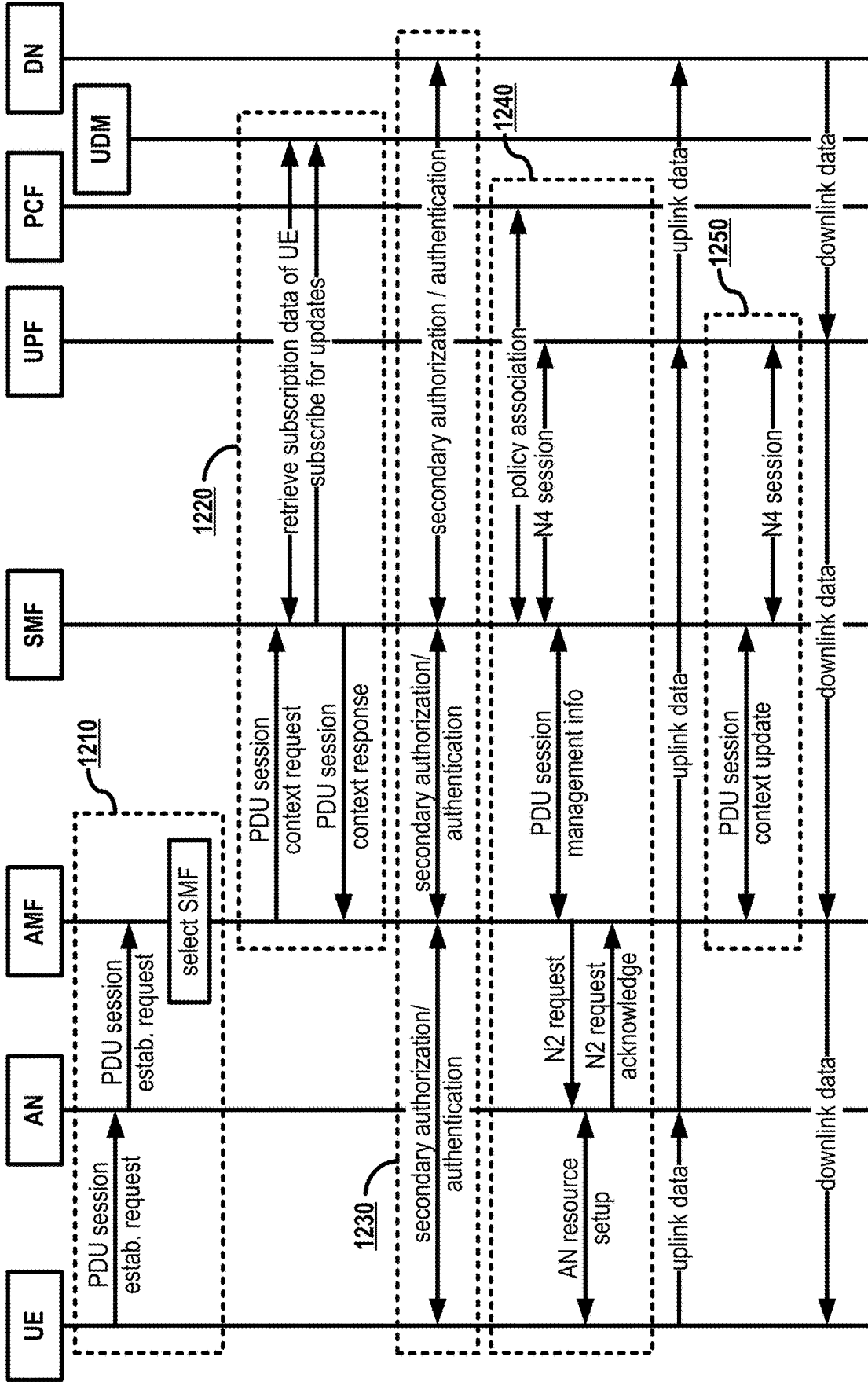


FIG. 12

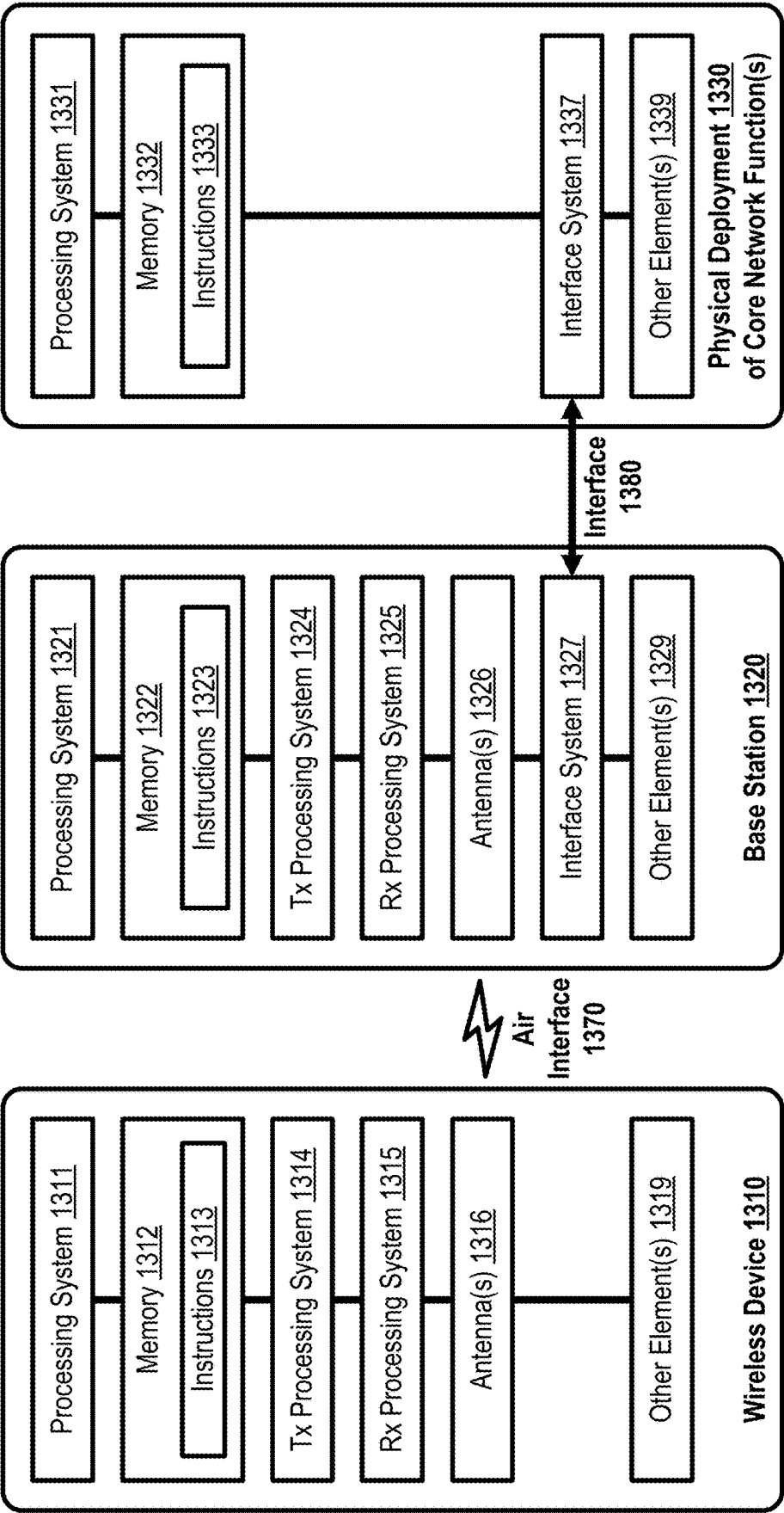


FIG. 13

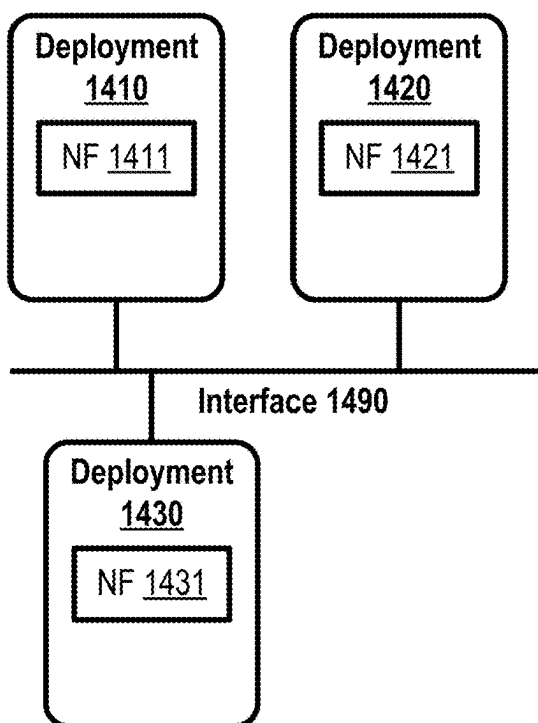


FIG. 14A

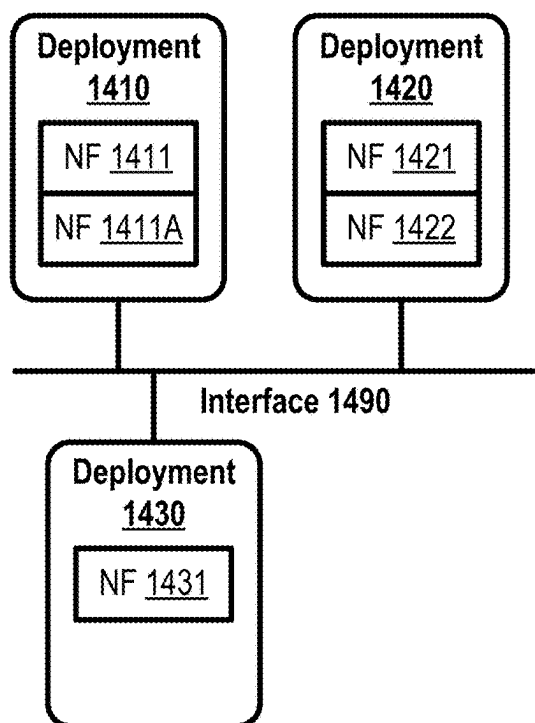


FIG. 14B

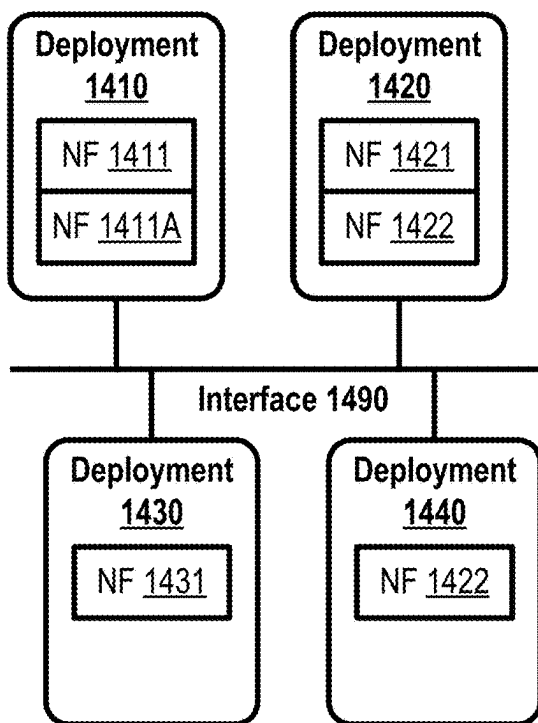


FIG. 14C

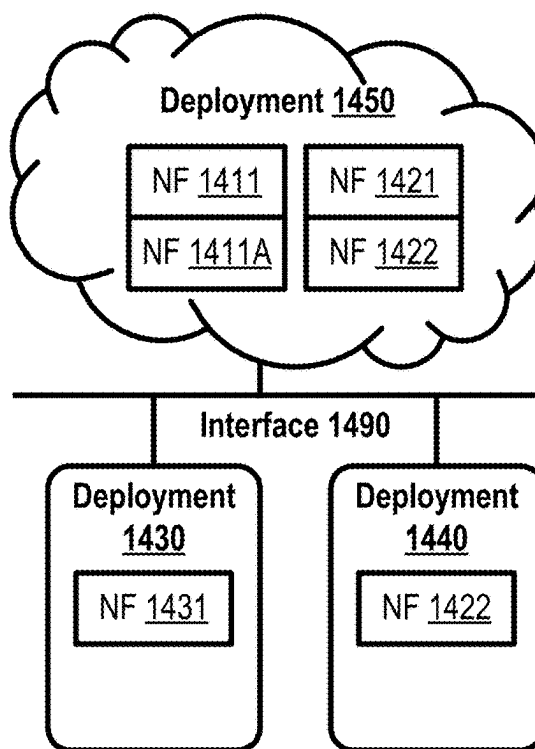


FIG. 14D

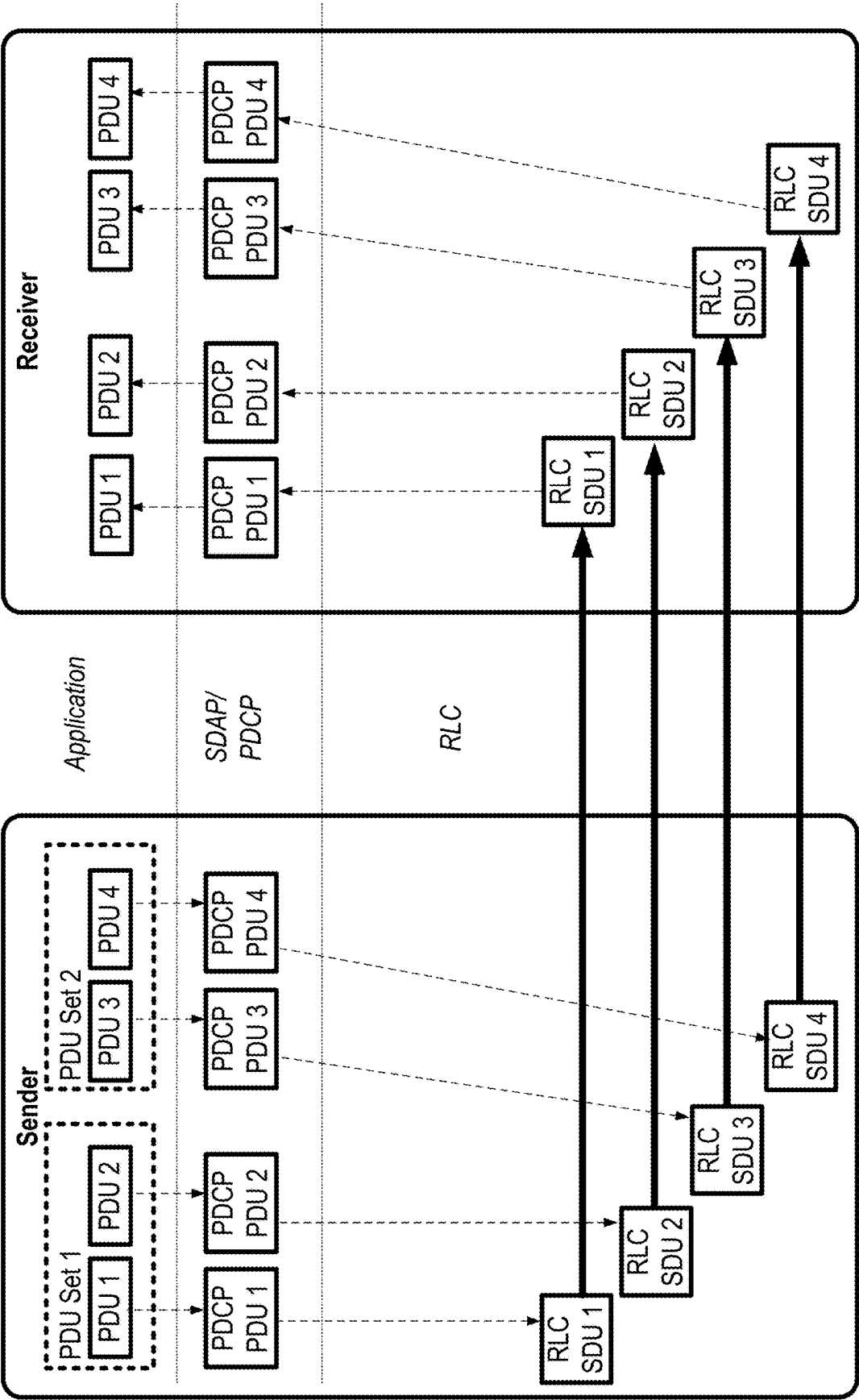


FIG. 15

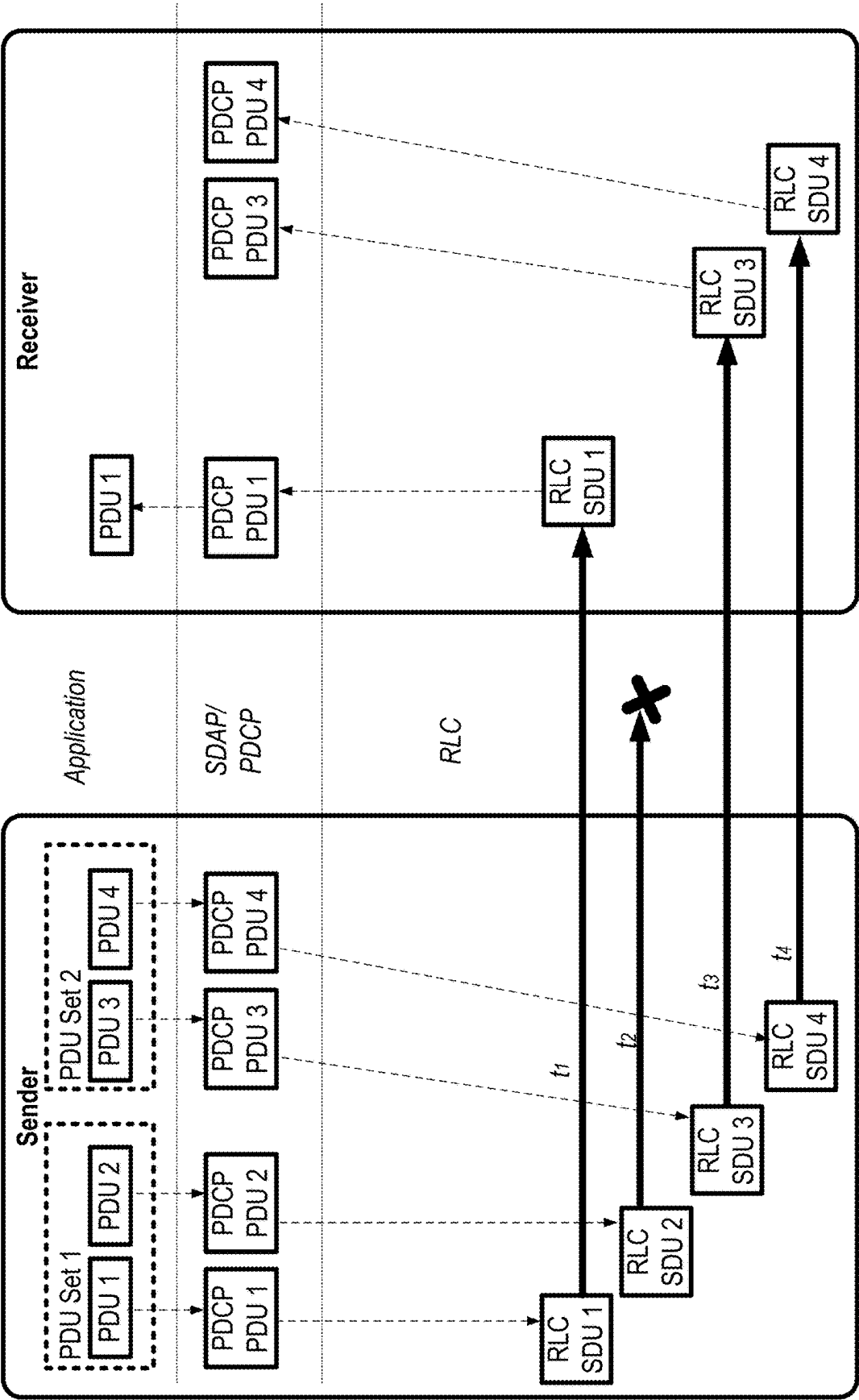


FIG. 16

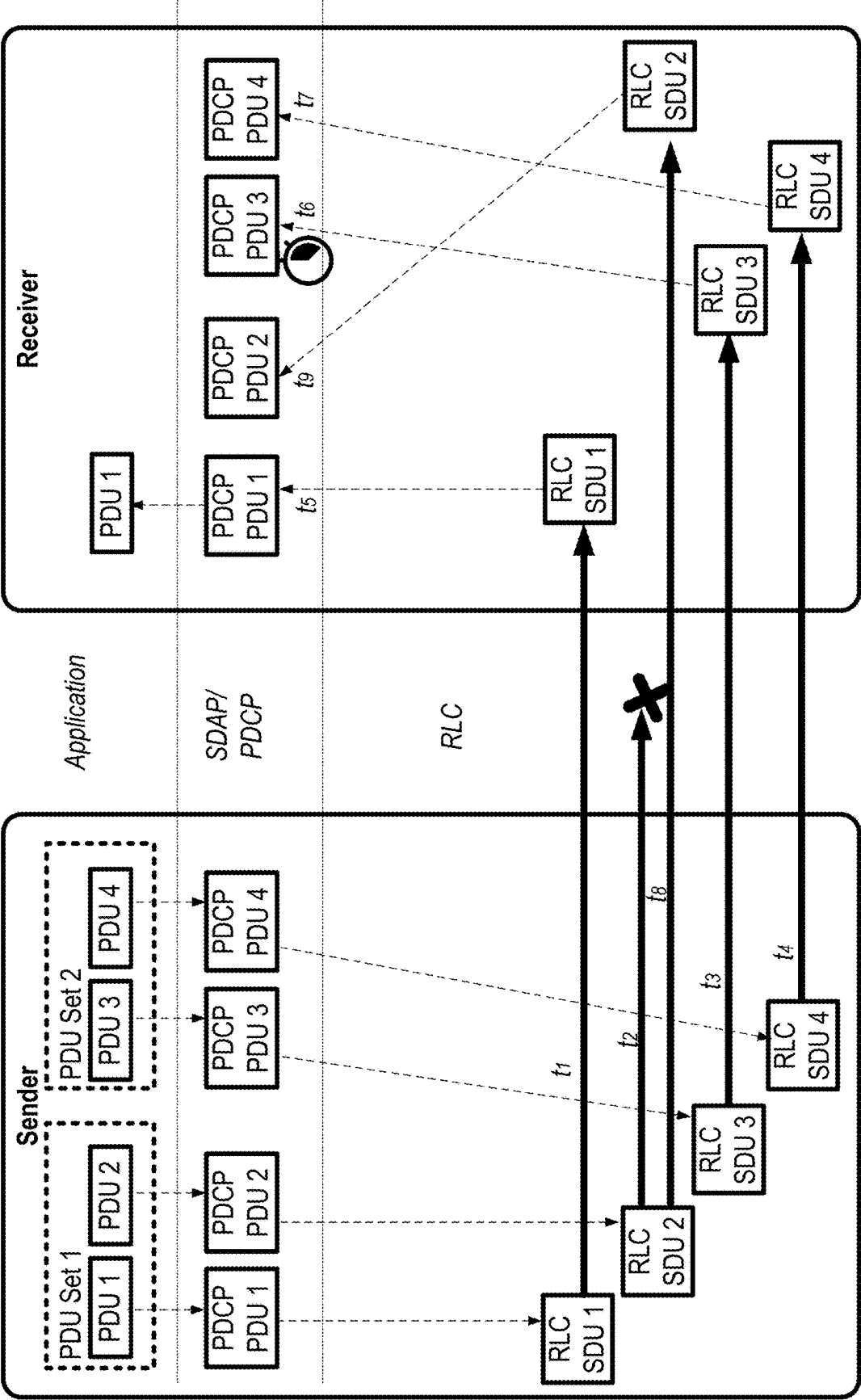


FIG. 17

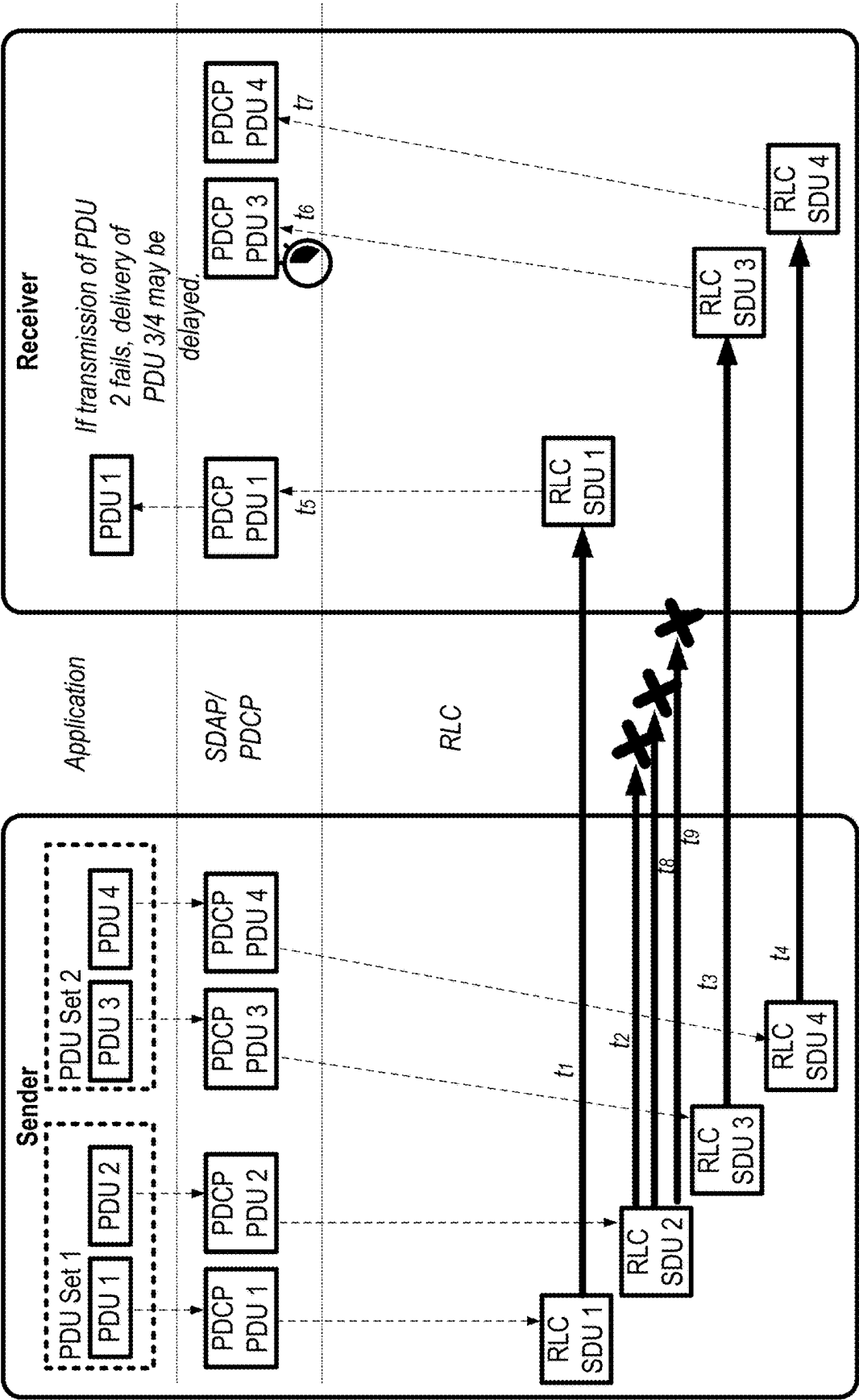


FIG. 18

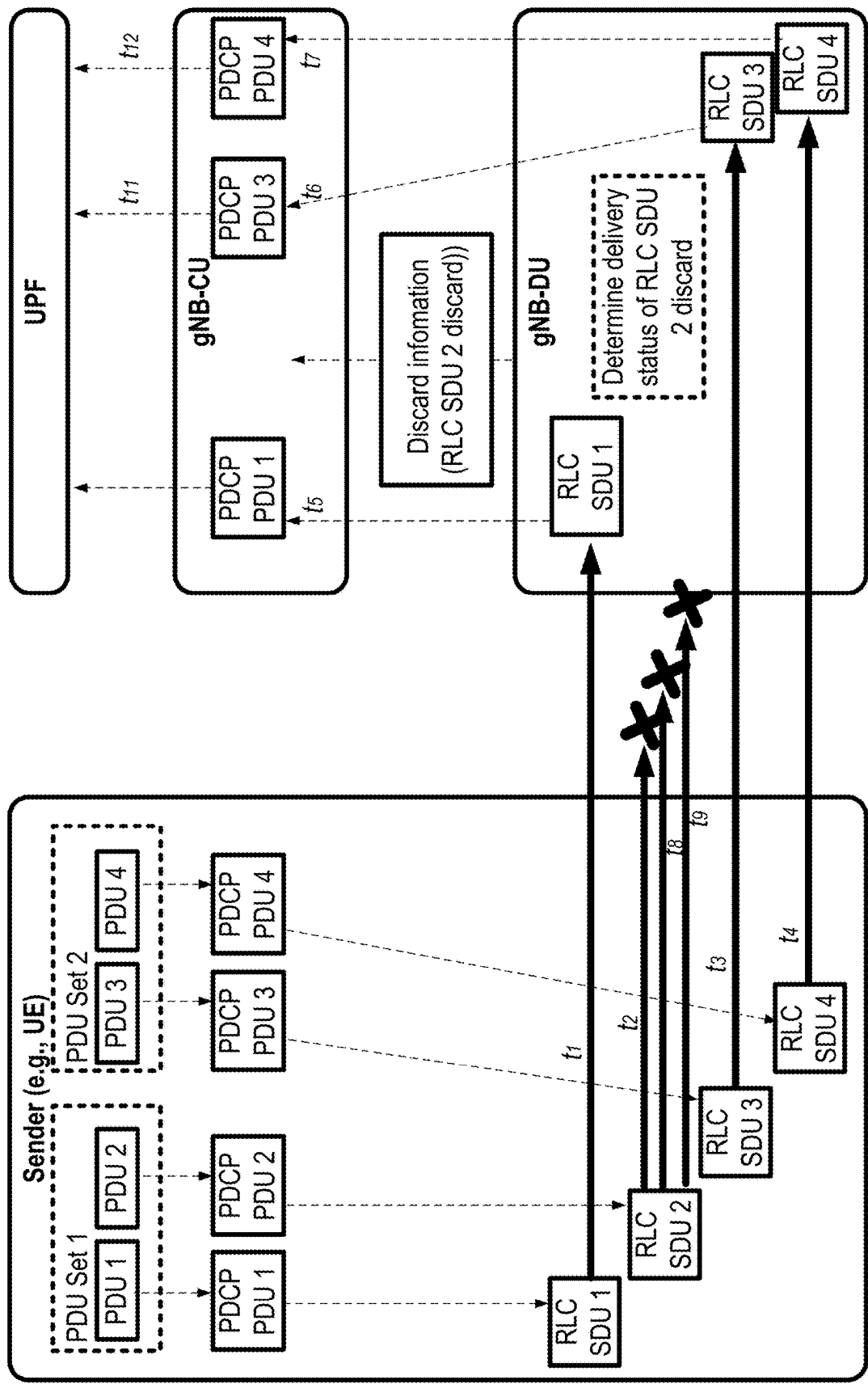


FIG. 19

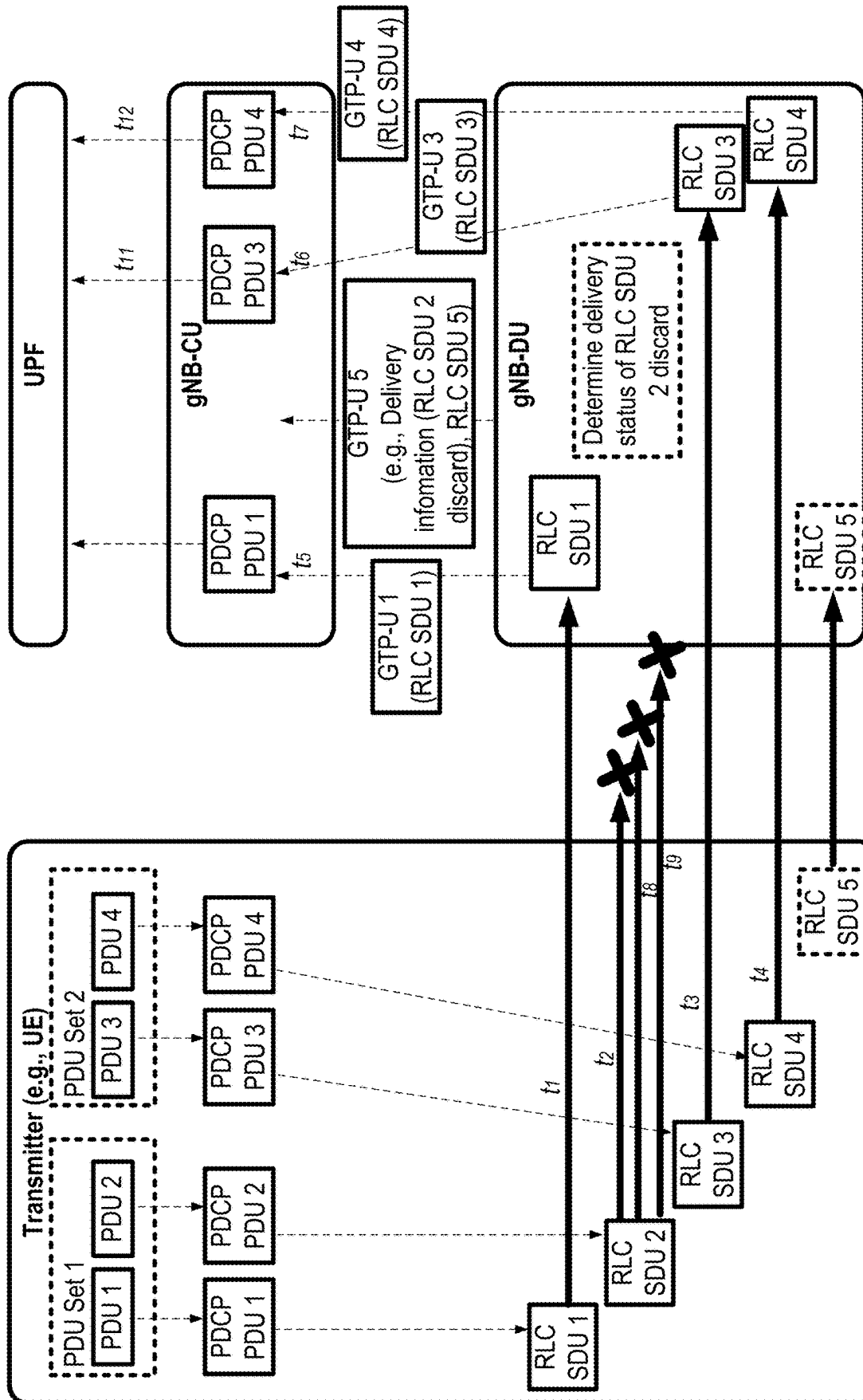


FIG. 20

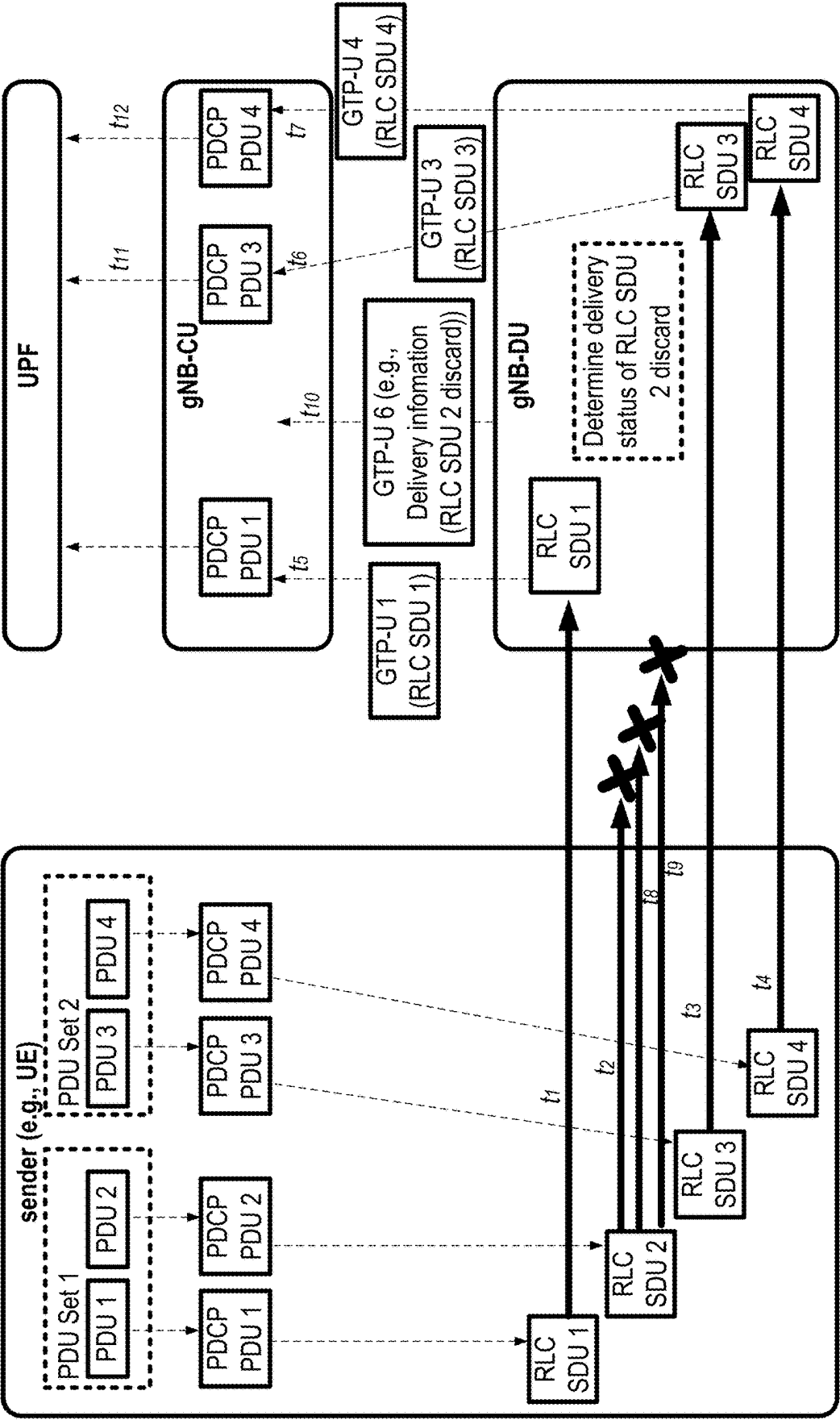


FIG. 21

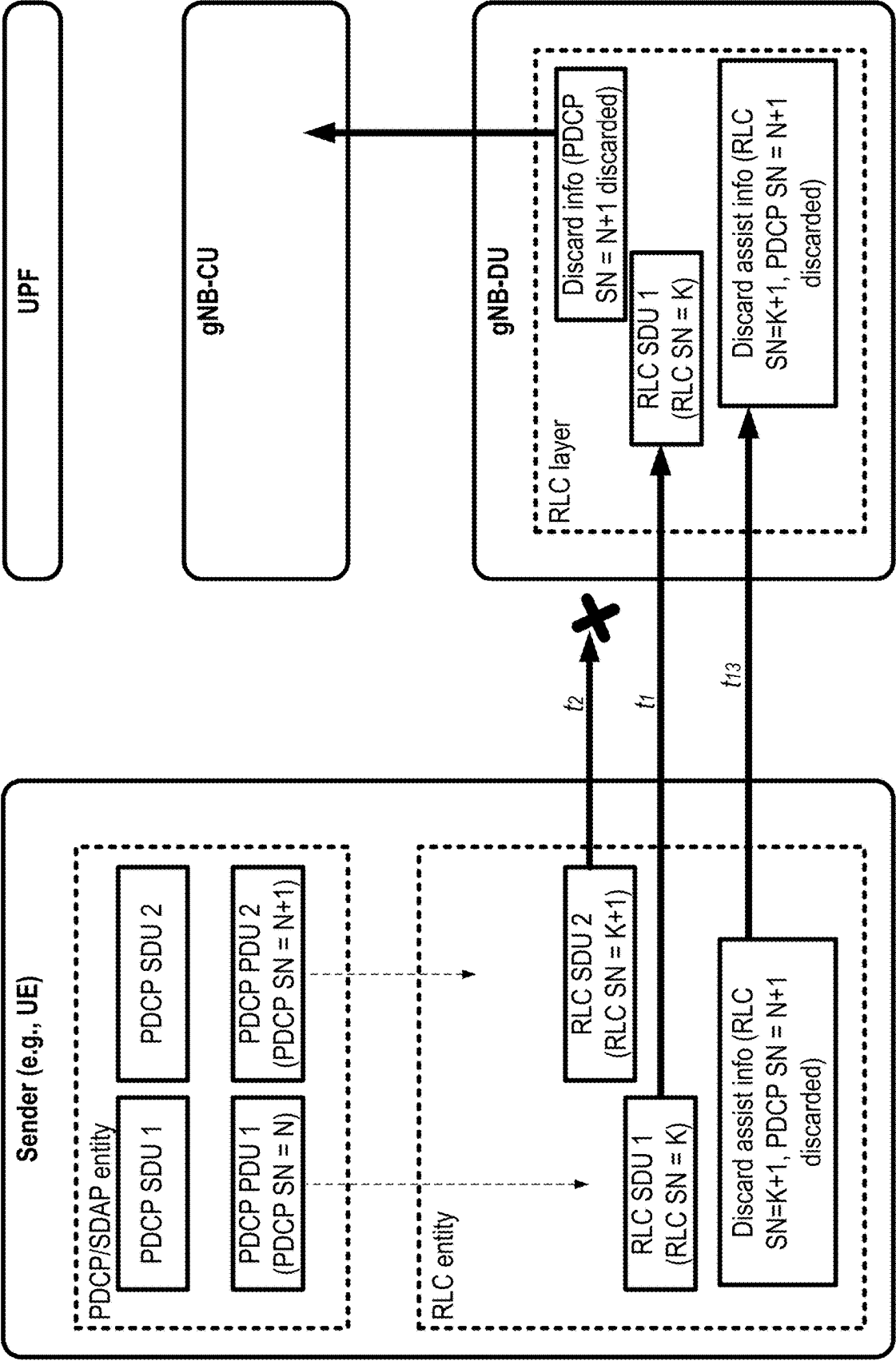


FIG. 22

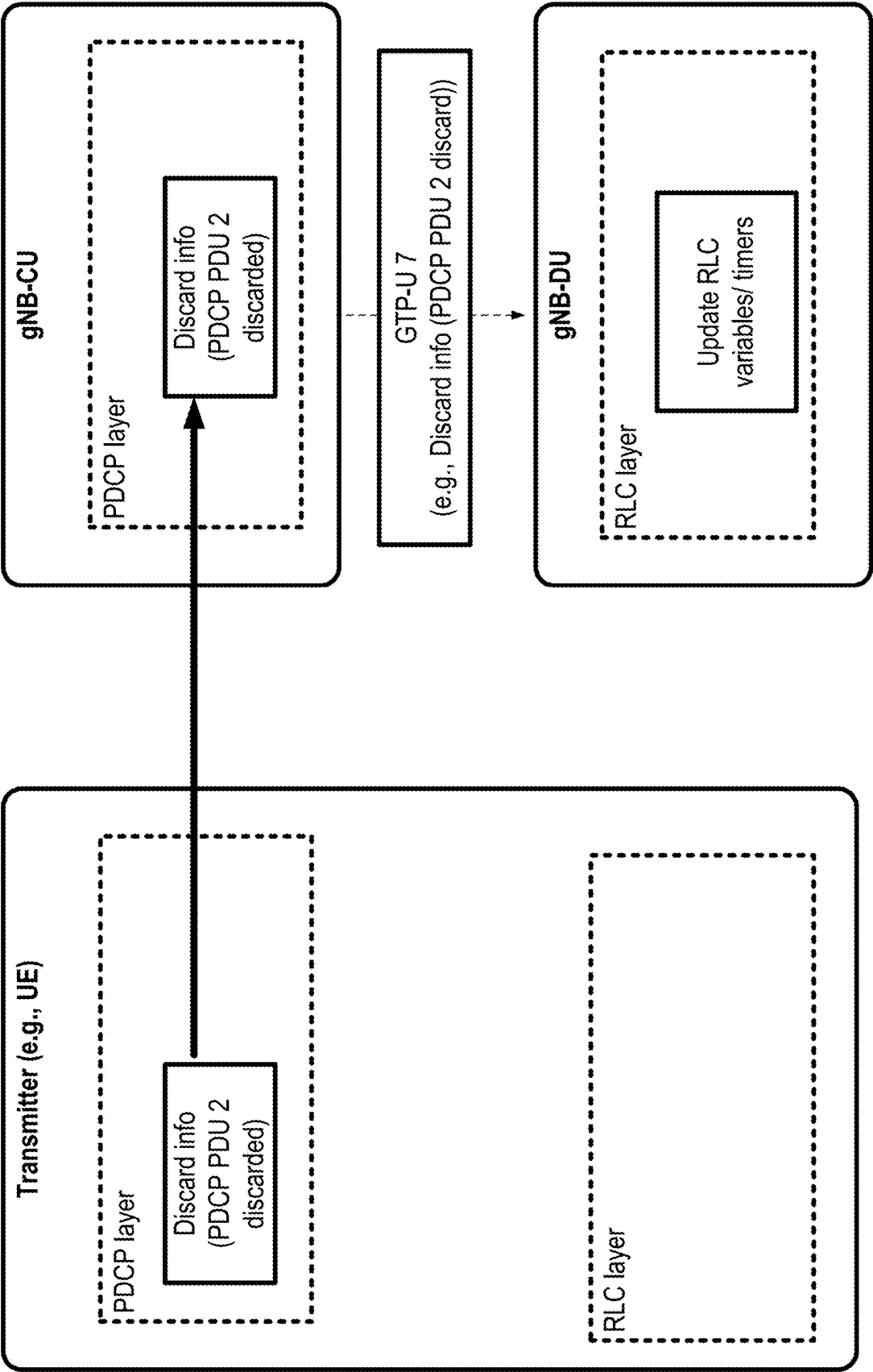


FIG. 23

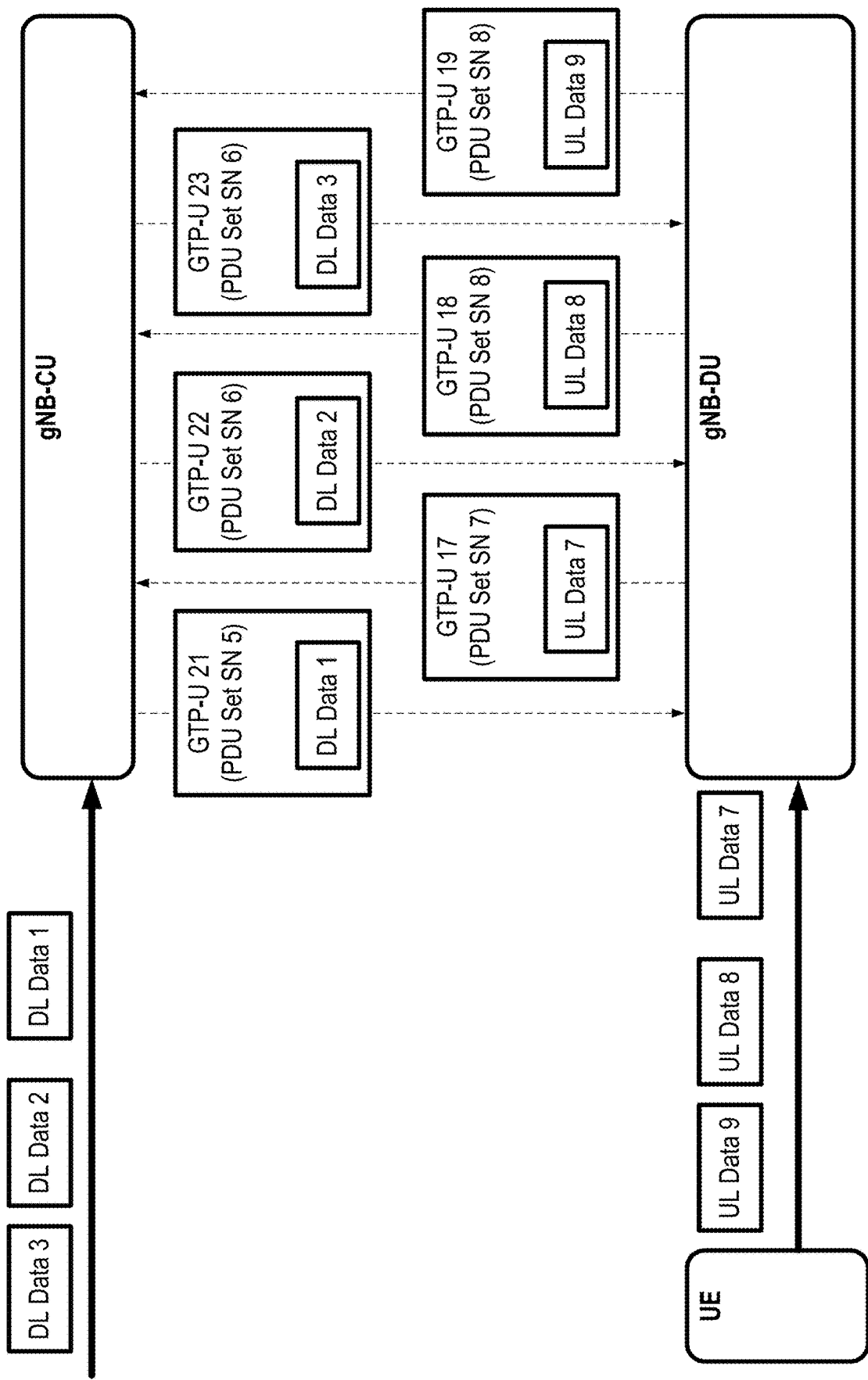


FIG. 24

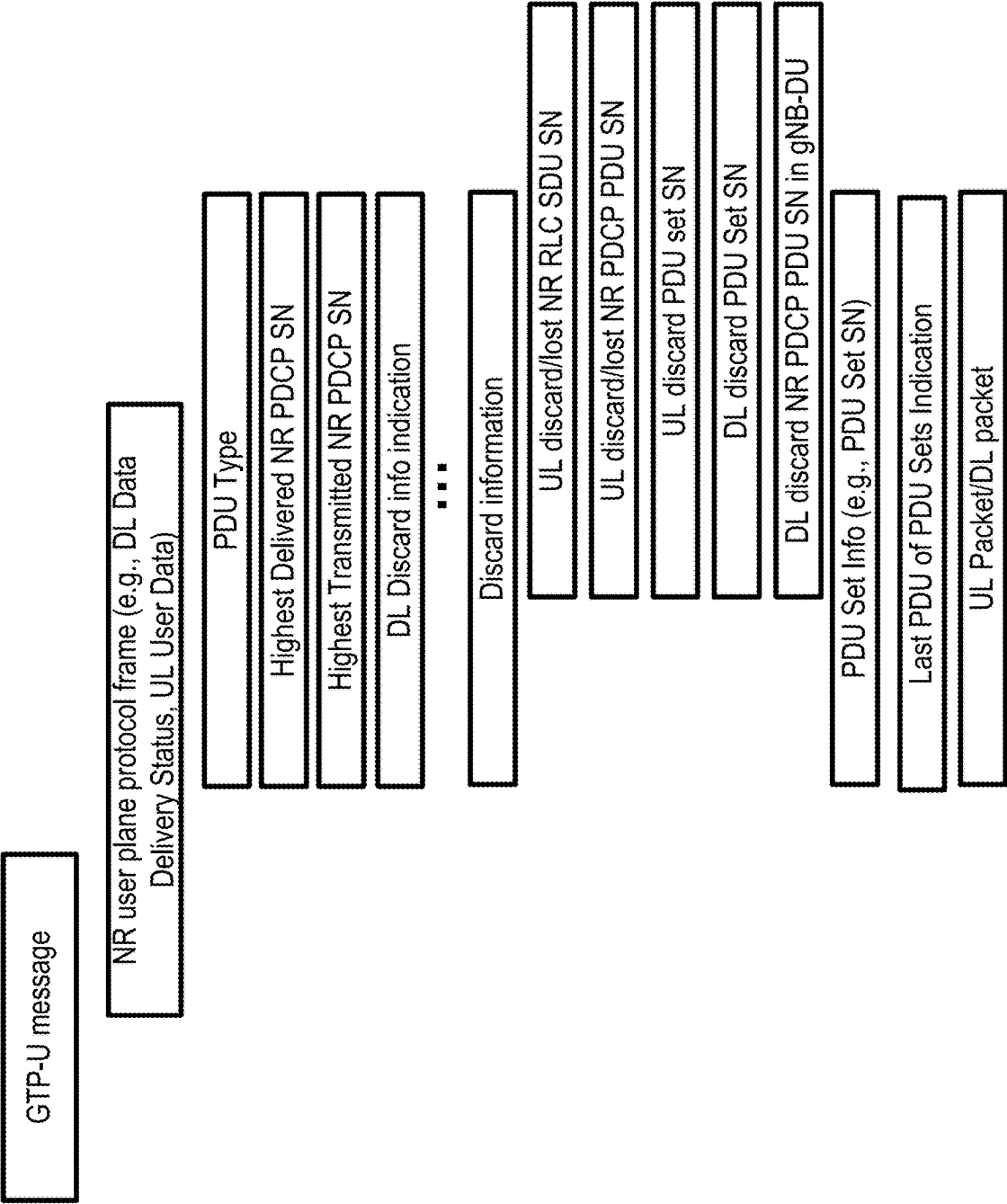


FIG. 25

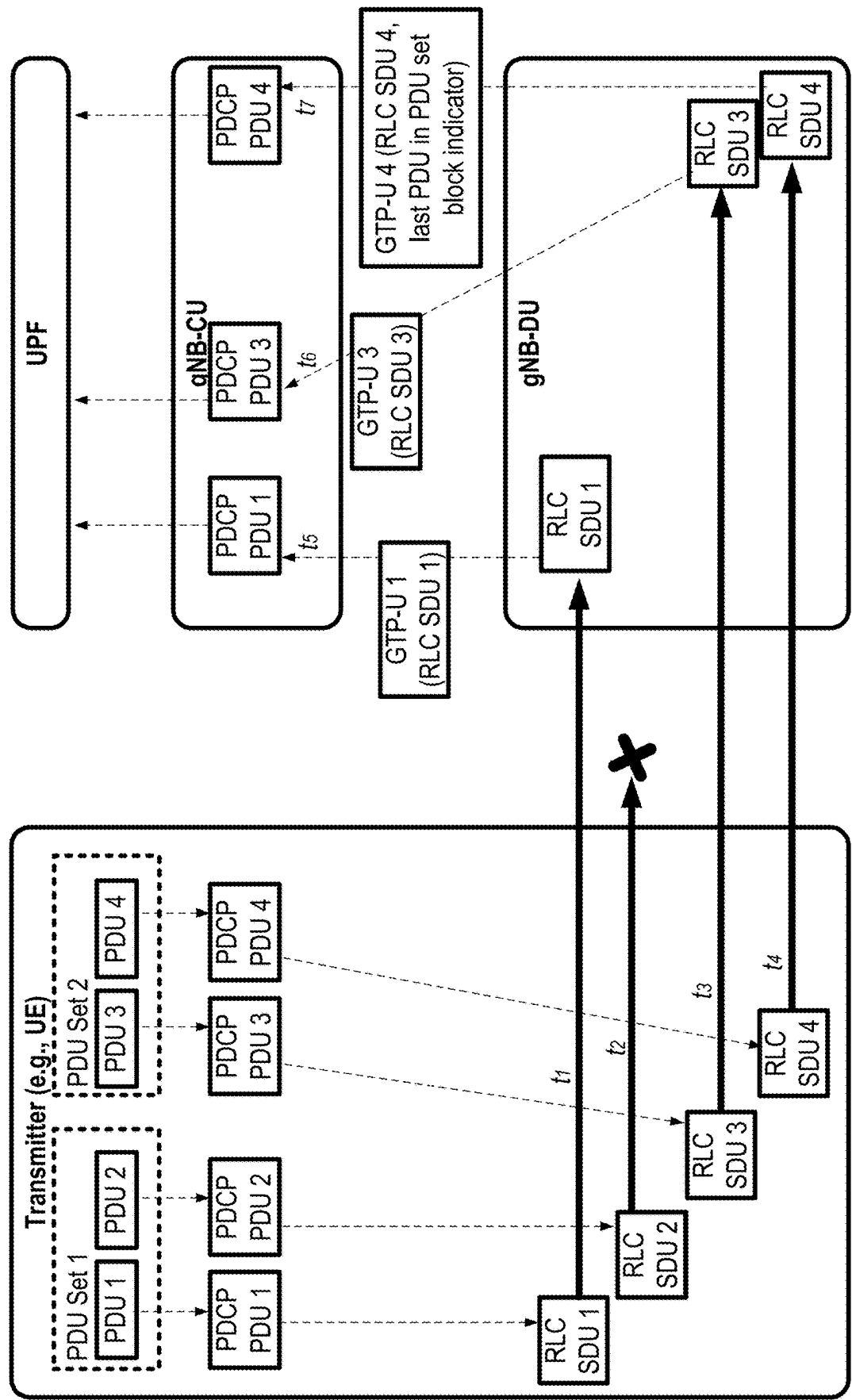


FIG. 26

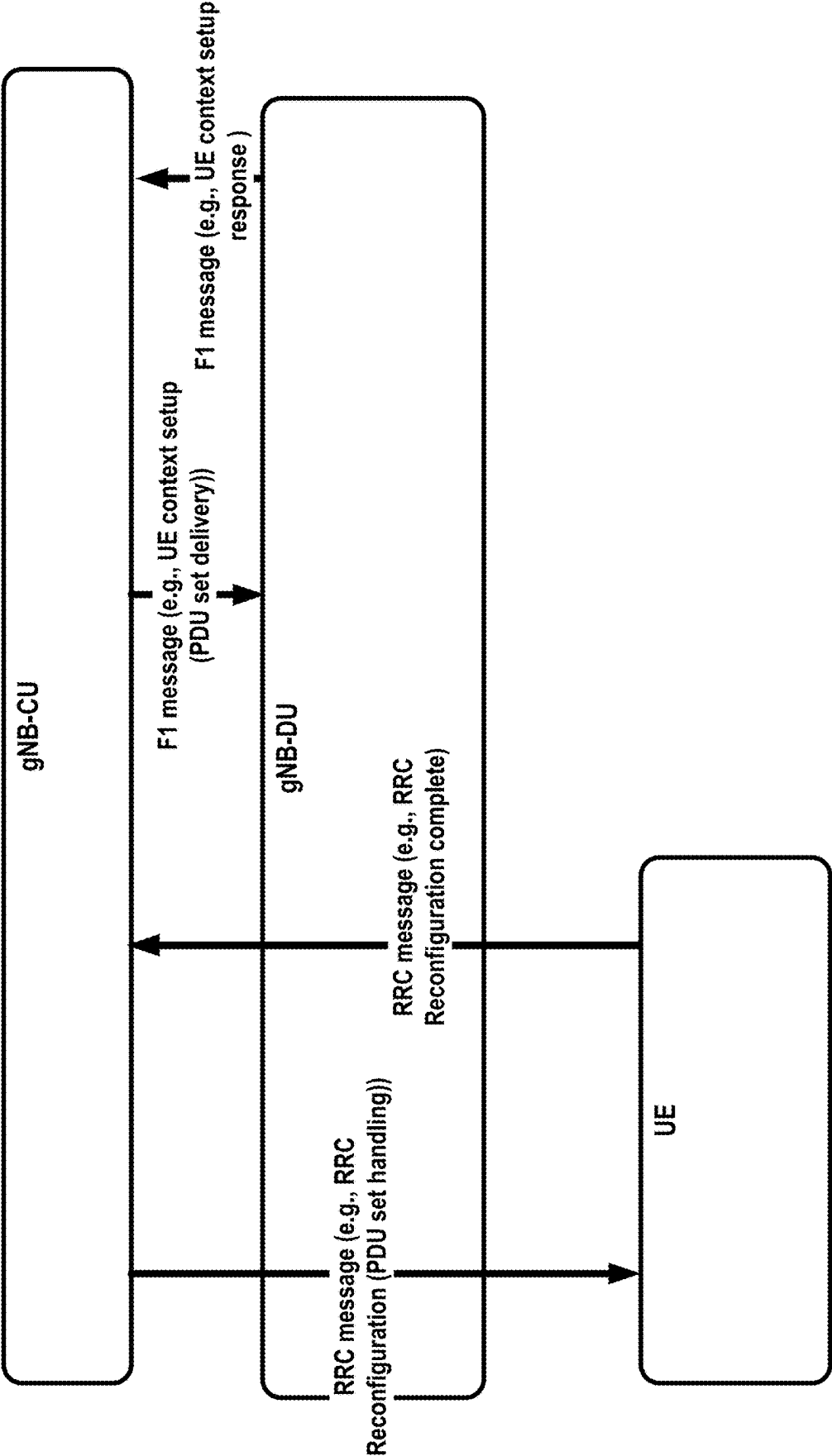


FIG. 27

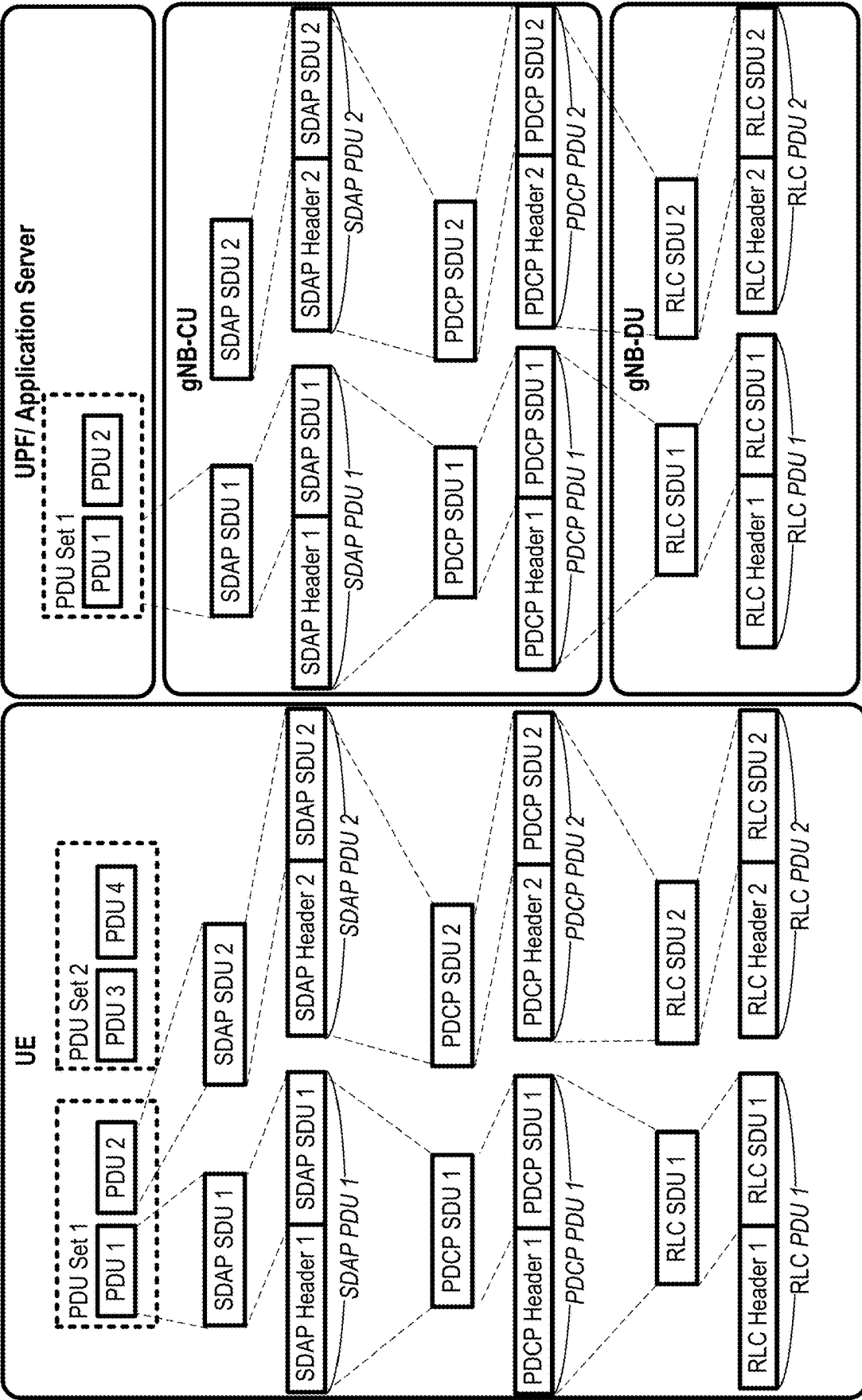


FIG. 28

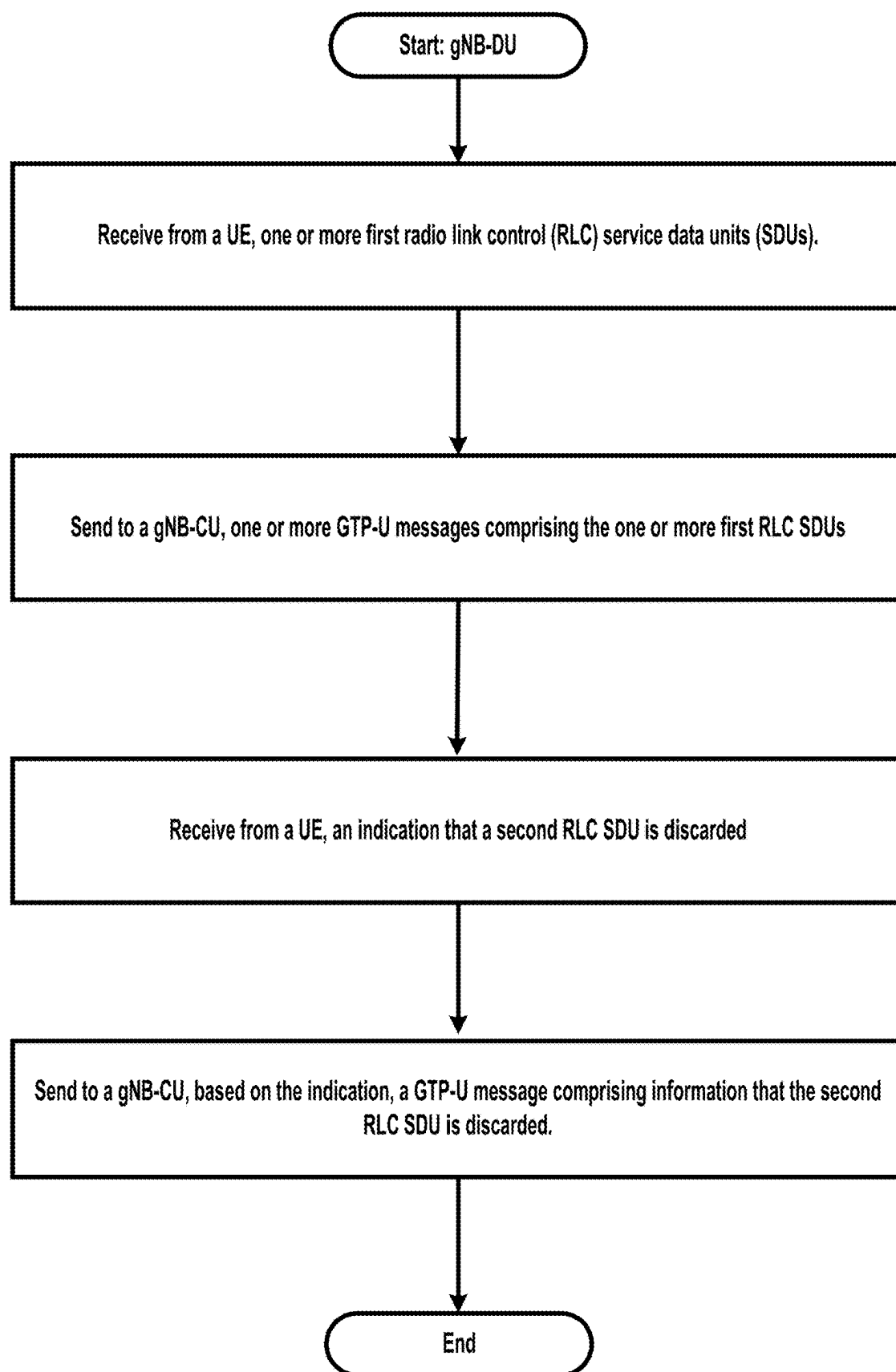
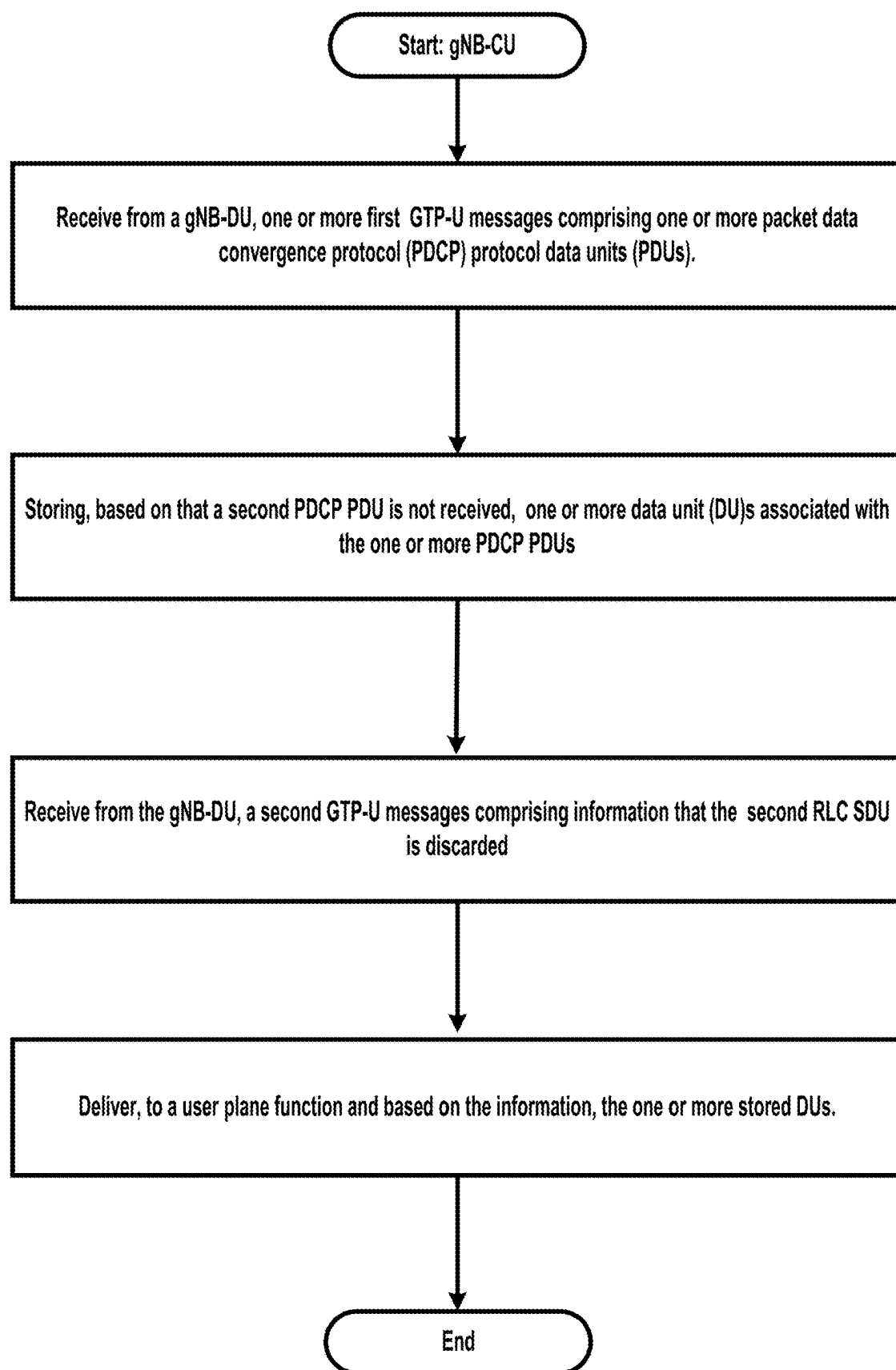


FIG. 29

**FIG. 30**

MEDIA DATA DELIVERY

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of International Application No. PCT/US2023/035401, filed Oct. 18, 2023, which claims the benefit of U.S. Provisional Application No. 63/421,289, filed Nov. 1, 2022, all of which are hereby incorporated by reference in their entireties.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Examples of several of the various embodiments of the present disclosure are described herein with reference to the drawings.

[0003] FIG. 1A and FIG. 1B illustrate example communication networks including an access network and a core network.

[0004] FIG. 2A, FIG. 2B, FIG. 2C, and FIG. 2D illustrate various examples of a framework for a service-based architecture within a core network.

[0005] FIG. 3 illustrates an example communication network including core network functions.

[0006] FIG. 4A and FIG. 4B illustrate example of core network architecture with multiple user plane functions and untrusted access.

[0007] FIG. 5 illustrates an example of a core network architecture for a roaming scenario.

[0008] FIG. 6 illustrates an example of network slicing.

[0009] FIG. 7A, FIG. 7B, and FIG. 7C illustrate a user plane protocol stack, a control plane protocol stack, and services provided between protocol layers of the user plane protocol stack.

[0010] FIG. 8 illustrates an example of a quality of service model for data exchange.

[0011] FIG. 9A, FIG. 9B, FIG. 9C, and FIG. 9D illustrate example states and state transitions of a wireless device.

[0012] FIG. 10 illustrates an example of a registration procedure for a wireless device.

[0013] FIG. 11 illustrates an example of a service request procedure for a wireless device.

[0014] FIG. 12 illustrates an example of a protocol data unit session establishment procedure for a wireless device.

[0015] FIG. 13 illustrates examples of components of the elements in a communications network.

[0016] FIG. 14A, FIG. 14B, FIG. 14C, and FIG. 14D illustrate various examples of physical core network deployments, each having one or more network functions or portions thereof.

[0017] FIG. 15 is a diagram of an aspect of an example embodiment of the present disclosure.

[0018] FIG. 16 is a diagram of an aspect of an example embodiment of the present disclosure.

[0019] FIG. 17 is a diagram of an aspect of an example embodiment of the present disclosure.

[0020] FIG. 18 is a diagram of an aspect of an example embodiment of the present disclosure.

[0021] FIG. 19 is a diagram of an aspect of an example embodiment of the present disclosure.

[0022] FIG. 20 is a diagram of an aspect of an example embodiment of the present disclosure.

[0023] FIG. 21 is a diagram of an aspect of an example embodiment of the present disclosure.

[0024] FIG. 22 is a diagram of an aspect of an example embodiment of the present disclosure.

[0025] FIG. 23 is a diagram of an aspect of an example embodiment of the present disclosure.

[0026] FIG. 24 is a diagram of an aspect of an example embodiment of the present disclosure.

[0027] FIG. 25 is a diagram of an aspect of an example embodiment of the present disclosure.

[0028] FIG. 26 is a diagram of an aspect of an example embodiment of the present disclosure.

[0029] FIG. 27 is a diagram of an aspect of an example embodiment of the present disclosure.

[0030] FIG. 28 is a diagram of an aspect of an example embodiment of the present disclosure.

[0031] FIG. 29 is a diagram of an aspect of an example embodiment of the present disclosure.

[0032] FIG. 30 is a diagram of an aspect of an example embodiment of the present disclosure.

DETAILED DESCRIPTION

[0033] In the present disclosure, various embodiments are presented as examples of how the disclosed techniques may be implemented and/or how the disclosed techniques may be practiced in environments and scenarios. It will be apparent to persons skilled in the relevant art that various changes in form and detail can be made therein without departing from the scope. In fact, after reading the description, it will be apparent to one skilled in the relevant art how to implement alternative embodiments. The present embodiments should not be limited by any of the described exemplary embodiments. The embodiments of the present disclosure will be described with reference to the accompanying drawings. Limitations, features, and/or elements from the disclosed example embodiments may be combined to create further embodiments within the scope of the disclosure. Any figures which highlight the functionality and advantages, are presented for example purposes only. The disclosed architecture is sufficiently flexible and configurable, such that it may be utilized in ways other than that shown. For example, the actions listed in any flowchart may be re-ordered or only optionally used in some embodiments.

[0034] Embodiments may be configured to operate as needed. The disclosed mechanism may be performed when certain criteria are met, for example, in a wireless device, a base station, a radio environment, a network, a combination of the above, and/or the like. Example criteria may be based, at least in part, on for example, wireless device or network node configurations, traffic load, initial system set up, packet sizes, traffic characteristics, a combination of the above, and/or the like. When the one or more criteria are met, various example embodiments may be applied. Therefore, it may be possible to implement example embodiments that selectively implement disclosed protocols.

[0035] A base station may communicate with a mix of wireless devices. Wireless devices and/or base stations may support multiple technologies, and/or multiple releases of the same technology. Wireless devices may have one or more specific capabilities. When this disclosure refers to a base station communicating with a plurality of wireless devices, this disclosure may refer to a subset of the total wireless devices in a coverage area. This disclosure may refer to, for example, a plurality of wireless devices of a given LTE or 5G release with a given capability and in a given sector of the base station. The plurality of wireless

devices in this disclosure may refer to a selected plurality of wireless devices, and/or a subset of total wireless devices in a coverage area which perform according to disclosed methods, and/or the like. There may be a plurality of base stations or a plurality of wireless devices in a coverage area that may not comply with the disclosed methods, for example, those wireless devices or base stations may perform based on older releases of LTE or 5G technology.

[0036] In this disclosure, “a” and “an” and similar phrases refer to a single instance of a particular element, but should not be interpreted to exclude other instances of that element. For example, a bicycle with two wheels may be described as having “a wheel”. Any term that ends with the suffix “(s)” is to be interpreted as “at least one” and/or “one or more.” In this disclosure, the term “may” is to be interpreted as “may, for example.” In other words, the term “may” is indicative that the phrase following the term “may” is an example of one of a multitude of suitable possibilities that may, or may not, be employed by one or more of the various embodiments. The terms “comprises” and “consists of”, as used herein, enumerate one or more components of the element being described. The term “comprises” is interchangeable with “includes” and does not exclude unenumerated components from being included in the element being described. By contrast, “consists of” provides a complete enumeration of the one or more components of the element being described.

[0037] The phrases “based on”, “in response to”, “depending on”, “employing”, “using”, and similar phrases indicate the presence and/or influence of a particular factor and/or condition on an event and/or action, but do not exclude unenumerated factors and/or conditions from also being present and/or influencing the event and/or action. For example, if action X is performed “based on” condition Y, this is to be interpreted as the action being performed “based at least on” condition Y. For example, if the performance of action X is performed when conditions Y and Z are both satisfied, then the performing of action X may be described as being “based on Y”.

[0038] The term “configured” may relate to the capacity of a device whether the device is in an operational or non-operational state. Configured may refer to specific settings in a device that effect the operational characteristics of the device whether the device is in an operational or non-operational state. In other words, the hardware, software, firmware, registers, memory values, and/or the like may be “configured” within a device, whether the device is in an operational or nonoperational state, to provide the device with specific characteristics. Terms such as “a control message to cause in a device” may mean that a control message has parameters that may be used to configure specific characteristics or may be used to implement certain actions in the device, whether the device is in an operational or non-operational state.

[0039] In this disclosure, a parameter may comprise one or more information objects, and an information object may comprise one or more other objects. For example, if parameter J comprises parameter K, and parameter K comprises parameter L, and parameter L comprises parameter M, then J comprises L, and J comprises M. A parameter may be referred to as a field or information element. In an example embodiment, when one or more messages comprise a plurality of parameters, it implies that a parameter in the

plurality of parameters is in at least one of the one or more messages, but does not have to be in each of the one or more messages.

[0040] This disclosure may refer to possible combinations of enumerated elements. For the sake of brevity and legibility, the present disclosure does not explicitly recite each and every permutation that may be obtained by choosing from a set of optional features. The present disclosure is to be interpreted as explicitly disclosing all such permutations. For example, the seven possible combinations of enumerated elements A, B, C consist of: (1) “A”; (2) “B”; (3) “C”; (4) “A and B”; (5) “A and C”; (6) “B and C”; and (7) “A, B, and C”. For the sake of brevity and legibility, these seven possible combinations may be described using any of the following interchangeable formulations: “at least one of A, B, and C”; “at least one of A, B, or C”; “one or more of A, B, and C”; “one or more of A, B, or C”; “A, B, and/or C”. It will be understood that impossible combinations are excluded. For example, “X and/or not-X” should be interpreted as “X or not-X”. It will be further understood that these formulations may describe alternative phrasings of overlapping and/or synonymous concepts, for example, “identifier, identification, and/or ID number”.

[0041] This disclosure may refer to sets and/or subsets. As an example, set X may be a set of elements comprising one or more elements. If every element of X is also an element of Y, then X may be referred to as a subset of Y. In this disclosure, only non-empty sets and subsets are considered. For example, if Y consists of the elements Y1, Y2, and Y3, then the possible subsets of Y are {Y1, Y2, Y3}, {Y1, Y2}, {Y1, Y3}, {Y2, Y3}, {Y1}, {Y2}, and {Y3}.

[0042] FIG. 1A illustrates an example of a communication network 100 in which embodiments of the present disclosure may be implemented. The communication network 100 may comprise, for example, a public land mobile network (PLMN) run by a network operator. As illustrated in FIG. 1A, the communication network 100 includes a wireless device 101, an access network (AN) 102, a core network (CN) 105, and one or more data network (DN)s 108.

[0043] The wireless device 101 may communicate with DN)s 108 via AN 102 and CN 105. In the present disclosure, the term wireless device may refer to and encompass any mobile device or fixed (non-mobile) device for which wireless communication is needed or usable. For example, a wireless device may be a telephone, smart phone, tablet, computer, laptop, sensor, meter, wearable device, Internet of Things (IoT) device, vehicle road side unit (RSU), relay node, automobile, unmanned aerial vehicle, urban air mobility, and/or any combination thereof. The term wireless device encompasses other terminology, including user equipment (UE), user terminal (UT), access terminal (AT), mobile station, handset, wireless transmit and receive unit (WTRU), and/or wireless communication device.

[0044] The AN 102 may connect wireless device 101 to CN 105 in any suitable manner. The communication direction from the AN 102 to the wireless device 101 is known as the downlink and the communication direction from the wireless device 101 to AN 102 is known as the uplink. Downlink transmissions may be separated from uplink transmissions using frequency division duplexing (FDD), time-division duplexing (TDD), and/or some combination of the two duplexing techniques. The AN 102 may connect to wireless device 101 through radio communications over an air interface. An access network that at least partially oper-

ates over the air interface may be referred to as a radio access network (RAN). The CN **105** may set up one or more end-to-end connection between wireless device **101** and the one or more DNs **108**. The CN **105** may authenticate wireless device **101** and provide charging functionality.

[0045] In the present disclosure, the term base station may refer to and encompass any element of AN **102** that facilitates communication between wireless device **101** and AN **102**. Access networks and base stations have many different names and implementations. The base station may be a terrestrial base station fixed to the earth. The base station may be a mobile base station with a moving coverage area. The base station may be in space, for example, on board a satellite. For example, WiFi and other standards may use the term access point. As another example, the Third-Generation Partnership Project (3GPP) has produced specifications for three generations of mobile networks, each of which uses different terminology. Third Generation (3G) and/or Universal Mobile Telecommunications System (UMTS) standards may use the term Node B. 4G, Long Term Evolution (LTE), and/or Evolved Universal Terrestrial Radio Access (E-UTRA) standards may use the term Evolved Node B (eNB). 5G and/or New Radio (NR) standards may describe AN **102** as a next-generation radio access network (NG-RAN) and may refer to base stations as Next Generation eNB (ng-eNB) and/or Generation Node B (gNB). Future standards (for example, 6G, 7G, 8G) may use new terminology to refer to the elements which implement the methods described in the present disclosure (e.g., wireless devices, base stations, ANs, CNs, and/or components thereof). A base station may be implemented as a repeater or relay node used to extend the coverage area of a donor node. A repeater node may amplify and rebroadcast a radio signal received from a donor node. A relay node may perform the same/similar functions as a repeater node but may decode the radio signal received from the donor node to remove noise before amplifying and rebroadcasting the radio signal.

[0046] The AN **102** may include one or more base stations, each having one or more coverage areas. The geographical size and/or extent of a coverage area may be defined in terms of a range at which a receiver of AN **102** can successfully receive transmissions from a transmitter (e.g., wireless device **101**) operating within the coverage area (and/or vice-versa). The coverage areas may be referred to as sectors or cells (although in some contexts, the term cell refers to the carrier frequency used in a particular coverage area, rather than the coverage area itself). Base stations with large coverage areas may be referred to as macrocell base stations. Other base stations cover smaller areas, for example, to provide coverage in areas with weak macrocell coverage, or to provide additional coverage in areas with high traffic (sometimes referred to as hotspots). Examples of small cell base stations include, in order of decreasing coverage area, microcell base stations, picocell base stations, and femtocell base stations or home base stations. Together, the coverage areas of the base stations may provide radio coverage to wireless device **101** over a wide geographic area to support wireless device mobility.

[0047] A base station may include one or more sets of antennas for communicating with the wireless device **101** over the air interface. Each set of antennas may be separately controlled by the base station. Each set of antennas may have a corresponding coverage area. As an example, a base station may include three sets of antennas to respectively

control three coverage areas on three different sides of the base station. The entirety of the base station (and its corresponding antennas) may be deployed at a single location. Alternatively, a controller at a central location may control one or more sets of antennas at one or more distributed locations. The controller may be, for example, a baseband processing unit that is part of a centralized or cloud RAN architecture. The baseband processing unit may be either centralized in a pool of baseband processing units or virtualized. A set of antennas at a distributed location may be referred to as a remote radio head (RRH).

[0048] FIG. 1B illustrates another example communication network **150** in which embodiments of the present disclosure may be implemented. The communication network **150** may comprise, for example, a PLMN run by a network operator. As illustrated in FIG. 1B, communication network **150** includes UEs **151**, a next generation radio access network (NG-RAN) **152**, a 5G core network (5G-CN) **155**, and one or more DNs **158**. The NG-RAN **152** includes one or more base stations, illustrated as generation node Bs (gNBs) **152A** and next generation evolved Node Bs (ng eNBs) **152B**. The 5G-CN **155** includes one or more network functions (NFs), including control plane functions **155A** and user plane functions **155B**. The one or more DNs **158** may comprise public DNS (e.g., the Internet), private DNs, and/or intra-operator DNs. Relative to corresponding components illustrated in FIG. 1A, these components may represent specific implementations and/or terminology.

[0049] The base stations of the NG-RAN **152** may be connected to the UEs **151** via Uu interfaces. The base stations of the NG-RAN **152** may be connected to each other via Xn interfaces. The base stations of the NG-RAN **152** may be connected to 5G CN **155** via NG interfaces. The Uu interface may include an air interface. The NG and Xn interfaces may include an air interface, or may consist of direct physical connections and/or indirect connections over an underlying transport network (e.g., an internet protocol (IP) transport network).

[0050] Each of the Uu, Xn, and NG interfaces may be associated with a protocol stack. The protocol stacks may include a user plane (UP) and a control plane (CP). Generally, user plane data may include data pertaining to users of the UEs **151**, for example, internet content downloaded via a web browser application, sensor data uploaded via a tracking application, or email data communicated to or from an email server. Control plane data, by contrast, may comprise signaling and messages that facilitate packaging and routing of user plane data so that it can be exchanged with the DN(s). The NG interface, for example, may be divided into an NG user plane interface (NG-U) and an NG control plane interface (NG-C). The NG-U interface may provide delivery of user plane data between the base stations and the one or more user plane network functions **155B**. The NG-C interface may be used for control signaling between the base stations and the one or more control plane network functions **155A**. The NG-C interface may provide, for example, NG interface management, UE context management, UE mobility management, transport of NAS messages, paging, PDU session management, and configuration transfer and/or warning message transmission. In some cases, the NG-C interface may support transmission of user data (for example, a small data transmission for an IoT device).

[0051] One or more of the base stations of the NG-RAN **152** may be split into a central unit (CU) and one or more

distributed units (DUs). A CU may be coupled to one or more DUs via an F1 interface. The CU may handle one or more upper layers in the protocol stack and the DU may handle one or more lower layers in the protocol stack. For example, the CU may handle RRC, PDCP, and SDAP, and the DU may handle RLC, MAC, and PHY. The one or more DUs may be in geographically diverse locations relative to the CU and/or each other. Accordingly, the CU/DU split architecture may permit increased coverage and/or better coordination.

[0052] The gNBs 152A and ng-eNBs 152B may provide different user plane and control plane protocol termination towards the UEs 151. For example, the gNB 154A may provide new radio (NR) protocol terminations over a Uu interface associated with a first protocol stack. The ng-eNBs 152B may provide Evolved UMTS Terrestrial Radio Access (E-UTRA) protocol terminations over a Uu interface associated with a second protocol stack.

[0053] The 5G-CN 155 may authenticate UEs 151, set up end-to-end connections between UEs 151 and the one or more DN 158, and provide charging functionality. The 5G-CN 155 may be based on a service-based architecture, in which the NFs making up the 5G-CN 155 offer services to each other and to other elements of the communication network 150 via interfaces. The 5G-CN 155 may include any number of other NFs and any number of instances of each NF.

[0054] FIG. 2A, FIG. 2B, FIG. 2C, and FIG. 2D illustrate various examples of a framework for a service-based architecture within a core network. In a service-based architecture, a service may be sought by a service consumer and provided by a service producer. Prior to obtaining a particular service, an NF may determine where such a service can be obtained. To discover a service, the NF may communicate with a network repository function (NRF). As an example, an NF that provides one or more services may register with a network repository function (NRF). The NRF may store data relating to the one or more services that the NF is prepared to provide to other NFs in the service-based architecture. A consumer NF may query the NRF to discover a producer NF (for example, by obtaining from the NRF a list of NF instances that provide a particular service).

[0055] In the example of FIG. 2A, an NF 211 (a consumer NF in this example) may send a request 221 to an NF 212 (a producer NF). The request 221 may be a request for a particular service and may be sent based on a discovery that NF 212 is a producer of that service. The request 221 may comprise data relating to NF 211 and/or the requested service. The NF 212 may receive request 221, perform one or more actions associated with the requested service (e.g., retrieving data), and provide a response 221. The one or more actions performed by the NF 212 may be based on request data included in the request 221, data stored by NF 212, and/or data retrieved by NF 212. The response 222 may notify NF 211 that the one or more actions have been completed. The response 222 may comprise response data relating to NF 212, the one or more actions, and/or the requested service.

[0056] In the example of FIG. 2B, an NF 231 sends a request 241 to an NF 232. In this example, part of the service produced by NF 232 is to send a request 242 to an NF 233. The NF 233 may perform one or more actions and provide a response 243 to NF 232. Based on response 243, NF 232 may send a response 244 to NF 231. It will be understood

from FIG. 2B that a single NF may perform the role of producer of services, consumer of services, or both. A particular NF service may include any number of nested NF services produced by one or more other NFs.

[0057] FIG. 2C illustrates examples of subscribe-notify interactions between a consumer NF and a producer NF. In FIG. 2C, an NF 251 sends a subscription 261 to an NF 252. An NF 253 sends a subscription 262 to the NF 252. Two NFs are shown in FIG. 2C for illustrative purposes (to demonstrate that the NF 252 may provide multiple subscription services to different NFs), but it will be understood that a subscribe-notify interaction only requires one subscriber. The NFs 251, 253 may be independent from one another. For example, the NFs 251, 253 may independently discover NF 252 and/or independently determine to subscribe to the service offered by NF 252. In response to receipt of a subscription, the NF 252 may provide a notification to the subscribing NF. For example, NF 252 may send a notification 263 to NF 251 based on subscription 261 and may send a notification 264 to NF 253 based on subscription 262.

[0058] As shown in the example illustration of FIG. 2C, the sending of the notifications 263, 264 may be based on a determination that a condition has occurred. For example, the notifications 263, 264 may be based on a determination that a particular event has occurred, a determination that a particular condition is outstanding, and/or a determination that a duration of time associated with the subscription has elapsed (for example, a period associated with a subscription for periodic notifications). As shown in the example illustration of FIG. 2C, NF 252 may send notifications 263, 264 to NFs 251, 253 simultaneously and/or in response to the same condition. However, it will be understood that the NF 252 may provide notifications at different times and/or in response to different notification conditions. In an example, the NF 251 may request a notification when a certain parameter, as measured by the NF 252, exceeds a first threshold, and the NF 252 may request a notification when the parameter exceeds a second threshold different from the first threshold. In an example, a parameter of interest and/or a corresponding threshold may be indicated in the subscriptions 261, 262.

[0059] FIG. 2D illustrates another example of a subscribe-notify interaction. In FIG. 2D, an NF 271 sends a subscription 281 to an NF 272. In response to receipt of subscription 281 and/or a determination that a notification condition has occurred, NF 272 may send a notification 284. The notification 284 may be sent to an NF 273. Unlike the example in FIG. 2C (in which a notification is sent to the subscribing NF), FIG. 2D demonstrates that a subscription and its corresponding notification may be associated with different NFs. For example, NF 271 may subscribe to the service provided by NF 272 on behalf of NF 273.

[0060] FIG. 3 illustrates another example communication network 300 in which embodiments of the present disclosure may be implemented. Communication network 300 includes a user equipment (UE) 301, an access network (AN) 302, and a data network (DN) 308. The remaining elements depicted in FIG. 3 may be included in and/or associated with a core network. Each element of the core network may be referred to as a network function (NF).

[0061] The NFs depicted in FIG. 3 include a user plane function (UPF) 305, an access and mobility management function (AMF) 312, a session management function (SMF) 314, a policy control function (PCF) 320, a network reposi-

tory function (NRF) 330, a network exposure function (NEF) 340, a unified data management (UDM) 350, an authentication server function (AUSF) 360, a network slice selection function (NSSF) 370, a charging function (CHF) 380, a network data analytics function (NWDAF) 390, and an application function (AF) 399. The UPF 305 may be a user-plane core network function, whereas the NFs 312, 314, and 320-390 may be control-plane core network functions. Although not shown in the example of FIG. 3, the core network may include additional instances of any of the NFs depicted and/or one or more different NF types that provide different services. Other examples of NF type include a gateway mobile location center (GMLC), a location management function (LMF), an operations, administration, and maintenance function (OAM), a public warning system (PWS), a short message service function (SMSF), a unified data repository (UDR), and an unstructured data storage function (UDSF).

[0062] Each element depicted in FIG. 3 has an interface with at least one other element. The interface may be a logical connection rather than, for example, a direct physical connection. Any interface may be identified using a reference point representation and/or a service-based representation. In a reference point representation, the letter 'N' is followed by a numeral, indicating an interface between two specific elements. For example, as shown in FIG. 3, AN 302 and UPF 305 interface via 'N3', whereas UPF 305 and DN 308 interface via 'N6'. By contrast, in a service-based representation, the letter 'N' is followed by letters. The letters identify an NF that provides services to the core network. For example, PCF 320 may provide services via interface 'Npcf'. The PCF 320 may provide services to any NF in the core network via 'Npcf'. Accordingly, a service-based representation may correspond to a bundle of reference point representations. For example, the Npcf interface between PCF 320 and the core network generally may correspond to an N7 interface between PCF 320 and SMF 314, an N30 interface between PCF 320 and NEF 340, etc.

[0063] The UPF 305 may serve as a gateway for user plane traffic between AN 302 and DN 308. The UE 301 may connect to UPF 305 via a Uu interface and an N3 interface (also described as NG-U interface). The UPF 305 may connect to DN 308 via an N6 interface. The UPF 305 may connect to one or more other UPFs (not shown) via an N9 interface. The UE 301 may be configured to receive services through a protocol data unit (PDU) session, which is a logical connection between UE 301 and DN 308. The UPF 305 (or a plurality of UPFs if desired) may be selected by SMF 314 to handle a particular PDU session between UE 301 and DN 308. The SMF 314 may control the functions of UPF 305 with respect to the PDU session. The SMF 314 may connect to UPF 305 via an N4 interface. The UPF 305 may handle any number of PDU sessions associated with any number of UEs (via any number of ANs). For purposes of handling the one or more PDU sessions, UPF 305 may be controlled by any number of SMFs via any number of corresponding N4 interfaces.

[0064] The AMF 312 depicted in FIG. 3 may control UE access to the core network. The UE 301 may register with the network via AMF 312. It may be necessary for UE 301 to register prior to establishing a PDU session. The AMF 312 may manage a registration area of UE 301, enabling the network to track the physical location of UE 301 within the network. For a UE in connected mode, AMF 312 may

manage UE mobility, for example, handovers from one AN or portion thereof to another. For a UE in idle mode, AMF 312 may perform registration updates and/or page the UE to transition the UE to connected mode.

[0065] The AMF 312 may receive, from UE 301, non-access stratum (NAS) messages transmitted in accordance with NAS protocol. NAS messages relate to communications between UE 301 and the core network. Although NAS messages may be relayed to AMF 312 via AN 302, they may be described as communications via the N1 interface. NAS messages may facilitate UE registration and mobility management, for example, by authenticating, identifying, configuring, and/or managing a connection of UE 301. NAS messages may support session management procedures for maintaining user plane connectivity and quality of service (QoS) of a session between UE 301 and DN 309. If the NAS message involves session management, AMF 312 may send the NAS message to SMF 314. NAS messages may be used to transport messages between UE 301 and other components of the core network (e.g., core network components other than AMF 312 and SMF 314). The AMF 312 may act on a particular NAS message itself, or alternatively, forward the NAS message to an appropriate core network function (e.g., SMF 314, etc.)

[0066] The SMF 314 depicted in FIG. 3 may establish, modify, and/or release a PDU session based on messaging received UE 301. The SMF 314 may allocate, manage, and/or assign an IP address to UE 301, for example, upon establishment of a PDU session. There may be multiple SMFs in the network, each of which may be associated with a respective group of wireless devices, base stations, and/or UPFs. A UE with multiple PDU sessions may be associated with a different SMF for each PDU session. As noted above, SMF 314 may select one or more UPFs to handle a PDU session and may control the handling of the PDU session by the selected UPF by providing rules for packet handling (PDR, FAR, QER, etc.). Rules relating to QoS and/or charging for a particular PDU session may be obtained from PCF 320 and provided to UPF 305.

[0067] The PCF 320 may provide, to other NFs, services relating to policy rules. The PCF 320 may use subscription data and information about network conditions to determine policy rules and then provide the policy rules to a particular NF which may be responsible for enforcement of those rules. Policy rules may relate to policy control for access and mobility, and may be enforced by the AMF. Policy rules may relate to session management, and may be enforced by the SMF 314. Policy rules may be, for example, network-specific, wireless device-specific, session-specific, or data flow-specific.

[0068] The NRF 330 may provide service discovery. The NRF 330 may belong to a particular PLMN. The NRF 330 may maintain NF profiles relating to other NFs in the communication network 300. The NF profile may include, for example, an address, PLMN, and/or type of the NF, a slice identifier, a list of the one or more services provided by the NF, and the authorization required to access the services.

[0069] The NEF 340 depicted in FIG. 3 may provide an interface to external domains, permitting external domains to selectively access the control plane of the communication network 300. The external domain may comprise, for example, third-party network functions, application functions, etc. The NEF 340 may act as a proxy between external elements and network functions such as AMF 312, SMF

314, PCF **320**, UDM **350**, etc. As an example, NEF **340** may determine a location or reachability status of UE **301** based on reports from AMF **312**, and provide status information to an external element. As an example, an external element may provide, via NEF **340**, information that facilitates the setting of parameters for establishment of a PDU session. The NEF **340** may determine which data and capabilities of the control plane are exposed to the external domain. The NEF **340** may provide secure exposure that authenticates and/or authorizes an external entity to which data or capabilities of the communication network **300** are exposed. The NEF **340** may selectively control the exposure such that the internal architecture of the core network is hidden from the external domain.

[0070] The UDM **350** may provide data storage for other NFs. The UDM **350** may permit a consolidated view of network information that may be used to ensure that the most relevant information can be made available to different NFs from a single resource. The UDM **350** may store and/or retrieve information from a unified data repository (UDR). For example, UDM **350** may obtain user subscription data relating to UE **301** from the UDR.

[0071] The AUSF **360** may support mutual authentication of UE **301** by the core network and authentication of the core network by UE **301**. The AUSF **360** may perform key agreement procedures and provide keying material that can be used to improve security.

[0072] The NSSF **370** may select one or more network slices to be used by the UE **301**. The NSSF **370** may select a slice based on slice selection information. For example, the NSSF **370** may receive Single Network Slice Selection Assistance Information (S-NSSAI) and map the S-NSSAI to a network slice instance identifier (NSI).

[0073] The CHF **380** may control billing-related tasks associated with UE **301**. For example, UPF **305** may report traffic usage associated with UE **301** to SMF **314**. The SMF **314** may collect usage data from UPF **305** and one or more other UPFs. The usage data may indicate how much data is exchanged, what DN the data is exchanged with, a network slice associated with the data, or any other information that may influence billing. The SMF **314** may share the collected usage data with the CHF. The CHF may use the collected usage data to perform billing-related tasks associated with UE **301**. The CHF may, depending on the billing status of UE **301**, instruct SMF **314** to limit or influence access of UE **301** and/or to provide billing-related notifications to UE **301**.

[0074] The NWDAF **390** may collect and analyze data from other network functions and offer data analysis services to other network functions. As an example, NWDAF **390** may collect data relating to a load level for a particular network slice instance from UPF **305**, AMF **312**, and/or SMF **314**. Based on the collected data, NWDAF **390** may provide load level data to the PCF **320** and/or NSSF **370**, and/or notify the PC **220** and/or NSSF **370** if load level for a slice reaches and/or exceeds a load level threshold.

[0075] The AF **399** may be outside the core network, but may interact with the core network to provide information relating to the QoS requirements or traffic routing preferences associated with a particular application. The AF **399** may access the core network based on the exposure constraints imposed by the NEF **340**. However, an operator of the core network may consider the AF **399** to be a trusted domain that can access the network directly.

[0076] FIGS. **4A**, **4B**, and **5** illustrate other examples of core network architectures that are analogous in some respects to the core network architecture **300** depicted in FIG. **3**. For conciseness, some of the core network elements depicted in FIG. **3** are omitted. Many of the elements depicted in FIGS. **4A**, **4B**, and **5** are analogous in some respects to elements depicted in FIG. **3**. For conciseness, some of the details relating to their functions or operation are omitted.

[0077] FIG. **4A** illustrates an example of a core network architecture **400A** comprising an arrangement of multiple UPFs. Core network architecture **400A** includes a UE **401**, an AN **402**, an AMF **412**, and an SMF **414**. Unlike previous examples of core network architectures described above, FIG. **4A** depicts multiple UPFs, including a UPF **405**, a UPF **406**, and a UPF **407**, and multiple DNs, including a DN **408** and a DN **409**. Each of the multiple UPFs **405**, **406**, **407** may communicate with the SMF **414** via an N4 interface. The DNs **408**, **409** communicate with the UPFs **405**, **406**, respectively, via N6 interfaces. As shown in FIG. **4A**, the multiple UPFs **405**, **406**, **407** may communicate with one another via N9 interfaces.

[0078] The UPFs **405**, **406**, **407** may perform traffic detection, in which the UPFs identify and/or classify packets. Packet identification may be performed based on packet detection rules (PDR) provided by the SMF **414**. A PDR may include packet detection information comprising one or more of: a source interface, a UE IP address, core network (CN) tunnel information (e.g., a CN address of an N3/N9 tunnel corresponding to a PDU session), a network instance identifier, a quality of service flow identifier (QFI), a filter set (for example, an IP packet filter set or an ethernet packet filter set), and/or an application identifier.

[0079] In addition to indicating how a particular packet is to be detected, a PDR may further indicate rules for handling the packet upon detection thereof. The rules may include, for example, forwarding action rules (FARs), multi-access rules (MARs), usage reporting rules (URRs), QoS enforcement rules (QERs), etc. For example, the PDR may comprise one or more FAR identifiers, MAR identifiers, URR identifiers, and/or QER identifiers. These identifiers may indicate the rules that are prescribed for the handling of a particular detected packet.

[0080] The UPF **405** may perform traffic forwarding in accordance with a FAR. For example, the FAR may indicate that a packet associated with a particular PDR is to be forwarded, duplicated, dropped, and/or buffered. The FAR may indicate a destination interface, for example, “access” for downlink or “core” for uplink. If a packet is to be buffered, the FAR may indicate a buffering action rule (BAR). As an example, UPF **405** may perform data buffering of a certain number downlink packets if a PDU session is deactivated.

[0081] The UPF **405** may perform QoS enforcement in accordance with a QER. For example, the QER may indicate a guaranteed bitrate that is authorized and/or a maximum bitrate to be enforced for a packet associated with a particular PDR. The QER may indicate that a particular guaranteed and/or maximum bitrate may be for uplink packets and/or downlink packets. The UPF **405** may mark packets belonging to a particular QoS flow with a corresponding QFI. The marking may enable a recipient of the packet to determine a QoS of the packet.

[0082] The UPF 405 may provide usage reports to the SMF 414 in accordance with a URR. The URR may indicate one or more triggering conditions for generation and reporting of the usage report, for example, immediate reporting, periodic reporting, a threshold for incoming uplink traffic, or any other suitable triggering condition. The URR may indicate a method for measuring usage of network resources, for example, data volume, duration, and/or event.

[0083] As noted above, the DNs 408, 409 may comprise public DNS (e.g., the Internet), private DNs (e.g., private, internal corporate-owned DNs), and/or intra-operator DNs. Each DN may provide an operator service and/or a third-party service. The service provided by a DN may be the Internet, an IP multimedia subsystem (IMS), an augmented or virtual reality network, an edge computing or mobile edge computing (MEC) network, etc. Each DN may be identified using a data network name (DNN). The UE 401 may be configured to establish a first logical connection with DN 408 (a first PDU session), a second logical connection with DN 409 (a second PDU session), or both simultaneously (first and second PDU sessions).

[0084] Each PDU session may be associated with at least one UPF configured to operate as a PDU session anchor (PSA, or “anchor”). The anchor may be a UPF that provides an N6 interface with a DN.

[0085] In the example of FIG. 4A, UPF 405 may be the anchor for the first PDU session between UE 401 and DN 408, whereas the UPF 406 may be the anchor for the second PDU session between UE 401 and DN 409. The core network may use the anchor to provide service continuity of a particular PDU session (for example, IP address continuity) as UE 401 moves from one access network to another. For example, suppose that UE 401 establishes a PDU session using a data path to the DN 408 using an access network other than AN 402. The data path may include UPF 405 acting as anchor. Suppose further that the UE 401 later moves into the coverage area of the AN 402. In such a scenario, SMF 414 may select a new UPF (UPF 407) to bridge the gap between the newly-entered access network (AN 402) and the anchor UPF (UPF 405). The continuity of the PDU session may be preserved as any number of UPFs are added or removed from the data path. When a UPF is added to a data path, as shown in FIG. 4A, it may be described as an intermediate UPF and/or a cascaded UPF.

[0086] As noted above, UPF 406 may be the anchor for the second PDU session between UE 401 and DN 409. Although the anchor for the first and second PDU sessions are associated with different UPFs in FIG. 4A, it will be understood that this is merely an example. It will also be understood that multiple PDU sessions with a single DN may correspond to any number of anchors. When there are multiple UPFs, a UPF at the branching point (UPF 407 in FIG. 4) may operate as an uplink classifier (UL-CL). The UL-CL may divert uplink user plane traffic to different UPFs.

[0087] The SMF 414 may allocate, manage, and/or assign an IP address to UE 401, for example, upon establishment of a PDU session. The SMF 414 may maintain an internal pool of IP addresses to be assigned. The SMF 414 may, if necessary, assign an IP address provided by a dynamic host configuration protocol (DHCP) server or an authentication, authorization, and accounting (AAA) server. IP address management may be performed in accordance with a session and service continuity (SSC) mode. In SSC mode 1, an IP address of UE 401 may be maintained (and the same anchor

UPF may be used) as the wireless device moves within the network. In SSC mode 2, the IP address of UE 401 changes as UE 401 moves within the network (e.g., the old IP address and UPF may be abandoned and a new IP address and anchor UPF may be established). In SSC mode 3, it may be possible to maintain an old IP address (similar to SSC mode 1) temporarily while establishing a new IP address (similar to SSC mode 2), thus combining features of SSC modes 1 and 2. Applications that are sensitive to IP address changes may operate in accordance with SSC mode 1.

[0088] UPF selection may be controlled by SMF 414. For example, upon establishment and/or modification of a PDU session between UE 401 and DN 408, SMF 414 may select UPF 405 as the anchor for the PDU session and/or UPF 407 as an intermediate UPF. Criteria for UPF selection include path efficiency and/or speed between AN 402 and DN 408. The reliability, load status, location, slice support and/or other capabilities of candidate UPFs may also be considered.

[0089] FIG. 4B illustrates an example of a core network architecture 400B that accommodates untrusted access. Similar to FIG. 4A, UE 401 as depicted in FIG. 4B connects to DN 408 via AN 402 and UPF 405. The AN 402 and UPF 405 constitute trusted (e.g., 3GPP) access to the DN 408. By contrast, UE 401 may also access DN 408 using an untrusted access network, AN 403, and a non-3GPP interworking function (N3IWF) 404.

[0090] The AN 403 may be, for example, a wireless land area network (WLAN) operating in accordance with the IEEE 802.11 standard. The UE 401 may connect to AN 403, via an interface Y1, in whatever manner is prescribed for AN 403. The connection to AN 403 may or may not involve authentication. The UE 401 may obtain an IP address from AN 403. The UE 401 may determine to connect to core network 400B and select untrusted access for that purpose. The AN 403 may communicate with N3IWF 404 via a Y2 interface. After selecting untrusted access, the UE 401 may provide N3IWF 404 with sufficient information to select an AMF. The selected AMF may be, for example, the same AMF that is used by UE 401 for 3GPP access (AMF 412 in the present example). The N3IWF 404 may communicate with AMF 412 via an N2 interface. The UPF 405 may be selected and N3IWF 404 may communicate with UPF 405 via an N3 interface. The UPF 405 may be a PDU session anchor (PSA) and may remain the anchor for the PDU session even as UE 401 shifts between trusted access and untrusted access.

[0091] FIG. 5 illustrates an example of a core network architecture 500 in which a UE 501 is in a roaming scenario. In a roaming scenario, UE 501 is a subscriber of a first PLMN (a home PLMN, or HPLMN) but attaches to a second PLMN (a visited PLMN, or VPLMN). Core network architecture 500 includes UE 501, an AN 502, a UPF 505, and a DN 508. The AN 502 and UPF 505 may be associated with a VPLMN. The VPLMN may manage the AN 502 and UPF 505 using core network elements associated with the VPLMN, including an AMF 512, an SMF 514, a PCF 520, an NRF 530, an NEF 540, and an NSSF 570. An AF 599 may be adjacent the core network of the VPLMN.

[0092] The UE 501 may not be a subscriber of the VPLMN. The AMF 512 may authorize UE 501 to access the network based on, for example, roaming restrictions that apply to UE 501. In order to obtain network services provided by the VPLMN, it may be necessary for the core network of the VPLMN to interact with core network

elements of a HPLMN of UE 501, in particular, a PCF 521, an NRF 531, an NEF 541, a UDM 551, and/or an AUSF 561. The VPLMN and HPLMN may communicate using an N32 interface connecting respective security edge protection proxies (SEPPs). In FIG. 5, the respective SEPPs are depicted as a VSEPP 590 and an HSEPP 591.

[0093] The VSEPP 590 and the HSEPP 591 communicate via an N32 interface for defined purposes while concealing information about each PLMN from the other. The SEPPs may apply roaming policies based on communications via the N32 interface. The PCF 520 and PCF 521 may communicate via the SEPPs to exchange policy-related signaling. The NRF 530 and NRF 531 may communicate via the SEPPs to enable service discovery of NFs in the respective PLMNs. The VPLMN and HPLMN may independently maintain NEF 540 and NEF 541. The NSSF 570 and NSSF 571 may communicate via the SEPPs to coordinate slice selection for UE 501. The HPLMN may handle all authentication and subscription related signaling. For example, when the UE 501 registers or requests service via the VPLMN, the VPLMN may authenticate UE 501 and/or obtain subscription data of UE 501 by accessing, via the SEPPs, the UDM 551 and AUSF 561 of the HPLMN.

[0094] The core network architecture 500 depicted in FIG. 5 may be referred to as a local breakout configuration, in which UE 501 accesses DN 508 using one or more UPFs of the VPLMN (i.e., UPF 505). However, other configurations are possible. For example, in a home-routed configuration (not shown in FIG. 5), UE 501 may access a DN using one or more UPFs of the HPLMN. In the home-routed configuration, an N9 interface may run parallel to the N32 interface, crossing the frontier between the VPLMN and the HPLMN to carry user plane data. One or more SMFs of the respective PLMNs may communicate via the N32 interface to coordinate session management for UE 501. The SMFs may control their respective UPFs on either side of the frontier.

[0095] FIG. 6 illustrates an example of network slicing. Network slicing may refer to division of shared infrastructure (e.g., physical infrastructure) into distinct logical networks. These distinct logical networks may be independently controlled, isolated from one another, and/or associated with dedicated resources.

[0096] Network architecture 600A illustrates an un-sliced physical network corresponding to a single logical network. The network architecture 600A comprises a user plane wherein UEs 601A, 601B, 601C (collectively, UEs 601) have a physical and logical connection to a DN 608 via an AN 602 and a UPF 605. The network architecture 600A comprises a control plane wherein an AMF 612 and a SMF 614 control various aspects of the user plane.

[0097] The network architecture 600A may have a specific set of characteristics (e.g., relating to maximum bit rate, reliability, latency, bandwidth usage, power consumption, etc.). This set of characteristics may be affected by the nature of the network elements themselves (e.g., processing power, availability of free memory, proximity to other network elements, etc.) or the management thereof (e.g., optimized to maximize bit rate or reliability, reduce latency or power bandwidth usage, etc.). The characteristics of network architecture 600A may change over time, for example, by upgrading equipment or by modifying procedures to target a particular characteristic. However, at any given time, network architecture 600A will have a single set of characteristics that may or may not be optimized for a particular use

case. For example, UEs 601A, 601B, 601C may have different requirements, but network architecture 600A can only be optimized for one of the three.

[0098] Network architecture 600B is an example of a sliced physical network divided into multiple logical networks. In FIG. 6, the physical network is divided into three logical networks, referred to as slice A, slice B, and slice C. For example, UE 601A may be served by AN 602A, UPF 605A, AMF 612, and SMF 614A. UE 601B may be served by AN 602B, UPF 605B, AMF 612, and SMF 614B. UE 601C may be served by AN 602C, UPF 605C, AMF 612, and SMF 614C. Although the respective UEs 601 communicate with different network elements from a logical perspective, these network elements may be deployed by a network operator using the same physical network elements.

[0099] Each network slice may be tailored to network services having different sets of characteristics. For example, slice A may correspond to enhanced mobile broadband (eMBB) service. Mobile broadband may refer to internet access by mobile users, commonly associated with smartphones. Slice B may correspond to ultra-reliable low-latency communication (URLLC), which focuses on reliability and speed. Relative to eMBB, URLLC may improve the feasibility of use cases such as autonomous driving and telesurgery. Slice C may correspond to massive machine type communication (mMTC), which focuses on low-power services delivered to a large number of users. For example, slice C may be optimized for a dense network of battery-powered sensors that provide small amounts of data at regular intervals. Many mMTC use cases would be prohibitively expensive if they operated using an eMBB or URLLC network.

[0100] If the service requirements for one of the UEs 601 changes, then the network slice serving that UE can be updated to provide better service. Moreover, the set of network characteristics corresponding to eMBB, URLLC, and mMTC may be varied, such that differentiated species of eMBB, URLLC, and mMTC are provided. Alternatively, network operators may provide entirely new services in response to, for example, customer demand.

[0101] In FIG. 6, each of the UEs 601 has its own network slice. However, it will be understood that a single slice may serve any number of UEs and a single UE may operate using any number of slices. Moreover, in the example network architecture 600B, the AN 602, UPF 605 and SMF 614 are separated into three separate slices, whereas the AMF 612 is unsliced. However, it will be understood that a network operator may deploy any architecture that selectively utilizes any mix of sliced and unsliced network elements, with different network elements divided into different numbers of slices. Although FIG. 6 only depicts three core network functions, it will be understood that other core network functions may be sliced as well. A PLMN that supports multiple network slices may maintain a separate network repository function (NRF) for each slice, enabling other NFs to discover network services associated with that slice.

[0102] Network slice selection may be controlled by an AMF, or alternatively, by a separate network slice selection function (NSSF). For example, a network operator may define and implement distinct network slice instances (NSIs). Each NSI may be associated with single network slice selection assistance information (S-NSSAI). The S-NSSAI may include a particular slice/service type (SST) indicator (indicating eMBB, URLLC, mMTC, etc.). as an

example, a particular tracking area may be associated with one or more configured S-NSSAIs. UEs may identify one or more requested and/or subscribed S-NSSAIs (e.g., during registration). The network may indicate to the UE one or more allowed and/or rejected S-NSSAIs.

[0103] The S-NSSAI may further include a slice differentiator (SD) to distinguish between different tenants of a particular slice and/or service type. For example, a tenant may be a customer (e.g., vehicle manufacture, service provider, etc.) of a network operator that obtains (for example, purchases) guaranteed network resources and/or specific policies for handling its subscribers. The network operator may configure different slices and/or slice types, and use the SD to determine which tenant is associated with a particular slice.

[0104] FIG. 7A, FIG. 7B, and FIG. 7C illustrate a user plane (UP) protocol stack, a control plane (CP) protocol stack, and services provided between protocol layers of the UP protocol stack.

[0105] The layers may be associated with an open system interconnection (OSI) model of computer networking functionality. In the OSI model, layer 1 may correspond to the bottom layer, with higher layers on top of the bottom layer. Layer 1 may correspond to a physical layer, which is concerned with the physical infrastructure used for transfer of signals (for example, cables, fiber optics, and/or radio frequency transceivers). In New Radio (NR), layer 1 may comprise a physical layer (PHY). Layer 2 may correspond to a data link layer. Layer 2 may be concerned with packaging of data (into, e.g., data frames) for transfer, between nodes of the network, using the physical infrastructure of layer 1. In NR, layer 2 may comprise a media access control layer (MAC), a radio link control layer (RLC), a packet data convergence layer (PDCP), and a service data application protocol layer (SDAP).

[0106] Layer 3 may correspond to a network layer. Layer 3 may be concerned with routing of the data which has been packaged in layer 2. Layer 3 may handle prioritization of data and traffic avoidance. In NR, layer 3 may comprise a radio resource control layer (RRC) and a non-access stratum layer (NAS). Layers 4 through 7 may correspond to a transport layer, a session layer, a presentation layer, and an application layer. The application layer interacts with an end user to provide data associated with an application. In an example, an end user implementing the application may generate data associated with the application and initiate sending of that information to a targeted data network (e.g., the Internet, an application server, etc.). Starting at the application layer, each layer in the OSI model may manipulate and/or repackage the information and deliver it to a lower layer. At the lowest layer, the manipulated and/or repackaged information may be exchanged via physical infrastructure (for example, electrically, optically, and/or electromagnetically). As it approaches the targeted data network, the information will be unpackaged and provided to higher and higher layers, until it once again reaches the application layer in a form that is usable by the targeted data network (e.g., the same form in which it was provided by the end user). To respond to the end user, the data network may perform this procedure in reverse.

[0107] FIG. 7A illustrates a user plane protocol stack. The user plane protocol stack may be a new radio (NR) protocol stack for a Uu interface between a UE 701 and a gNB 702. In layer 1 of the UP protocol stack, the UE 701 may

implement PHY 731 and the gNB 702 may implement PHY 732. In layer 2 of the UP protocol stack, the UE 701 may implement MAC 741, RLC 751, PDCP 761, and SDAP 771. The gNB 702 may implement MAC 742, RLC 752, PDCP 762, and SDAP 772.

[0108] FIG. 7B illustrates a control plane protocol stack. The control plane protocol stack may be an NR protocol stack for the Uu interface between the UE 701 and the gNB 702 and/or an N1 interface between the UE 701 and an AMF 712. In layer 1 of the CP protocol stack, the UE 701 may implement PHY 731 and the gNB 702 may implement PHY 732. In layer 2 of the CP protocol stack, the UE 701 may implement MAC 741, RLC 751, PDCP 761, RRC 781, and NAS 791. The gNB 702 may implement MAC 742, RLC 752, PDCP 762, and RRC 782. The AMF 712 may implement NAS 792.

[0109] The NAS may be concerned with the non-access stratum, in particular, communication between the UE 701 and the core network (e.g., the AMF 712). Lower layers may be concerned with the access stratum, for example, communication between the UE 701 and the gNB 702. Messages sent between the UE 701 and the core network may be referred to as NAS messages. In an example, a NAS message may be relayed by the gNB 702, but the content of the NAS message (e.g., information elements of the NAS message) may not be visible to the gNB 702.

[0110] FIG. 7C illustrates an example of services provided between protocol layers of the NR user plane protocol stack illustrated in FIG. 7A. The UE 701 may receive services through a PDU session, which may be a logical connection between the UE 701 and a data network (DN). The UE 701 and the DN may exchange data packets associated with the PDU session. The PDU session may comprise one or more quality of service (QoS) flows. SDAP 771 and SDAP 772 may perform mapping and/or demapping between the one or more QoS flows of the PDU session and one or more radio bearers (e.g., data radio bearers). The mapping between the QoS flows and the data radio bearers may be determined in the SDAP 772 by the gNB 702, and the UE 701 may be notified of the mapping (e.g., based on control signaling and/or reflective mapping). For reflective mapping, the SDAP 772 of the gNB 702 may mark downlink packets with a QoS flow indicator (QFI) and deliver the downlink packets to the UE 701. The UE 701 may determine the mapping based on the QFI of the downlink packets.

[0111] PDCP 761 and PDCP 762 may perform header compression and/or decompression. Header compression may reduce the amount of data transmitted over the physical layer. The PDCP 761 and PDCP 762 may perform ciphering and/or deciphering. Ciphering may reduce unauthorized decoding of data transmitted over the physical layer (e.g., intercepted on an air interface), and protect data integrity (e.g., to ensure control messages originate from intended sources). The PDCP 761 and PDCP 762 may perform retransmissions of undelivered packets, in-sequence delivery and reordering of packets, duplication of packets, and/or identification and removal of duplicate packets. In a dual connectivity scenario, PDCP 761 and PDCP 762 may perform mapping between a split radio bearer and RLC channels.

[0112] RLC 751 and RLC 752 may perform segmentation, retransmission through Automatic Repeat Request (ARQ). The RLC 751 and RLC 752 may perform removal of duplicate data units received from MAC 741 and MAC 742,

respectively. The RLCs 213 and 223 may provide RLC channels as a service to PDCPs 214 and 224, respectively.

[0113] MAC 741 and MAC 742 may perform multiplexing and/or demultiplexing of logical channels. MAC 741 and MAC 742 may map logical channels to transport channels. In an example, UE 701 may, in MAC 741, multiplex data units of one or more logical channels into a transport block. The UE 701 may transmit the transport block to the gNB 702 using PHY 731. The gNB 702 may receive the transport block using PHY 732 and demultiplex data units of the transport blocks back into logical channels. MAC 741 and MAC 742 may perform error correction through Hybrid Automatic Repeat Request (HARQ), logical channel prioritization, and/or padding.

[0114] PHY 731 and PHY 732 may perform mapping of transport channels to physical channels. PHY 731 and PHY 732 may perform digital and analog signal processing functions (e.g., coding/decoding and modulation/demodulation) for sending and receiving information (e.g., transmission via an air interface). PHY 731 and PHY 732 may perform multi-antenna mapping.

[0115] FIG. 8 illustrates an example of a quality of service (QoS) model for differentiated data exchange. In the QoS model of FIG. 8, there are a UE 801, a AN 802, and a UPF 805. The QoS model facilitates prioritization of certain packet or protocol data units (PDUs), also referred to as packets. For example, higher-priority packets may be exchanged faster and/or more reliably than lower-priority packets. The network may devote more resources to exchange of high-QoS packets.

[0116] In the example of FIG. 8, a PDU session 810 is established between UE 801 and UPF 805. The PDU session 810 may be a logical connection enabling the UE 801 to exchange data with a particular data network (for example, the Internet). The UE 801 may request establishment of the PDU session 810. At the time that the PDU session 810 is established, the UE 801 may, for example, identify the targeted data network based on its data network name (DNN). The PDU session 810 may be managed, for example, by a session management function (SMF, not shown). In order to facilitate exchange of data associated with the PDU session 810, between the UE 801 and the data network, the SMF may select the UPF 805 (and optionally, one or more other UPFs, not shown).

[0117] One or more applications associated with UE 801 may generate uplink packets 812A-812E associated with the PDU session 810. In order to work within the QoS model, UE 801 may apply QoS rules 814 to uplink packets 812A-812E. The QoS rules 814 may be associated with PDU session 810 and may be determined and/or provided to the UE 801 when PDU session 810 is established and/or modified. Based on QoS rules 814, UE 801 may classify uplink packets 812A-812E, map each of the uplink packets 812A-812E to a QoS flow, and/or mark uplink packets 812A-812E with a QoS flow indicator (QFI). As a packet travels through the network, and potentially mixes with other packets from other UEs having potentially different priorities, the QFI indicates how the packet should be handled in accordance with the QoS model. In the present illustration, uplink packets 812A, 812B are mapped to QoS flow 816A, uplink packet 812C is mapped to QoS flow 816B, and the remaining packets are mapped to QoS flow 816C.

[0118] The QoS flows may be the finest granularity of QoS differentiation in a PDU session. In the figure, three QoS

flows 816A-816C are illustrated. However, it will be understood that there may be any number of QoS flows. Some QoS flows may be associated with a guaranteed bit rate (GBR QoS flows) and others may have bit rates that are not guaranteed (non-GBR QoS flows). QoS flows may also be subject to per-UE and per-session aggregate bit rates. One of the QoS flows may be a default QoS flow. The QoS flows may have different priorities. For example, QoS flow 816A may have a higher priority than QoS flow 816B, which may have a higher priority than QoS flow 816C. Different priorities may be reflected by different QoS flow characteristics. For example, QoS flows may be associated with flow bit rates. A particular QoS flow may be associated with a guaranteed flow bit rate (GFBR) and/or a maximum flow bit rate (MFBR). QoS flows may be associated with specific packet delay budgets (PDBs), packet error rates (PERs), and/or maximum packet loss rates. QoS flows may also be subject to per-UE and per-session aggregate bit rates.

[0119] In order to work within the QoS model, UE 801 may apply resource mapping rules 818 to the QoS flows 816A-816C. The air interface between UE 801 and AN 802 may be associated with resources 820. In the present illustration, QoS flow 816A is mapped to resource 820A, whereas QoS flows 816B, 816C are mapped to resource 820B. The resource mapping rules 818 may be provided by the AN 802. In order to meet QoS requirements, the resource mapping rules 818 may designate more resources for relatively high-priority QoS flows. With more resources, a high-priority QoS flow such as QoS flow 816A may be more likely to obtain the high flow bit rate, low packet delay budget, or other characteristic associated with QoS rules 814. The resources 820 may comprise, for example, radio bearers. The radio bearers (e.g., data radio bearers) may be established between the UE 801 and the AN 802. The radio bearers in 5G, between the UE 801 and the AN 802, may be distinct from bearers in LTE, for example, Evolved Packet System (EPS) bearers between a UE and a packet data network gateway (PGW), S1 bearers between an eNB and a serving gateway (SGW), and/or an S5/S8 bearer between an SGW and a PGW.

[0120] Once a packet associated with a particular QoS flow is received at AN 802 via resource 820A or resource 820B, AN 802 may separate packets into respective QoS flows 856A-856C based on QoS profiles 828. The QoS profiles 828 may be received from an SMF. Each QoS profile may correspond to a QFI, for example, the QFI marked on the uplink packets 812A-812E. Each QoS profile may include QoS parameters such as 5G QoS identifier (5QI) and an allocation and retention priority (ARP). The QoS profile for non-GBR QoS flows may further include additional QoS parameters such as a reflective QoS attribute (RQA). The QoS profile for GBR QoS flows may further include additional QoS parameters such as a guaranteed flow bit rate (GFBR), a maximum flow bit rate (MFBR), and/or a maximum packet loss rate. The 5QI may be a standardized 5QI which have one-to-one mapping to a standardized combination of 5G QoS characteristics per well-known services. The 5QI may be a dynamically assigned 5QI which the standardized 5QI values are not defined. The 5QI may represent 5G QoS characteristics. The 5QI may comprise a resource type, a default priority level, a packet delay budget (PDB), a packet error rate (PER), a maximum data burst volume, and/or an averaging window. The resource type may indicate a non-GBR QoS flow, a GBR QoS flow or a

delay-critical GBR QoS flow. The averaging window may represent a duration over which the GFBR and/or MFBR is calculated. ARP may be a priority level comprising pre-emption capability and a pre-emption vulnerability. Based on the ARP, the AN **802** may apply admission control for the QoS flows in a case of resource limitations.

[0121] The AN **802** may select one or more N3 tunnels **850** for transmission of the QoS flows **856A-856C**. After the packets are divided into QoS flows **856A-856C**, the packet may be sent to UPF **805** (e.g., towards a DN) via the selected one or more N3 tunnels **850**. The UPF **805** may verify that the QFIs of the uplink packets **812A-812E** are aligned with the QoS rules **814** provided to the UE **801**. The UPF **805** may measure and/or count packets and/or provide packet metrics to, for example, a PCF.

[0122] The figure also illustrates a process for downlink. In particular, one or more applications may generate downlink packets **852A-852E**. The UPF **805** may receive downlink packets **852A-852E** from one or more DNs and/or one or more other UPFs. As per the QoS model, UPF **805** may apply packet detection rules (PDRs) **854** to downlink packets **852A-852E**. Based on PDRs **854**, UPF **805** may map packets **852A-852E** into QoS flows. In the present illustration, downlink packets **852A**, **852B** are mapped to QoS flow **856A**, downlink packet **852C** is mapped to QoS flow **856B**, and the remaining packets are mapped to QoS flow **856C**.

[0123] The QoS flows **856A-856C** may be sent to AN **802**. The AN **802** may apply resource mapping rules to the QoS flows **856A-856C**. In the present illustration, QoS flow **856A** is mapped to resource **820A**, whereas QoS flows **856B**, **856C** are mapped to resource **820B**. In order to meet QoS requirements, the resource mapping rules may designate more resources to high-priority QoS flows.

[0124] FIGS. 9A-9D illustrate example states and state transitions of a wireless device (e.g., a UE). At any given time, the wireless device may have a radio resource control (RRC) state, a registration management (RM) state, and a connection management (CM) state.

[0125] FIG. 9A is an example diagram showing RRC state transitions of a wireless device (e.g., a UE). The UE may be in one of three RRC states: RRC idle **910**, (e.g., RRC_IDLE), RRC inactive **920** (e.g., RRC_INACTIVE), or RRC connected **930** (e.g., RRC_CONNECTED). The UE may implement different RAN-related control-plane procedures depending on its RRC state. Other elements of the network, for example, a base station, may track the RRC state of one or more UEs and implement RAN-related control-plane procedures appropriate to the RRC state of each.

[0126] In RRC connected **930**, it may be possible for the UE to exchange data with the network (for example, the base station). The parameters necessary for exchange of data may be established and known to both the UE and the network. The parameters may be referred to and/or included in an RRC context of the UE (sometimes referred to as a UE context). These parameters may include, for example: one or more AS contexts; one or more radio link configuration parameters; bearer configuration information (e.g., relating to a data radio bearer, signaling radio bearer, logical channel, QoS flow, and/or PDU session); security information; and/or PHY, MAC, RLC, PDCP, and/or SDAP layer configuration information. The base station with which the UE is connected may store the RRC context of the UE.

[0127] While in RRC connected **930**, mobility of the UE may be managed by the access network, whereas the UE

itself may manage mobility while in RRC idle **910** and/or RRC inactive **920**. While in RRC connected **930**, the UE may manage mobility by measuring signal levels (e.g., reference signal levels) from a serving cell and neighboring cells and reporting these measurements to the base station currently serving the UE. The network may initiate handover based on the reported measurements. The RRC state may transition from RRC connected **930** to RRC idle **910** through a connection release procedure **930** or to RRC inactive **920** through a connection inactivation procedure **932**.

[0128] In RRC idle **910**, an RRC context may not be established for the UE. In RRC idle **910**, the UE may not have an RRC connection with a base station. While in RRC idle **910**, the UE may be in a sleep state for a majority of the time (e.g., to conserve battery power). The UE may wake up periodically (e.g., once in every discontinuous reception cycle) to monitor for paging messages from the access network. Mobility of the UE may be managed by the UE through a procedure known as cell reselection. The RRC state may transition from RRC idle **910** to RRC connected **930** through a connection establishment procedure **913**, which may involve a random access procedure, as discussed in greater detail below.

[0129] In RRC inactive **920**, the RRC context previously established is maintained in the UE and the base station. This may allow for a fast transition to RRC connected **930** with reduced signaling overhead as compared to the transition from RRC idle **910** to RRC connected **930**. The RRC state may transition to RRC connected **930** through a connection resume procedure **923**. The RRC state may transition to RRC idle **910** through a connection release procedure **921** that may be the same as or similar to connection release procedure **931**.

[0130] An RRC state may be associated with a mobility management mechanism. In RRC idle **910** and RRC inactive **920**, mobility may be managed by the UE through cell reselection. The purpose of mobility management in RRC idle **910** and/or RRC inactive **920** is to allow the network to be able to notify the UE of an event via a paging message without having to broadcast the paging message over the entire mobile communications network. The mobility management mechanism used in RRC idle **910** and/or RRC inactive **920** may allow the network to track the UE on a cell-group level so that the paging message may be broadcast over the cells of the cell group that the UE currently resides within instead of the entire communication network. Tracking may be based on different granularities of grouping. For example, there may be three levels of cell-grouping granularity: individual cells; cells within a RAN area identified by a RAN area identifier (RAI); and cells within a group of RAN areas, referred to as a tracking area and identified by a tracking area identifier (TAI).

[0131] Tracking areas may be used to track the UE at the CN level. The CN may provide the UE with a list of TAIs associated with a UE registration area. If the UE moves, through cell reselection, to a cell associated with a TAI not included in the list of TAIs associated with the UE registration area, the UE may perform a registration update with the CN to allow the CN to update the UE's location and provide the UE with a new the UE registration area.

[0132] RAN areas may be used to track the UE at the RAN level. For a UE in RRC inactive **920** state, the UE may be assigned a RAN notification area. A RAN notification area may comprise one or more cell identities, a list of RAIs,

and/or a list of TAIs. In an example, a base station may belong to one or more RAN notification areas. In an example, a cell may belong to one or more RAN notification areas. If the UE moves, through cell reselection, to a cell not included in the RAN notification area assigned to the UE, the UE may perform a notification area update with the RAN to update the UE's RAN notification area.

[0133] A base station storing an RRC context for a UE or a last serving base station of the UE may be referred to as an anchor base station. An anchor base station may maintain an RRC context for the UE at least during a period of time that the UE stays in a RAN notification area of the anchor base station and/or during a period of time that the UE stays in RRC inactive **920**.

[0134] FIG. 9B is an example diagram showing registration management (RM) state transitions of a wireless device (e.g., a UE). The states are RM deregistered **940**, (e.g., RM-DEREGISTERED) and RM registered **950** (e.g., RM-REGISTERED).

[0135] In RM deregistered **940**, the UE is not registered with the network, and the UE is not reachable by the network. In order to be reachable by the network, the UE must perform an initial registration. As an example, the UE may register with an AMF of the network. If registration is rejected (registration reject **944**), then the UE remains in RM deregistered **940**. If registration is accepted (registration accept **945**), then the UE transitions to RM registered **950**. While the UE is RM registered **950**, the network may store, keep, and/or maintain a UE context for the UE. The UE context may be referred to as wireless device context. The UE context corresponding to network registration (maintained by the core network) may be different from the RRC context corresponding to RRC state (maintained by an access network, e.g., a base station). The UE context may comprise a UE identifier and a record of various information relating to the UE, for example, UE capability information, policy information for access and mobility management of the UE, lists of allowed or established slices or PDU sessions, and/or a registration area of the UE (i.e., a list of tracking areas covering the geographical area where the wireless device is likely to be found).

[0136] While the UE is RM registered **950**, the network may store the UE context of the UE, and if necessary use the UE context to reach the UE. Moreover, some services may not be provided by the network unless the UE is registered. The UE may update its UE context while remaining in RM registered **950** (registration update accept **955**). For example, if the UE leaves one tracking area and enters another tracking area, the UE may provide a tracking area identifier to the network. The network may deregister the UE, or the UE may deregister itself (deregistration **954**). For example, the network may automatically deregister the wireless device if the wireless device is inactive for a certain amount of time. Upon deregistration, the UE may transition to RM deregistered **940**.

[0137] FIG. 9C is an example diagram showing connection management (CM) state transitions of a wireless device (e.g., a UE), shown from a perspective of the wireless device. The UE may be in CM idle **960** (e.g., CM-IDLE) or CM connected **970** (e.g., CM-CONNECTED).

[0138] In CM idle **960**, the UE does not have a non access stratum (NAS) signaling connection with the network. As a result, the UE can not communicate with core network functions. The UE may transition to CM connected **970** by

establishing an AN signaling connection (AN signaling connection establishment **967**). This transition may be initiated by sending an initial NAS message. The initial NAS message may be a registration request (e.g., if the UE is RM deregistered **940**) or a service request (e.g., if the UE is RM registered **950**). If the UE is RM registered **950**, then the UE may initiate the AN signaling connection establishment by sending a service request, or the network may send a page, thereby triggering the UE to send the service request.

[0139] In CM connected **970**, the UE can communicate with core network functions using NAS signaling. As an example, the UE may exchange NAS signaling with an AMF for registration management purposes, service request procedures, and/or authentication procedures. As another example, the UE may exchange NAS signaling, with an SMF, to establish and/or modify a PDU session. The network may disconnect the UE, or the UE may disconnect itself (AN signaling connection release **976**). For example, if the UE transitions to RM deregistered **940**, then the UE may also transition to CM idle **960**. When the UE transitions to CM idle **960**, the network may deactivate a user plane connection of a PDU session of the UE.

[0140] FIG. 9D is an example diagram showing CM state transitions of the wireless device (e.g., a UE), shown from a network perspective (e.g., an AMF). The CM state of the UE, as tracked by the AMF, may be in CM idle **980** (e.g., CM-IDLE) or CM connected **990** (e.g., CM-CONNECTED). When the UE transitions from CM idle **980** to CM connected **990**, the AMF may establish an N2 context of the UE (N2 context establishment **989**). When the UE transitions from CM connected **990** to CM idle **980**, the AMF may release the N2 context of the UE (N2 context release **998**).

[0141] FIGS. 10-12 illustrate example procedures for registering, service request, and PDU session establishment of a UE.

[0142] FIG. 10 illustrates an example of a registration procedure for a wireless device (e.g., a UE). Based on the registration procedure, the UE may transition from, for example, RM deregistered **940** to RM registered **950**.

[0143] Registration may be initiated by a UE for the purposes of obtaining authorization to receive services, enabling mobility tracking, enabling reachability, or other purposes. The UE may perform an initial registration as a first step toward connection to the network (for example, if the UE is powered on, airplane mode is turned off, etc.). Registration may also be performed periodically to keep the network informed of the UE's presence (for example, while in CM-IDLE state), or in response to a change in UE capability or registration area. Deregistration (not shown in FIG. 10) may be performed to stop network access.

[0144] At **1010**, the UE transmits a registration request to an AN. As an example, the UE may have moved from a coverage area of a previous AMF (illustrated as AMF #1) into a coverage area of a new AMF (illustrated as AMF #2). The registration request may be a NAS message. The registration request may include a UE identifier. The AN may select an AMF for registration of the UE. For example, the AN may select a default AMF. For example, the AN may select an AMF that is already mapped to the UE (e.g., a previous AMF). The NAS registration request may include a network slice identifier and the AN may select an AMF based on the requested slice. After the AMF is selected, the AN may send the registration request to the selected AMF.

[0145] At **1020**, the AMF that receives the registration request (AMF #2) performs a context transfer. The context may be a UE context, for example, an RRC context for the UE. As an example, AMF #2 may send AMF #1 a message requesting a context of the UE. The message may include the UE identifier. The message may be a Namf_Communication_UEContextTransfer message. AMF #1 may send to AMF #2 a message that includes the requested UE context. This message may be a Namf_Communication_UEContextTransfer message. After the UE context is received, the AMF #2 may coordinate authentication of the UE. After authentication is complete, AMF #2 may send to AMF #1 a message indicating that the UE context transfer is complete. This message may be a Namf_Communication_UEContextTransfer Response message.

[0146] Authentication may require participation of the UE, an AUSF, a UDM and/or a UDR (not shown). For example, the AMF may request that the AUSF authenticate the UE. For example, the AUSF may execute authentication of the UE. For example, the AUSF may get authentication data from UDM. For example, the AUSF may send a subscription permanent identifier (SUPI) to the AMF based on the authentication being successful. For example, the AUSF may provide an intermediate key to the AMF. The intermediate key may be used to derive an access-specific security key for the UE, enabling the AMF to perform security context management (SCM). The AUSF may obtain subscription data from the UDM. The subscription data may be based on information obtained from the UDM (and/or the UDR). The subscription data may include subscription identifiers, security credentials, access and mobility related subscription data and/or session related data.

[0147] At **1030**, the new AMF, AMF #2, registers and/or subscribes with the UDM. AMF #2 may perform registration using a UE context management service of the UDM (Nudm_UECM). AMF #2 may obtain subscription information of the UE using a subscriber data management service of the UDM (Nudm_SDM). AMF #2 may further request that the UDM notify AMF #2 if the subscription information of the UE changes. As the new AMF registers and subscribes, the old AMF, AMF #1, may deregister and unsubscribe. After deregistration, AMF #1 is free of responsibility for mobility management of the UE.

[0148] At **1040**, AMF #2 retrieves access and mobility (AM) policies from the PCF. As an example, the AMF #2 may provide subscription data of the UE to the PCF. The PCF may determine access and mobility policies for the UE based on the subscription data, network operator data, current network conditions, and/or other suitable information. For example, the owner of a first UE may purchase a higher level of service than the owner of a second UE. The PCF may provide the rules associated with the different levels of service. Based on the subscription data of the respective UEs, the network may apply different policies which facilitate different levels of service.

[0149] For example, access and mobility policies may relate to service area restrictions, RAT/frequency selection priority (RFSP, where RAT stands for radio access technology), authorization and prioritization of access type (e.g., LTE versus NR), and/or selection of non-3GPP access (e.g., Access Network Discovery and Selection Policy (ANDSP)). The service area restrictions may comprise a list of tracking areas where the UE is allowed to be served (or forbidden from being served). The access and mobility policies may

include a UE route selection policy (URSP) that influences routing to an established PDU session or a new PDU session. As noted above, different policies may be obtained and/or enforced based on subscription data of the UE, location of the UE (i.e., location of the AN and/or AMF), or other suitable factors.

[0150] At **1050**, AMF #2 may update a context of a PDU session. For example, if the UE has an existing PDU session, the AMF #2 may coordinate with an SMF to activate a user plane connection associated with the existing PDU session. The SMF may update and/or release a session management context of the PDU session (Nsmf_PDUSession_UpdateSMContext, Nsmf_PDUSession_ReleaseSMContext).

[0151] At **1060**, AMF #2 sends a registration accept message to the AN, which forwards the registration accept message to the UE. The registration accept message may include a new UE identifier and/or a new configured slice identifier. The UE may transmit a registration complete message to the AN, which forwards the registration complete message to the AMF #2. The registration complete message may acknowledge receipt of the new UE identifier and/or new configured slice identifier.

[0152] At **1070**, AMF #2 may obtain UE policy control information from the PCF. The PCF may provide an access network discovery and selection policy (ANDSP) to facilitate non-3GPP access. The PCF may provide a UE route selection policy (URSP) to facilitate mapping of particular data traffic to particular PDU session connectivity parameters. As an example, the URSP may indicate that data traffic associated with a particular application should be mapped to a particular SSC mode, network slice, PDU session type, or preferred access type (3GPP or non-3GPP).

[0153] FIG. 11 illustrates an example of a service request procedure for a wireless device (e.g., a UE). The service request procedure depicted in FIG. 11 is a network-triggered service request procedure for a UE in a CM-IDLE state. However, other service request procedures (e.g., a UE-triggered service request procedure) may also be understood by reference to FIG. 11, as will be discussed in greater detail below.

[0154] At **1110**, a UPF receives data. The data may be downlink data for transmission to a UE. The data may be associated with an existing PDU session between the UE and a DN. The data may be received, for example, from a DN and/or another UPF. The UPF may buffer the received data. In response to the receiving of the data, the UPF may notify an SMF of the received data. The identity of the SMF to be notified may be determined based on the received data. The notification may be, for example, an N4 session report. The notification may indicate that the UPF has received data associated with the UE and/or a particular PDU session associated with the UE. In response to receiving the notification, the SMF may send PDU session information to an AMF. The PDU session information may be sent in an N1N2 message transfer for forwarding to an AN. The PDU session information may include, for example, UPF tunnel endpoint information and/or QoS information.

[0155] At **1120**, the AMF determines that the UE is in a CM-IDLE state. The determining at **1120** may be in response to the receiving of the PDU session information. Based on the determination that the UE is CM-IDLE, the service request procedure may proceed to **1130** and **1140**, as depicted in FIG. 11. However, if the UE is not CM-IDLE

(e.g., the UE is CM-CONNECTED), then **1130** and **1140** may be skipped, and the service request procedure may proceed directly to **1150**.

[0156] At **1130**, the AMF pages the UE. The paging at **1130** may be performed based on the UE being CM-IDLE. To perform the paging, the AMF may send a page to the AN. The page may be referred to as a paging or a paging message. The page may be an N2 request message. The AN may be one of a plurality of ANs in a RAN notification area of the UE. The AN may send a page to the UE. The UE may be in a coverage area of the AN and may receive the page.

[0157] At **1140**, the UE may request service. The UE may transmit a service request to the AMF via the AN. As depicted in FIG. 11, the UE may request service at **1140** in response to receiving the paging at **1130**. However, as noted above, this is for the specific case of a network-triggered service request procedure. In some scenarios (for example, if uplink data becomes available at the UE), then the UE may commence a UE-triggered service request procedure. The UE-triggered service request procedure may commence starting at **1140**.

[0158] At **1150**, the network may authenticate the UE. Authentication may require participation of the UE, an AUSF, and/or a UDM, for example, similar to authentication described elsewhere in the present disclosure. In some cases (for example, if the UE has recently been authenticated), the authentication at **1150** may be skipped.

[0159] At **1160**, the AMF and SMF may perform a PDU session update. As part of the PDU session update, the SMF may provide the AMF with one or more UPF tunnel endpoint identifiers. In some cases (not shown in FIG. 11), it may be necessary for the SMF to coordinate with one or more other SMFs and/or one or more other UPFs to set up a user plane.

[0160] At **1170**, the AMF may send PDU session information to the AN. The PDU session information may be included in an N2 request message. Based on the PDU session information, the AN may configure a user plane resource for the UE. To configure the user plane resource, the AN may, for example, perform an RRC reconfiguration of the UE. The AN may acknowledge to the AMF that the PDU session information has been received. The AN may notify the AMF that the user plane resource has been configured, and/or provide information relating to the user plane resource configuration.

[0161] In the case of a UE-triggered service request procedure, the UE may receive, at **1170**, a NAS service accept message from the AMF via the AN. After the user plane resource is configured, the UE may transmit uplink data (for example, the uplink data that caused the UE to trigger the service request procedure).

[0162] At **1180**, the AMF may update a session management (SM) context of the PDU session. For example, the AMF may notify the SMF (and/or one or more other associated SMFs) that the user plane resource has been configured, and/or provide information relating to the user plane resource configuration. The AMF may provide the SMF (and/or one or more other associated SMFs) with one or more AN tunnel endpoint identifiers of the AN. After the SM context update is complete, the SMF may send an update SM context response message to the AMF.

[0163] Based on the update of the session management context, the SMF may update a PCF for purposes of policy

control. For example, if a location of the UE has changed, the SMF may notify the PCF of the UE's a new location.

[0164] Based on the update of the session management context, the SMF and UPF may perform a session modification. The session modification may be performed using N4 session modification messages. After the session modification is complete, the UPF may transmit downlink data (for example, the downlink data that caused the UPF to trigger the network-triggered service request procedure) to the UE. The transmitting of the downlink data may be based on the one or more AN tunnel endpoint identifiers of the AN.

[0165] FIG. 12 illustrates an example of a protocol data unit (PDU) session establishment procedure for a wireless device (e.g., a UE). The UE may determine to transmit the PDU session establishment request to create a new PDU session, to hand over an existing PDU session to a 3GPP network, or for any other suitable reason.

[0166] At **1210**, the UE initiates PDU session establishment. The UE may transmit a PDU session establishment request to an AMF via an AN. The PDU session establishment request may be a NAS message. The PDU session establishment request may indicate: a PDU session ID; a requested PDU session type (new or existing); a requested DN (DNN); a requested network slice (S-NSSAI); a requested SSC mode; and/or any other suitable information. The PDU session ID may be generated by the UE. The PDU session type may be, for example, an Internet Protocol (IP)-based type (e.g., IPv4, IPv6, or dual stack IPv4/IPv6), an Ethernet type, or an unstructured type.

[0167] The AMF may select an SMF based on the PDU session establishment request. In some scenarios, the requested PDU session may already be associated with a particular SMF. For example, the AMF may store a UE context of the UE, and the UE context may indicate that the PDU session ID of the requested PDU session is already associated with the particular SMF. In some scenarios, the AMF may select the SMF based on a determination that the SMF is prepared to handle the requested PDU session. For example, the requested PDU session may be associated with a particular DNN and/or S-NSSAI, and the SMF may be selected based on a determination that the SMF can manage a PDU session associated with the particular DNN and/or S-NSSAI.

[0168] At **1220**, the network manages a context of the PDU session. After selecting the SMF at **1210**, the AMF sends a PDU session context request to the SMF. The PDU session context request may include the PDU session establishment request received from the UE at **1210**. The PDU session context request may be a Nsmf_PDUSession_CreateSMContext Request and/or a Nsmf_PDUSession_UpdateSMContext Request. The PDU session context request may indicate identifiers of the UE; the requested DN; and/or the requested network slice. Based on the PDU session context request, the SMF may retrieve subscription data from a UDM. The subscription data may be session management subscription data of the UE. The SMF may subscribe for updates to the subscription data, so that the PCF will send new information if the subscription data of the UE changes. After the subscription data of the UE is obtained, the SMF may transmit a PDU session context response to the AMF. The PDU session context response may be a Nsmf_PDUSession_CreateSMContext Response and/or a Nsmf_

PDUSession_UpdateSMContext Response. The PDU session context response may include a session management context ID.

[0169] At 1230, secondary authorization/authentication may be performed, if necessary. The secondary authorization/authentication may involve the UE, the AMF, the SMF, and the DN. The SMF may access the DN via a Data Network Authentication, Authorization and Accounting (DN AAA) server.

[0170] At 1240, the network sets up a data path for uplink data associated with the PDU session. The SMF may select a PCF and establish a session management policy association. Based on the association, the PCF may provide an initial set of policy control and charging rules (PCC rules) for the PDU session. When targeting a particular PDU session, the PCF may indicate, to the SMF, a method for allocating an IP address to the PDU Session, a default charging method for the PDU session, an address of the corresponding charging entity, triggers for requesting new policies, etc. The PCF may also target a service data flow (SDF) comprising one or more PDU sessions. When targeting an SDF, the PCF may indicate, to the SMF, policies for applying QoS requirements, monitoring traffic (e.g., for charging purposes), and/or steering traffic (e.g., by using one or more particular N6 interfaces).

[0171] The SMF may determine and/or allocate an IP address for the PDU session. The SMF may select one or more UPFs (a single UPF in the example of FIG. 12) to handle the PDU session. The SMF may send an N4 session message to the selected UPF. The N4 session message may be an N4 Session Establishment Request and/or an N4 Session Modification Request. The N4 session message may include packet detection, enforcement, and reporting rules associated with the PDU session. In response, the UPF may acknowledge by sending an N4 session establishment response and/or an N4 session modification response.

[0172] The SMF may send PDU session management information to the AMF. The PDU session management information may be a session service request (e.g., Namf_Communication_N1N2MessageTransfer) message. The PDU session management information may include the PDU session ID. The PDU session management information may be a NAS message. The PDU session management information may include N1 session management information and/or N2 session management information. The N1 session management information may include a PDU session establishment accept message. The PDU session establishment accept message may include tunneling endpoint information of the UPF and quality of service (QoS) information associated with the PDU session.

[0173] The AMF may send an N2 request to the AN. The N2 request may include the PDU session establishment accept message. Based on the N2 request, the AN may determine AN resources for the UE. The AN resources may be used by the UE to establish the PDU session, via the AN, with the DN. The AN may determine resources to be used for the PDU session and indicate the determined resources to the UE. The AN may send the PDU session establishment accept message to the UE. For example, the AN may perform an RRC reconfiguration of the UE. After the AN resources are set up, the AN may send an N2 request acknowledge to the AMF. The N2 request acknowledge may

include N2 session management information, for example, the PDU session ID and tunneling endpoint information of the AN.

[0174] After the data path for uplink data is set up at 1240, the UE may optionally send uplink data associated with the PDU session. As shown in FIG. 12, the uplink data may be sent to a DN associated with the PDU session via the AN and the UPF.

[0175] At 1250, the network may update the PDU session context. The AMF may transmit a PDU session context update request to the SMF. The PDU session context update request may be a Nsmf_PDUSession_UpdateSMContext Request. The PDU session context update request may include the N2 session management information received from the AN. The SMF may acknowledge the PDU session context update. The acknowledgement may be a Nsmf_PDUSession_UpdateSMContext Response. The acknowledgement may include a subscription requesting that the SMF be notified of any UE mobility event. Based on the PDU session context update request, the SMF may send an N4 session message to the UPF. The N4 session message may be an N4 Session Modification Request. The N4 session message may include tunneling endpoint information of the AN. The N4 session message may include forwarding rules associated with the PDU session. In response, the UPF may acknowledge by sending an N4 session modification response.

[0176] After the UPF receives the tunneling endpoint information of the AN, the UPF may relay downlink data associated with the PDU session. As shown in FIG. 12, the downlink data may be received from a DN associated with the PDU session via the AN and the UPF.

[0177] FIG. 13 illustrates examples of components of the elements in a communications network. FIG. 13 includes a wireless device 1310, a base station 1320, and a physical deployment of one or more network functions 1330 (henceforth “deployment 1330”). Any wireless device described in the present disclosure may have similar components and may be implemented in a similar manner as the wireless device 1310. Any other base station described in the present disclosure (or any portion thereof, depending on the architecture of the base station) may have similar components and may be implemented in a similar manner as the base station 1320. Any physical core network deployment in the present disclosure (or any portion thereof, depending on the architecture of the base station) may have similar components and may be implemented in a similar manner as the deployment 1330.

[0178] The wireless device 1310 may communicate with base station 1320 over an air interface 1370. The communication direction from wireless device 1310 to base station 1320 over air interface 1370 is known as uplink, and the communication direction from base station 1320 to wireless device 1310 over air interface 1370 is known as downlink. Downlink transmissions may be separated from uplink transmissions using FDD, TDD, and/or some combination of duplexing techniques. FIG. 13 shows a single wireless device 1310 and a single base station 1320, but it will be understood that wireless device 1310 may communicate with any number of base stations or other access network components over air interface 1370, and that base station 1320 may communicate with any number of wireless devices over air interface 1370.

[0179] The wireless device 1310 may comprise a processing system 1311 and a memory 1312. The memory 1312 may comprise one or more computer-readable media, for example, one or more non-transitory computer readable media. The memory 1312 may include instructions 1313. The processing system 1311 may process and/or execute instructions 1313. Processing and/or execution of instructions 1313 may cause wireless device 1310 and/or processing system 1311 to perform one or more functions or activities. The memory 1312 may include data (not shown). One of the functions or activities performed by processing system 1311 may be to store data in memory 1312 and/or retrieve previously-stored data from memory 1312. In an example, downlink data received from base station 1320 may be stored in memory 1312, and uplink data for transmission to base station 1320 may be retrieved from memory 1312. As illustrated in FIG. 13, the wireless device 1310 may communicate with base station 1320 using a transmission processing system 1314 and/or a reception processing system 1315. Alternatively, transmission processing system 1314 and reception processing system 1315 may be implemented as a single processing system, or both may be omitted and all processing in the wireless device 1310 may be performed by the processing system 1311. Although not shown in FIG. 13, transmission processing system 1314 and/or reception processing system 1315 may be coupled to a dedicated memory that is analogous to but separate from memory 1312, and comprises instructions that may be processed and/or executed to carry out one or more of their respective functionalities. The wireless device 1310 may comprise one or more antennas 1316 to access air interface 1370.

[0180] The wireless device 1310 may comprise one or more other elements 1319. The one or more other elements 1319 may comprise software and/or hardware that provide features and/or functionalities, for example, a speaker, a microphone, a keypad, a display, a touchpad, a satellite transceiver, a universal serial bus (USB) port, a hands-free headset, a frequency modulated (FM) radio unit, a media player, an Internet browser, an electronic control unit (e.g., for a motor vehicle), and/or one or more sensors (e.g., an accelerometer, a gyroscope, a temperature sensor, a radar sensor, a lidar sensor, an ultrasonic sensor, a light sensor, a camera, a global positioning sensor (GPS) and/or the like). The wireless device 1310 may receive user input data from and/or provide user output data to the one or more one or more other elements 1319. The one or more other elements 1319 may comprise a power source. The wireless device 1310 may receive power from the power source and may be configured to distribute the power to the other components in wireless device 1310. The power source may comprise one or more sources of power, for example, a battery, a solar cell, a fuel cell, or any combination thereof.

[0181] The wireless device 1310 may transmit uplink data to and/or receive downlink data from base station 1320 via air interface 1370. To perform the transmission and/or reception, one or more of the processing system 1311, transmission processing system 1314, and/or reception system 1315 may implement open systems interconnection (OSI) functionality. As an example, transmission processing system 1314 and/or reception system 1315 may perform layer 1 OSI functionality, and processing system 1311 may perform higher layer functionality. The wireless device 1310 may transmit and/or receive data over air interface 1370

using one or more antennas 1316. For scenarios where the one or more antennas 1316 include multiple antennas, the multiple antennas may be used to perform one or more multi-antenna techniques, such as spatial multiplexing (e.g., single-user multiple-input multiple output (MIMO) or multi-user MIMO), transmit/receive diversity, and/or beamforming.

[0182] The base station 1320 may comprise a processing system 1321 and a memory 1322. The memory 1322 may comprise one or more computer-readable media, for example, one or more non-transitory computer readable media. The memory 1322 may include instructions 1323. The processing system 1321 may process and/or execute instructions 1323. Processing and/or execution of instructions 1323 may cause base station 1320 and/or processing system 1321 to perform one or more functions or activities. The memory 1322 may include data (not shown). One of the functions or activities performed by processing system 1321 may be to store data in memory 1322 and/or retrieve previously-stored data from memory 1322. The base station 1320 may communicate with wireless device 1310 using a transmission processing system 1324 and a reception processing system 1325. Although not shown in FIG. 13, transmission processing system 1324 and/or reception processing system 1325 may be coupled to a dedicated memory that is analogous to but separate from memory 1322, and comprises instructions that may be processed and/or executed to carry out one or more of their respective functionalities. The wireless device 1320 may comprise one or more antennas 1326 to access air interface 1370.

[0183] The base station 1320 may transmit downlink data to and/or receive uplink data from wireless device 1310 via air interface 1370. To perform the transmission and/or reception, one or more of the processing system 1321, transmission processing system 1324, and/or reception system 1325 may implement OSI functionality. As an example, transmission processing system 1324 and/or reception system 1325 may perform layer 1 OSI functionality, and processing system 1321 may perform higher layer functionality. The base station 1320 may transmit and/or receive data over air interface 1370 using one or more antennas 1326. For scenarios where the one or more antennas 1326 include multiple antennas, the multiple antennas may be used to perform one or more multi-antenna techniques, such as spatial multiplexing (e.g., single-user multiple-input multiple output (MIMO) or multi-user MIMO), transmit/receive diversity, and/or beamforming.

[0184] The base station 1320 may comprise an interface system 1327. The interface system 1327 may communicate with one or more base stations and/or one or more elements of the core network via an interface 1380. The interface 1380 may be wired and/or wireless and interface system 1327 may include one or more components suitable for communicating via interface 1380. In FIG. 13, interface 1380 connects base station 1320 to a single deployment 1330, but it will be understood that wireless device 1310 may communicate with any number of base stations and/or CN deployments over interface 1380, and that deployment 1330 may communicate with any number of base stations and/or other CN deployments over interface 1380. The base station 1320 may comprise one or more other elements 1329 analogous to one or more of the one or more other elements 1319.

[0185] The deployment 1330 may comprise any number of portions of any number of instances of one or more network functions (NFs). The deployment 1330 may comprise a processing system 1331 and a memory 1332. The memory 1332 may comprise one or more computer-readable media, for example, one or more non-transitory computer readable media. The memory 1332 may include instructions 1333. The processing system 1331 may process and/or execute instructions 1333. Processing and/or execution of instructions 1333 may cause the deployment 1330 and/or processing system 1331 to perform one or more functions or activities. The memory 1332 may include data (not shown). One of the functions or activities performed by processing system 1331 may be to store data in memory 1332 and/or retrieve previously-stored data from memory 1332. The deployment 1330 may access the interface 1380 using an interface system 1337. The deployment 1330 may comprise one or more other elements 1339 analogous to one or more of the one or more other elements 1319.

[0186] One or more of the systems 1311, 1314, 1315, 1321, 1324, 1325, and/or 1331 may comprise one or more controllers and/or one or more processors. The one or more controllers and/or one or more processors may comprise, for example, a general-purpose processor, a digital signal processor (DSP), a microcontroller, an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) and/or other programmable logic device, discrete gate and/or transistor logic, discrete hardware components, an on-board unit, or any combination thereof. One or more of the systems 1311, 1314, 1315, 1321, 1324, 1325, and/or 1331 may perform signal coding/processing, data processing, power control, input/output processing, and/or any other functionality that may enable wireless device 1310, base station 1320, and/or deployment 1330 to operate in a mobile communications system.

[0187] Many of the elements described in the disclosed embodiments may be implemented as modules. A module is defined here as an element that performs a defined function and has a defined interface to other elements. The modules described in this disclosure may be implemented in hardware, software in combination with hardware, firmware, wetware (e.g. hardware with a biological element) or a combination thereof, which may be behaviorally equivalent. For example, modules may be implemented as a software routine written in a computer language configured to be executed by a hardware machine (such as C, C++, Fortran, Java, Basic, Matlab and/or the like) or a modeling/simulation program such as Simulink, Stateflow, GNU Octave, or LabVIEWMathScript. It may be possible to implement modules using physical hardware that incorporates discrete or programmable analog, digital and/or quantum hardware. Examples of programmable hardware comprise computers, microcontrollers, microprocessors, DSPs, ASICs, FPGAs, and complex programmable logic devices (CPLDs). Computers, microcontrollers and microprocessors may be programmed using languages such as assembly, C, C++ and/or the like. FPGAs, ASICs and CPLDs are often programmed using hardware description languages (HDL) such as VHDL or Verilog that configure connections between internal hardware modules with lesser functionality on a programmable device. The mentioned technologies are often used in combination to achieve the result of a functional module.

[0188] The wireless device 1310, base station 1320, and/or deployment 1330 may implement timers and/or counters. A timer/counter may start at an initial value. As used herein, starting may comprise restarting. Once started, the timer/counter may run. Running of the timer/counter may be associated with an occurrence. When the occurrence occurs, the value of the timer/counter may change (for example, increment or decrement). The occurrence may be, for example, an exogenous event (for example, a reception of a signal, a measurement of a condition, etc.), an endogenous event (for example, a transmission of a signal, a calculation, a comparison, a performance of an action or a decision to so perform, etc.), or any combination thereof. In the case of a timer, the occurrence may be the passage of a particular amount of time. However, it will be understood that a timer may be described and/or implemented as a counter that counts the passage of a particular unit of time. A timer/counter may run in a direction of a final value until it reaches the final value. The reaching of the final value may be referred to as expiration of the timer/counter. The final value may be referred to as a threshold. A timer/counter may be paused, wherein the present value of the timer/counter is held, maintained, and/or carried over, even upon the occurrence of one or more occurrences that would otherwise cause the value of the timer/counter to change. The timer/counter may be un-paused or continued, wherein the value that was held, maintained, and/or carried over begins changing again when the one or more occurrence occur. A timer/counter may be set and/or reset. As used herein, setting may comprise resetting. When the timer/counter sets and/or resets, the value of the timer/counter may be set to the initial value. A timer/counter may be started and/or restarted. As used herein, starting may comprise restarting. In some embodiments, when the timer/counter restarts, the value of the timer/counter may be set to the initial value and the timer/counter may begin to run.

[0189] FIGS. 14A, 14B, 14C, and 14D illustrate various example arrangements of physical core network deployments, each having one or more network functions or portions thereof. The core network deployments comprise a deployment 1410, a deployment 1420, a deployment 1430, a deployment 1440, and/or a deployment 1450. Each deployment may be analogous to, for example, the deployment 1330 depicted in FIG. 13. In particular, each deployment may comprise a processing system for performing one or more functions or activities, memory for storing data and/or instructions, and an interface system for communicating with other network elements (for example, other core network deployments). Each deployment may comprise one or more network functions (NFs). The term NF may refer to a particular set of functionalities and/or one or more physical elements configured to perform those functionalities (e.g., a processing system and memory comprising instructions that, when executed by the processing system, cause the processing system to perform the functionalities). For example, in the present disclosure, when a network function is described as performing X, Y, and Z, it will be understood that this refers to the one or more physical elements configured to perform X, Y, and Z, no matter how or where the one or more physical elements are deployed. The term NF may refer to a network node, network element, and/or network device.

[0190] As will be discussed in greater detail below, there are many different types of NF and each type of NF may be associated with a different set of functionalities. A plurality

of different NFs may be flexibly deployed at different locations (for example, in different physical core network deployments) or in a same location (for example, co-located in a same deployment). A single NF may be flexibly deployed at different locations (implemented using different physical core network deployments) or in a same location. Moreover, physical core network deployments may also implement one or more base stations, application functions (AFs), data networks (DNs), or any portions thereof. NFs may be implemented in many ways, including as network elements on dedicated or shared hardware, as software instances running on dedicated or shared hardware, or as virtualized functions instantiated on a platform (e.g., a cloud-based platform).

[0191] FIG. 14A illustrates an example arrangement of core network deployments in which each deployment comprises one network function. A deployment 1410 comprises an NF 1411, a deployment 1420 comprises an NF 1421, and a deployment 1430 comprises an NF 1431. The deployments 1410, 1420, 1430 communicate via an interface 1490. The deployments 1410, 1420, 1430 may have different physical locations with different signal propagation delays relative to other network elements. The diversity of physical locations of deployments 1410, 1420, 1430 may enable provision of services to a wide area with improved speed, coverage, security, and/or efficiency.

[0192] FIG. 14B illustrates an example arrangement wherein a single deployment comprises more than one NF. Unlike FIG. 14A, where each NF is deployed in a separate deployment, FIG. 14B illustrates multiple NFs in deployments 1410, 1420. In an example, deployments 1410, 1420 may implement a software-defined network (SDN) and/or a network function virtualization (NFV).

[0193] For example, deployment 1410 comprises an additional network function, NF 1411A. The NFs 1411, 1411A may consist of multiple instances of the same NF type, co-located at a same physical location within the same deployment 1410. The NFs 1411, 1411A may be implemented independently from one another (e.g., isolated and/or independently controlled). For example, the NFs 1411, 1411A may be associated with different network slices. A processing system and memory associated with the deployment 1410 may perform all of the functionalities associated with the NF 1411 in addition to all of the functionalities associated with the NF 1411A. In an example, NFs 1411, 1411A may be associated with different PLMNs, but deployment 1410, which implements NFs 1411, 1411A, may be owned and/or operated by a single entity.

[0194] Elsewhere in FIG. 14B, deployment 1420 comprises NF 1421 and an additional network function, NF 1422. The NFs 1421, 1422 may be different NF types. Similar to NFs 1411, 1411A, the NFs 1421, 1422 may be co-located within the same deployment 1420, but separately implemented. As an example, a first PLMN may own and/or operate deployment 1420 having NFs 1421, 1422. As another example, the first PLMN may implement NF 1421 and a second PLMN may obtain from the first PLMN (e.g., rent, lease, procure, etc.) at least a portion of the capabilities of deployment 1420 (e.g., processing power, data storage, etc.) in order to implement NF 1422. As yet another example, the deployment may be owned and/or operated by one or more third parties, and the first PLMN and/or second PLMN may procure respective portions of the capabilities of the deployment 1420. When multiple NFs are provided at a

single deployment, networks may operate with greater speed, coverage, security, and/or efficiency.

[0195] FIG. 14C illustrates an example arrangement of core network deployments in which a single instance of an NF is implemented using a plurality of different deployments. In particular, a single instance of NF 1422 is implemented at deployments 1420, 1440. As an example, the functionality provided by NF 1422 may be implemented as a bundle or sequence of subservices. Each subservice may be implemented independently, for example, at a different deployment. Each subservice may be implemented in a different physical location. By distributing implementation of subservices of a single NF across different physical locations, the mobile communications network may operate with greater speed, coverage, security, and/or efficiency.

[0196] FIG. 14D illustrates an example arrangement of core network deployments in which one or more network functions are implemented using a data processing service. In FIG. 14D, NFs 1411, 1411A, 1421, 1422 are included in a deployment 1450 that is implemented as a data processing service. The deployment 1450 may comprise, for example, a cloud network and/or data center. The deployment 1450 may be owned and/or operated by a PLMN or by a non-PLMN third party. The NFs 1411, 1411A, 1421, 1422 that are implemented using the deployment 1450 may belong to the same PLMN or to different PLMNs. The PLMN(s) may obtain (e.g., rent, lease, procure, etc.) at least a portion of the capabilities of the deployment 1450 (e.g., processing power, data storage, etc.). By providing one or more NFs using a data processing service, the mobile communications network may operate with greater speed, coverage, security, and/or efficiency.

[0197] As shown in the figures, different network elements (e.g., NFs) may be located in different physical deployments, or co-located in a single physical deployment. It will be understood that in the present disclosure, the sending and receiving of messages among different network elements is not limited to inter-deployment transmission or intra-deployment transmission, unless explicitly indicated.

[0198] In an example, a deployment may be a 'black box' that is preconfigured with one or more NFs and preconfigured to communicate, in a prescribed manner, with other 'black box' deployments (e.g., via the interface 1490). Additionally or alternatively, a deployment may be configured to operate in accordance with open-source instructions (e.g., software) designed to implement NFs and communicate with other deployments in a transparent manner. The deployment may operate in accordance with open RAN (O-RAN) standards.

[0199] An example embodiment depicted in FIG. 15 illustrates how data generated by an application is delivered from a sender to a receiver. The unit of data generated by the application may be an application data unit (ADU). The ADU may comprise, for example, a picture file, a video frame, text file and so on. The ADU may, for example, be generated and/or created by a first instance of a particular application, for use and/or enjoyment by a second instance of the application, or for processing by an application server of the application. To reliably deliver the ADU and/or to process the ADU efficiently, the ADU may be divided into one or more smaller units. For example, the one or more smaller units may be one or more protocol data units (PDUs). One or more first PDUs (e.g., PDU 1, PDU 2) for a first ADU may be of a first PDU set (e.g., PDU set 1). One

or more second PDUs (e.g., PDU 3, PDU 4) for a second ADU may be of a second PDU set (e.g., PDU set 2).

[0200] In an example, the application may deliver the one or more first PDUs and/or the one or more second PDUs to an SDAP/PDCP entity (e.g., a SDAP entity, a PDCP entity, and/or both a SDAP entity and a PDCP entity). The first PDU (e.g., PDU 1) may be delivered from the application to the SDAP/PDCP entity. In the SDAP/PDCP entity, the first PDU may be a first SDAP SDU, a first SDAP PDU, a first PDCP SDU, and/or a first PDCP PDU. The second PDU (e.g., PDU 2) may be delivered from the application to the SDAP/PDCP entity. In the SDAP/PDCP entity, the second PDU may be a second SDAP SDU, a second SDAP PDU, a second PDCP SDU, and/or a second PDCP PDU. Similarly, the PDU 3 may be a third PDCP PDU (e.g., PDCP PDU 3) and/or the PDU 4 may be a fourth PDCP PDU (e.g., PDCP PDU 4).

[0201] In an example, one or more PDCP PDUs (e.g., PDCP PDU 1, 2, 3, 4) may be delivered from the SDAP/PDCP entity to a RLC entity. The RLC layer may provide functionality of forwarding the one or more packets, for example, over a particular interface, from one node to another, using a MAC entity and/or a PHY entity.

[0202] As depicted in FIG. 15, for example, the application in the sender may generate one or more PDU sets. For example, the one or more PDU sets comprise the first PDU set and/or the second PDU set. The application in the sender may deliver the one or more PDU sets to the SDAP/PDCP entity of the sender. The SDAP/PDCP entity may classify the one or more PDUs of the one or more PDU sets, may apply header compression to the one or more PDUs to reduce size of headers of the one or more PDUs, may apply ciphering to the one or more PDUs to provide security, and/or may generate one or more PDCP PDUs.

[0203] In an example, the SDAP/PDCP entity of the sender delivers the generated one or more PDCP PDUs to the RLC entity. The RLC entity may be responsible for transferring data between a UE and a NG-RAN, using the MAC entity and/or the PHY entity. For example, the RLC entity of the sender may process and generate one or more RLC PDUs for the one or more PDCP PDUs (e.g., RLC SDUs) delivered from the PDCP/SDAP entity. For example, the RLC entity may generate a first RLC PDU from the first PDCP PDU (e.g., the first RLC SDU) and/or the RLC entity may generate a second RLC PDU from the second PDCP PDU (e.g., the second RLC SDU).

[0204] In an example, the one or more RLC PDUs generated by the RLC entity of the sender may be delivered to the MAC entity of the sender. The MAC entity of the sender may send the one or more RLC PDUs to a MAC entity of the receiver. The MAC entity of the receiver may deliver the one or more RLC PDUs to a RLC entity of the receiver. For example, the RLC entity of the receiver may receive the one or more RLC PDUs (e.g., RLC PDU 1, 2, 3, 4). The RLC entity of the receiver may recover the one or more RLC SDUs (e.g., PDCP PDUs) using the one or more RLC PDUs. The RLC entity may deliver the one or more recovered PDCP PDUs to a PDCP entity of the receiver. The PDCP entity of the receiver may process the one or more received PDCP PDUs, and/or may recover one or more PDUs from the one or more PDCP PDUs. To recover a PDCP SDU (or RLC SDU) from a PDCP PDU (or a RLC PDU) may be that the PDCP PDU is extracted from the PDCP PDU, that the PDCP PDU is re-assembled from the PDCP SDU.

[0205] FIG. 16 illustrates an example of data delivery where one or more SDUs are not delivered from a sender to a receiver. For brief illustration, in the FIG. 16, RLC PDU is not shown. For one or more RLC SDUs (e.g., RLC SDU 1, RLC SDU 2, RLC SDU 3, RLC SDU 4), a RLC entity may transmit the one or more RLC PDUs comprising the one or more RLC SDUs. When transmission of a RLC SDU (or a RLC PDU associated with the RLC SDU) fails, the RLC entity may perform re-transmission of the RLC SDU (or the RLC PDU). For example, if the RLC entity of a sender receives from a RLC entity of the receiver, a negative acknowledgement for the one or more RLC SDUs (or RLC PDUs), the RLC entity of the sender may perform re-transmission of the one or more RLC SDUs (or RLC PDUs).

[0206] In the example of FIG. 16, at time $t=t_1$, the RLC entity of the sender may transmit the RLC PDU 1 (or RLC SDU 1) and may receive acknowledgement for the RLC PDU 1 (or RLC SDU 1). At time $t=t_2$, the RLC entity may transmit next RLC PDU, which is RLC PDU 2 (or RLC SDU 2). The RLC entity may start transmission of RLC PDU 3 (or RLC SDU 3) at $t=t_3$, RLC PDU 4 (or RLC SDU 4) at $t=t_4$. For the received RLC PDUs (e.g., RLC PDU 1, RLC PDU 3 and RLC PDU 4), the receiver may perform assembly of one or more RLC SDUs (e.g., PDCP PDUs), using the received one or more RLC PDUs. The receiver may deliver the one or more assembled RLC SDUs to the next hop (e.g., UPF, NG-RAN, UE, internet router, upper layer).

[0207] In an example, due to e.g., temporary deterioration of radio condition, one or more RLC PDUs (RLC SDUs) sent by the sender may not be received by the receiver. In the example of FIG. 16, the RLC PDU 2 (RLC SDU 2) is not successfully received by the receiver. From the viewpoint of the PDCP entity of the receiver, while the PDCP PDU 1, the PDCP PDU 3 and the PDCP PDU 4 are received, the PDCP PDU 2 is not received. If the PDCP PDU 2 is not received, the application may not have some PDUs (e.g., PDU 2) of the first PDU set. In this case, even if the application receives the PDU 1, the PDU 1 may not be usable due to missing PDU 2. In other example, the usability of the second PDU set may be dependent on the availability of previous PDU set (e.g., PDU set 1). For example, in the audio playback, if the one or more PDUs of the PDU set 2 are generated based on the one or more PDUs of the PDU set 1, the application cannot play audio using the one of more PDUs of the PDU set 2, due to missing the one or more PDUs of the PDU set 1. In other words, the unsuccessful delivery of the RLC PDU 2 from the sender to the receiver impacts not only data included in the RLC PDU 2, but also data includes in other RLC PDUs. To address this problem, the sender may employ one or more retransmission mechanisms, as FIG. 17 shows.

[0208] FIG. 17 illustrates an example of data delivery where a retransmission of a RLC PDU is used, to enhance reliability of data delivery. For brief illustration, in the FIG. 17, RLC PDU is not shown. Similar to the previous figure (e.g., FIG. 16), the RLC entity of the sender may transmit the one or more RLC PDUs (e.g., RLC PDU 1, 2, 3, 4) to the RLC entity of the receiver. For brevity, redundant details will be omitted.

[0209] In an example, the RLC entity of the sender may transmit the RLC PDU 1, 2, 3 and 4 at $t=t_1$, t_2 , t_3 and t_4 . The RLC entity of the receiver may receive the RLC PDU 1 and may deliver the PDCP PDU 1 (included in the RLC PDU 1)

to the PDCP entity at $t=5$. The RLC entity of the receiver may receive the RLC PDU 3 (e.g., RLC SDU 3) and may deliver the PDCP PDU 3 (included in the RLC PDU 3) to the PDCP entity at $t=6$. The RLC entity of the receiver may receive the RLC PDU 4 (RLC SDU 4) and may deliver the PDCP PDU 4 (included in the RLC PDU 4) to the PDCP entity at $t=7$. The PDCP entity of the receiver may determine that the PDCP PDU 2 is not received. For example, based on receiving the PDCP PDU 3 and not receiving the PDCP PDU 2, the receiver may determine that the PDCP PDU 2 is not received. For the missing PDCP PDU 2, the PDCP entity of the receiver may start a time period to wait for the PDCP PDU 2 to be delivered. The PDCP entity of the receiver may associate the PDCP PDU 2 (e.g., sequence number of the PDCP PDU 2) to the time period. For example, based on receiving the PDCP PDU 3 and not receiving the PDCP PDU 2, the receiver may buffer the one or more PDCP PDUs (e.g., PDCP PDU 3, PDCP PDU 4), until the PDCP PDU 2 is received and/or until the time period expires.

[0210] Based on detecting transmission failure of the RLC PDU 2 (RLC SDU 2), the RLC entity of the sender may re-transmit the RLC PDU 2 (RLC SDU 2) at $t=8$. For example, the RLC entity of the sender may detect the transmission failure, if the RLC entity receives HARQ failure from a MAC entity and/or if the RLC entity receives a negative acknowledgement (NACK) from the RLC entity of the receiver.

[0211] In an example, the RLC entity of the receiver may receive the RLC PDU 2 at $t=8$, and deliver the PDCP PDU 2 (included in the RLC PDU 2) to the PDCP entity at $t=9$. The PDCP entity of the receiver may determine that the PDCP PDU 2 is associated with the time period and/or may recover one or more PDUs (e.g., PDU 2, PDU 3, PDU 4) from the received PDCP PDU 2 and/or one or more PDCP PDUs (e.g., PDCP PDU 3, PDCP PDU 4) buffered in the PDCP entity. Based on that the PDCP PDU 2 associated with the time period is received, the PDCP entity of the receiver may stop the time period.

[0212] As FIG. 17 shows, retransmission by the RLC entity and/or buffer management by the PDCP entity may help in reliable data delivery from the sender to the receiver. For example, as the PDCP entity waits for the delivery of the PDCP PDU 2 instead of immediate delivery of the PDCP PDU 1, 3 and 4, the application may be able to use the PDU 1, 3 and 4 of the PDCP PDU 1, 3 and 4. However, the using of the time period may cause unnecessary delay in data delivery.

[0213] FIG. 18 illustrates an example in which the RLC PDU 2 is not successfully delivered from the sender to the receiver, after one or more retransmission was performed. For brief illustration, in the FIG. 18, RLC PDU is not shown. Similar to the previous figure (e.g., FIG. 16, FIG. 17), the RLC entity of the sender may transmit the one or more RLC PDUs (e.g., RLC PDU 1, 2, 3, 4) to the RLC entity of the receiver. For brevity, redundant details will be omitted. Reverting back to FIG. 18, for the retransmission of the RLC PDU 2 at $t=8$, the RLC entity of the sender may determine that the RLC PDU 2 is not successfully delivered to the RLC entity of the receiver. For example, the RLC entity of the sender may receive a second negative acknowledgement of the RLC entity of the receiver. Based on the determination that the RLC PDU 2 is not successfully delivered to the RLC entity of the receiver, the RLC entity of the sender may

perform retransmission of the RLC PDU 2 at $t=9$. For the retransmission of the RLC PDU 2 at $t=9$, the RLC entity of the sender may determine that the RLC PDU 2 is not successfully delivered to the RLC entity of the receiver. Based on the determination, the RLC entity of the sender may determine that the transmission of the RLC PDU 2 fails, that the RLC entity of the sender may discard the RLC PDU 2 (e.g., RLC SDU 2, PDCP PDU 2), and/or that the RLC entity of the sender may abort transmission of the RLC PDU 2.

[0214] In an example, the PDCP entity of the receiver may start the time period at $t=6$, when the PDCP entity receives the PDCP PDU 3. For example, the time period may be set to expire at $t=10$. Until the expiration of the time period, the PDCP entity of the receiver may wait for the delivery of the PDCP PDU 2, may buffer the PDCP PDU 3, PDCP PDU 4 and/or may not forward the PDU 3 of the PDCP PDU 3 and PDU 4 of the PDCP PDU 4.

[0215] In an example, the RLC entity may decide not to transmit the RLC PDU 2 at $t=9$. In an example, the t_9 (e.g., 12:01 PM) may be earlier than t_{10} (e.g., 12:02 PM). However, in the existing technology, due to the running time period of the PDCP entity, the PDCP entity of the receiver may buffer the PDCP PDU 3 and the PDCP PDU 4, and/or may not deliver the PDU 3 of the PDCP 3 and the PDU 4 of the PDCP PDU 4, until $t=10$. In other words, after $t=9$, because the transmitter does not transmit the PDCP PDU 2, the receiver cannot receive the PDCP PDU 2. Accordingly, even if the receiver wait further from $t=9$ to $t=10$, the receiver cannot receive the PDCP PDU 2. The delay (or waiting) from $t=9$ to $t=10$ by the receiver does not bring any gain in data delivery. This unnecessary delay may cause degradation of service quality of the application. For example, the late arrival of frames for a video application does not help in improving quality of video service.

[0216] Example embodiments of the present disclosure cure the above issues and improve system efficiency by enhancement in operation of a network and/or a UE. In an example, a first network node (e.g., gNB-DU) may assist a second network node (e.g., gNB-CU) with information on one or more discarded data units. In another example, the second network node may assist the first network node in determining when to stop expecting to receive one or more data units. These signaling enhancements may reduce delay in data delivery and increase data rate between a sender and a receiver.

[0217] In the specification, the term “NG-RAN” may be interpreted as a base station, which may comprise at least one of a gNB, an eNB, a ng-eNB, a NodeB, an access node, an access point, an N3IWF, a relay node, a base station central unit (e.g., gNB-CU), a base station distributed unit (e.g., gNB-DU), and/or the like. In the specification, a gNB may be interpreted as a base station. In the specification, a gNB-CU may be interpreted as a base station central unit. In the specification, a gNB-DU may be interpreted as a base station distributed unit.

[0218] In the specification, the term “core network” node may be interpreted as a core network device, which may comprise at least one of an AMF, a SMF, a NSSF, a UPF, a NRF, a UDM, a PCF, a SoR-AF, an AF, an DDNMF, an MB-SMF, an MB-UPF and/or the like. A term of core network may be interpreted as a core network node. In the

specification, a term of an access node may be interpreted as a base station, which may comprise a NG-RAN, and/or the like.

[0219] In the specification, the term “network node” may be interpreted as a core network node, an access node, a UE, and/or the like. A network may comprise one or more network nodes.

[0220] In the specification, a protocol entity may be interpreted as an entity performing a set of specific functions related to a wireless access (e.g., LTE access, NR access) and/or a wireline access (e.g., Ethernet) and/or communication (e.g., TCP, IP). In an example, an entity may be interpreted as a protocol entity. In an example, the protocol entity of LTE and/or NR may be at least one of a SDAP entity, a PDCP entity, a RLC entity, a MAC entity and/or a PHY entity. In an example, a layer (e.g., a SDAP layer, a PDCP layer, a RLC layer, a MAC layer a PHY layer) may be interpreted as a protocol entity (e.g., SDAP entity, a PDCP entity, a RLC entity, a MAC entity, a PHY entity).

[0221] In the specification, a service data unit may be interpreted as a unit of a data, received by a protocol entity. In the specification, a protocol data unit may be interpreted as a unit of a data, sent by a protocol entity. A protocol entity may receive one or more service data units (SDUs) from other protocol entity, and the protocol entity may send one or more protocol service data units (PDUs) to another protocol entity of same host or another host. For example, a PDCP entity may receive one or more PDCP SDUs from a higher entity (e.g., an SDAP entity) and the PDCP entity may send one or more PDCP PDUs to a lower entity (e.g., an RLC entity). The lower entity (e.g., an RLC entity) may receive one or more SDUs (e.g., RLC SDUs) from the higher layer. The one or more SDUs received by the lower layer may be same as the one or more PDUs sent by the higher layer. FIG. 28 illustrates one example of relationship between one or more PDUs and/or one or more SDUs. For example, PDU 1 and PDU 2 may be generated by an application of a sender (a UE or in an application server). The PDU 1 and the PDU 2 may be delivered to a sending SDAP entity as a SDAP SDU 1 and 1 SDAP SDU 2. The sending SDAP entity may construct a SDAP PDU 1 from a SDAP header 1 and the SDAP SDU 1. The sending SDAP entity may deliver the SDAP PDU 1 to a sending PDCP entity. The sending PDCP entity may receive the SDAP PDU 1 as a PDCP SDU 1. The sending PDCP entity may construct a PDCP PDU 1 from a PDCP header 1 and the PDCP SDU 1. The sending PDCP entity may deliver the PDCP PDU 1 to a sending RLC entity. The sending RLC entity may receive the PDCP PDU 1 as a RLC SDU 1. The sending RLC entity may construct a RLC PDU 1 from a RLC header 1 and the RLC SDU 1. The sending RLC entity may deliver the RLC PDU 1 to a receiving RLC entity via a MAC/PHY entity. The receiving RLC entity may receive the RLC PDU 1. The receiving RLC entity may recover the RLC SDU 1 from the RLC PDU 1 and/or may deliver the RLC SDU 1 to a receiving PDCP entity. The receiving PDCP entity may receive the RLC SDU 1 as the PDCP SDU 1. The receiving PDCP entity may recover the PDCP SDU 1 from the PDCP PDU 1 and/or may deliver the PDCP SDU 1 to a receiving SDAP entity. The receiving SDAP entity may receive the PDCP SDU 1 as the SDAP PDU 1.

[0222] In the specification, for brief description, transmission of a RLC SDU may be interpreted as transmission of a RLC PDU. In the specification, for brief description, trans-

mission of a RLC PDU may be interpreted as transmission of a RLC SDU. If an amount of resource available in a MAC entity and/or in a PHY entity is limited, the resource may not enough to accommodate the amount of data of the RLC SDU. In that case, the RLC entity may segment the RLC SDU. For example, if the size of the RLC SDU is 100 bytes, and if the MAC entity supports maximum size of 50 bytes, then the RLC entity may segment the RLC SDU into smaller units. For example, the RLC SDU may be mapped into one or more RLC PDUs of 50 bytes. Alternatively, one or more RLC PDU segments (which are smaller than the RLC SDU and/or the RLC PDU) may be generated from the RLC SDU and/or the RLC PDU. The one or more RLC PDU segments or the one or more RLC PDUs may comprise at least a portion of the RLC SDU. This detail may complicate the description of the specification. In this specification, for the ease of description and brevity, the transmission of a RLC PDU may be interpreted as transmission of a RLC SDU. Similarly, the transmission of a RLC SDU may be interpreted as transmission of a RLC PDU. Similarly, reception of a RLC PDU may be interpreted as reception of a RLC SDU. Similarly, reception of a RLC SDU may be interpreted as reception of a RLC PDU.

[0223] In the specification, the term “AF (application function)” may be interpreted as a AS (application server), which may host and/or run one or more applications.

[0224] In the specification, the term “PDU set” may be interpreted as one or more PDUs carrying a payload of one unit of information generated at an application layer level (e.g., a frame or video slice). In some implementations all PDUs in a PDU Set may be needed by the application layer to use the corresponding unit of information. In other implementations, the application layer may be able to recover parts of the unit of information unit, when some PDUs are missing.

[0225] In the specification, the term “ADU” may be interpreted as one unit of information. The unit of information may be exchanged among one or more hosts serving an application. In an example, an application (e.g., an internet browser, an instant messaging application, a video-player application, etc.) may be running on a first host (e.g., a smartphone, computer, application server, etc.) and the same application may be running on a second host (e.g., another smartphone, computer, application server, etc.). The application on a first host may generate one or more units (e.g., a picture file, a text message, etc.) of information. Each of the one or more units of information may comprises one or more PDUs, and/or the one or more PDUs for a unit of information may be a PDU set.

[0226] FIG. 19 may depict one example embodiment of the present disclosure. By informing information of discarded units, gNB-CU may reduce unnecessary delay in data delivery.

[0227] In an example, an application of a sender (e.g., a UE) may generate one or more PDU sets. The one or more PDU sets may comprise a first PDU set and/or a second PDU set. The first PDU set may comprise a first PDU (e.g., PDU 1) and/or a second PDU (e.g., PDU 2). The second PDU set may comprise a third PDU (e.g., PDU 3) and/or a fourth PDU (e.g., PDU 4). The application may deliver one or more PDUs (e.g., PDU 1, 2, 3, 4) of the one or more PDU sets to a first entity (e.g., a PDCP entity, a SDAP entity, and/or both, not shown in the figure). The first entity may receive the one or more PDUs. For example, the SDAP entity (not shown in

the figure) may receive the first PDU as a first SDAP SDU (e.g., SDAP SDU 1), generate a first SDAP PDU (e.g., SDAP PDU 1), and/or deliver the first SDAP PDU to the PDCP entity. The PDCP entity may receive the first SDAP PDU as a first PDCP SDU (e.g., PDCP SDU 1), generate a first PDCP PDU (e.g., PDCP PDU 1), and/or deliver the first PDCP PDU to a second entity (e.g., a RLC entity). Similar handling may apply to the second PDU, the third PDU, and/or the fourth PDU.

[0228] In an example, the RLC entity may receive the one or more PDCP PDUs (e.g., PDCP PDU 1, 2, 3, 4) as one or more RLC SDUs (e.g., RLC SDU 1, 2, 3, 4). The RLC entity may process the one or more RLC SDUs, and/or may generate one or more RLC PDUs. For example, the RLC entity may generate a first RLC PDU (e.g., RLC PDU 1) from a first RLC SDU (e.g., RLC SDU 1, PDCP PDU 1), a second RLC PDU (e.g., RLC PDU 2) from a second RLC SDU (e.g., RLC SDU 2, PDCP PDU 2) and so on.

[0229] In an example, the RLC entity of the sender may transmit the one or more RLC PDUs to the receiver. The one or more RLC PDUs may comprises the one or more RLC SDUs. In the FIG. 19, it will be understood that the transmission of a RLC SDU is done via a RLC PDU (which is not shown in the figure). For example, the RLC entity may transmit the first RLC PDU (the first RLC SDU) at $t=t_1$, the second RLC PDU (the second RLC SDU) at $t=t_2$, the third RLC PDU (the third RLC SDU) at $t=t_3$, and/or the fourth RLC PDU (the fourth RLC SDU) at $t=t_4$.

[0230] In an example, the RLC entity of the receiver (e.g., gNB-DU) may receive the first RLC PDU (the first RLC SDU), the third RLC PDU (the third RLC SDU) and/or the fourth RLC PDU (the fourth RLC SDU).

[0231] In an example, the receiver (e.g., gNB-DU) may recover one or more RLC SDUs (e.g., RLC SDU 1, 3, 4) from the one or more received RLC PDUs (e.g., RLC PDU 1, 3, 4), and/or forward one or more RLC SDUs to a gNB-CU. The one or more RLC SDUs may be the one or more PDCP PDUs (e.g., PDCP PDU 1, 3, 4). The one or more RLC SDUs may be recovered from one or more RLC PDUs successfully received by the receiver. For example, the one or more RLC SDUs may comprise the first RLC SDU (e.g., the first PDCP PDU), the third RLC SDU (e.g., the third PDCP PDU), the fourth RLC SDU (e.g., the fourth PDCP PDU). For example, the gNB-DU may deliver the first RLC SDU (e.g., first PDCP PDU) to the gNB-CU at $t=t_5$, the third RLC SDU (e.g., third PDCP PDU) to the gNB-CU at $t=t_6$, and/or the fourth RLC SDU (e.g., fourth PDCP PDU) to the gNB-CU at $t=t_7$.

[0232] In an example, the gNB-CU may receive the one or more RLC SDUs (PDCP PDUs) from the gNB-DU. The gNB-CU may recover the first PDCP SDU (or the first SDAP SDU, the first PDU), and/or forward the first PDCP SDU (or the first SDAP SDU, the first PDU) to a UPF. In other example, if the gNB-CU is configured to forward per PDU set granularity, the gNB-CU may buffer the first PDCP PDU (or the first SDAP SDU, the first PDU) and/or may wait until all PDUs of the PDU set (which the first PDU belong to) are received from the gNB-DU. In this case, when all PDUs of the first PDU set are received from the gNB-DU, then all PDUs of the first PDU set may be forwarded to the UPF. In other example, if the gNB-CU is configured to forward per PDU granularity, the gNB-CU may not buffer the first PDCP PDU (or the first SDAP SDU, the first PDU),

and/or may forward the first PDU without waiting for receiving all PDUs of the first PDU set.

[0233] In an example, because the second RLC SDU (the second PDCP PDU) is missing, the gNB-CU may buffer the third RLC SDU (the third PDCP PDU) and/or the fourth RLC SDU (the fourth PDCP PDU), and/or may not forward the third PDU and/or the fourth PDU.

[0234] In an example, the RLC entity of the receiver (e.g., gNB-DU) may receive the first RLC PDU, the third RLC PDU and/or the fourth RLC PDU. Based on not receiving the second RLC PDU and/or receiving the first, third and fourth RLC PDUs, the RLC entity of the gNB-DU may determine that the second RLC PDU is not received (e.g., missing). The RLC entity of the gNB-DU may send a negative acknowledgement to the RLC entity of the sender (e.g., the UE). The RLC entity of the sender may perform re-transmission of the second RLC PDU at $t=t_8$. For example, when the RLC entity of the sender receives the negative acknowledgement of the second RLC PDU, the RLC entity of the sender may re-transmit the second RLC PDU. The RLC entity of the sender may perform one or more re-transmission of the second RLC PDU until the RLC entity of the sender receives acknowledgment for the second RLC PDU from the RLC entity of the receiver and/or until the RLC entity of the sender transmits the second RLC PDU for the configured maximum number for transmission times. For example, the configured maximum number of transmission may be three. The RLC entity of the sender may perform retransmission of the second RLC PDU at $t=t_9$. After $t=t_9$, because the number of transmission for the second RLC PDU reaches the configured maximum number of transmission, the RLC entity of the sender may not attempt transmission of the second RLC PDU anymore, and/or may discard the second RLC PDU. The discarding (or not transmitting) of the second RLC PDU may be discarding (or not transmitting) of the second RLC SDU, the second PDCP PDU, the second PDCP SDU, the second SDAP PDU, and/or the second SDAP SDU, which are associated with the second RLC PDU. Based on the discarding of the second RLC PDU (or the second RLC SDU), the sender may send an indication for the second RLC PDU, to the receiver. For example, the indication for the second RLC PDU may be that the sender discards the second RLC PDU.

[0235] In an example, the RLC entity of the receiver (e.g., gNB-DU) may determine delivery failure for the second RLC PDU (equivalently, the second RLC SDU, the second PDCP PDU, the second PDCP SDU, the second SDAP PDU, the second SDAP SDU, the second PDU). The delivery failure for the second PDCP PDU may be that the sender will not transmit the second RLC PDU anymore, and/or that the second RLC PDU (or the second RLC SDU) is discarded. For example, when the RLC entity of the receiver receives from the RLC entity of the sender the indication for the second RLC PDU, the RLC entity of the receiver may determine the delivery failure of the second RLC PDU. For example, the indication of the second RLC PDU may indicate that the transmission of the second RLC PDU (or the second RLC SDU) is aborted/stopped, that the second RLC PDU (or the second RLC SDU) is discarded, and/or that the second RLC PDU (or the second RLC SDU) may not be transmitted. For example, the indication of the second RLC PDU may be delivered via a RLC control PDU.

[0236] In an example, if the RLC entity of the gNB-DU determines the delivery failure of the second RLC PDU, the gNB-DU may send to the gNB-CU, a discard information. The discard information may comprise information of the one or more RLC PDUs (or RLC SDUs, PDCP PDUs) for which delivery failure occurs. For example, the discard information may comprise for each RLC PDU (or RLC SDU) of the one or more RLC PDUs (or RLC SDUs) for which the delivery failure occurs:

[0237] An identifier of the RLC PDU. For example, the identifier may comprise a RLC sequence number assigned for the RLC PDU.

[0238] An identifier of a RLC SDU associated with the RLC PDU. For example, the identifier may comprise a RLC sequence number assigned for the RLC SDU from which the RLC PDU is generated.

[0239] An identifier of a PDCP PDU associated with the RLC PDU. For example, the identifier may comprise a PDCP sequence number assigned for the PDCP PDU to which the RLC PDU is associated. For example, the RLC PDU comprises a data of the PDCP PDU.

[0240] An identifier of a PDCP SDU associated with the RLC PDU. For example, the identifier may comprise a PDCP sequence number assigned for the PDCP SDU to which the RLC PDU is associated. For example, the RLC PDU comprises a data of the PDCP SDU.

[0241] An identifier of a PDU set associated with the RLC PDU. For example, the identifier may comprise a PDU set sequence number to which the RLC PDU is associated. For example, the RLC PDU comprises a data of the PDU set.

[0242] An identifier of a PDU associated with the RLC PDU. For example, the identifier may comprise a PDU sequence number to which the RLC PDU is associated. For example, the RLC PDU comprises a data of the PDU.

[0243] In an example, the discard information may comprise the RLC sequence number (e.g., SN=2) of the second RLC PDU (or the RLC SDU). For example, the discard information may indicate that the second RLC PDU (or associated RLC SDU, PDCP PDU, PDCP SDU, PDU, PDU set) is discarded, is not transmitted, is not successfully received.

[0244] In an example, the gNB-CU may receive from the gNB-DU, the discard information. Based on the discard information, the gNB-CU may determine that the RLC SDU 2 (e.g., the second PDCP PDU) may not be delivered from the gNB-DU. Based on the determination, the gNB-CU may forward one or more buffered PDCP SDUs (or SDAP SDUs, PDUs) associated with the buffered RLC SDUs (or PDCP PDUs, PDCP SDUs) to the UPF. For example, the gNB-CU may recover the third PDCP SDU (or the third SDAP SDU, the third PDU) from the third RLC SDU, and/or may forward the third PDCP SDU (or the third SDAP SDU, the third PDU) to the UPF. For example, the gNB-CU may recover the fourth PDCP SDU (or the fourth SDAP SDU, the fourth PDU) from the fourth RLC SDU, and/or may forward the fourth PDCP SDU. In other example, if the gNB-CU is configured to forward one or more PDUs per PDU set granularity, because at least one PDU (e.g., PDU 2) of the first PDU set is not received, the gNB-CU may not forward the first PDU (associated with the first PDU set) to UPF and/or the gNB-CU may discard the first PDU.

[0245] FIG. 20 may depict one example embodiment of the present disclosure. By informing information of discarded units (e.g., PDCP PDUs, PDCP SDUs, SDAP SDUs, SDAP PDUs, RLC PDUs, RLC SDUs, PDUs), gNB-CU may reduce unnecessary delay in data delivery. Similar to the previous figure (e.g., FIG. 19), the sender (e.g., a UE) may transmit the one or more PDUs (e.g., PDU 1, 2, 3, 4) to the receiver (e.g., gNB-DU and/or gNB-CU). For brevity, redundant details will be omitted.

[0246] In an example, the sender may transmit the one or more RLC SDUs (e.g., RLC SDU 1, 2, 3, 4) via one or more RLC PDUs to the gNB-DU. The gNB-DU may receive the first RLC PDU at $t=t_1$, the third RLC PDU at $t=t_3$, and the fourth RLC PDU at $t=t_4$. From the one or more received RLC PDUs, the gNB-DU may recover the one or more RLC SDUs. For example, the gNB-DU may recover the first RLC SDU (e.g., PDCP PDU 1) from the first RLC PDU, the third RLC SDU (e.g., PDCP PDU 3) from the third RLC PDU, the fourth RLC SDU (e.g., PDCP PDU 4) from the fourth RLC PDU.

[0247] In an example, the gNB-DU may forward the one or more recovered RLC SDUs (e.g., RLC SDU 1, 3, 4) to the gNB-CU. For the delivery of the one or more recovered RLC SDUs from the gNB-DU to the gNB-CU, the gNB-DU may send one or more GTP-U messages. Each of the GTP-U messages may be at least one of:

[0248] GTP-U signaling message. This message may be used for user plane path management, or for user plane tunnel management.

[0249] G-PDU. This may carry an original packet (T-PDU) and/or a GTP-U header. For example, the original packet may be an IP datagram, Ethernet frame or unstructured PDU Data, from a UE, or from a network node in an external packet data network. For example, the original packet may be a ADU, a DU, a PDCP PDU, a PDCP SDU, a RLC SDU, a SDAP SDU, a SDAP PDU, a PDU of a PDU set and/or the like.

[0250] In an example, a GTP-U header may comprise at least one of:

[0251] Version field: This field may be used to determine a version of the GTP-U protocol.

[0252] Protocol Type (PT): This bit may be used as a protocol discriminator between GTP (when PT is '1') and GTP' (when PT is '0').

[0253] Message Type: This field may indicate the type of the GTP-U message.

[0254] One or more extension headers. This may indicate additional information between the sender and the receiver.

[0255] In an example, the one or more extension headers may comprise a NR-RAN container (e.g., NR user plane frame). The NR-RAN container may comprise at least one of:

[0256] PDU type: This may indicate the structure of the NR-RAN container (e.g., NR user plane frame).

[0257] NR-U sequence number: This parameter may indicate a NR-U sequence number assigned by the sender of the GTP-U message. For example, whenever the sender of the GTP-U message sends a new GTP-U message, the sender may assign a new number as a NR-U sequence number. This may help the sender and the receiver to manage connection between the sender and the receiver of the GTP-U message.

- [0258] DL discard NR PDCP PDU SN. This parameter may indicate a downlink (e.g., from the network to the UE, from the gNB-CU to gNB-DU) PDCP PDU sequence number up to and including which all PDCP PDUs should be discarded by the gNB-DU.
- [0259] Lost Packet Report: This parameter may indicate a presence of Number of lost NR-U Sequence Number ranges reported, Start of lost NR-U Sequence Number range and End of lost NR-U Sequence Number range, in the GTP-U header.
- [0260] Number of lost NR-U Sequence Number ranges reported: This parameter may indicate a number of NR-U sequence number ranges reported to be lost. For example, gNB-DU may receive one or more GTP-U messages from gNB-CU. If some of the one or more GTP-U messages are not received, the gNB-DU may report information of the not received GTP-U messages. For example, due to congestion of a link between the gNB-CU and the gNB-DU, one or more GTP-U messages may be lost during transfer from the gNB-CU to the gNB-DU.
- [0261] Highest successfully delivered NR PDCP Sequence Number: This parameter may be indicated by the gNB-DU to the gNB-CU, may indicate a highest sequence number of DL PDCP PDU successfully delivered from the gNB-DU to the UE. The DL PDCP PDU may be successfully delivered, if the receiver (e.g., the UE) receives the DL PDCP PDU sent by the gNB-DU.
- [0262] Highest transmitted NR PDCP Sequence Number: This parameter may be indicated by the gNB-DU to the gNB-CU, may indicate a highest sequence number of a PDCP PDU that the gNB-DU transmits from the gNB-DU to the UE.
- [0263] Assistance Information Type: This field may describe a type of radio quality assistance information provided from gNB-DU to gNB-CU. A DL Radio Quality Index may express a radio quality of a data radio bearer or a RLC entity in DL. A UL Radio Quality Index may express a radio quality of a data radio bearer or a RLC entity in UL.
- [0264] In an example, the gNB-DU may send the one or more GTP-U messages. The one or more GTP-U messages may comprise a first GTP-U message (GTP-U 1), a third GTP-U message (GTP-U 3), a fourth GTP-U message (GTP-U 4). The first GTP-U message may comprise a first GTP-U header and/or the first RLC SDU (the first PDCP PDU). The third GTP-U message may comprise a third GTP-U header and/or the third RLC SDU (the third PDCP PDU). The fourth GTP-U message may comprise a fourth GTP-U header and/or the fourth RLC SDU (the fourth PDCP PDU). For example, the gNB-DU may send the first GTP-U message at $t=t5$, the third GTP-U message at $t=t6$, the fourth GTP-U message at $t=t7$.
- [0265] In an example, the gNB-DU may determine that the delivery of the second RLC PDU (or the second RLC SDU) fails. Based on the determination, the gNB-DU may send a fifth GTP-U message to the gNB-CU. The fifth GTP-U message may comprise at least one of a fifth GTP-U header and/or a fifth RLC SDU (a fifth PDCP PDU). In an example, the fifth RLC SDU may comprise data of a fifth PDU (or a fifth PDCP SDU, a fifth SDAP SDU). In an example, the fifth GTP-U header may comprise a fifth NR-RAN container. In an example, the fifth NR-RAN container may comprise at least one of:
- [0266] the PDU type.
- [0267] the NR-U sequence number.
- [0268] the DL discard NR PDCP PDU SN.
- [0269] the Lost Packet Report.
- [0270] the Number of lost NR-U Sequence Number ranges reported.
- [0271] the Highest successfully delivered NR PDCP Sequence Number.
- [0272] the Highest transmitted NR PDCP Sequence Number.
- [0273] the Assistance Information Type.
- [0274] the discard information. For example, the discard information may indicate one or more UL (uplink, from the UE to a NG-RAN) sequence number. For example, each of the one or more UL sequence number may comprise at least one of a UL PDCP sequence number, a UL RLC sequence number, UL SDAP sequence number, and/or a sequence number of a PDU. The UL PDCP sequence number may comprise a sequence number of at least one of a PDCP PDU and/or a PDCP SDU. The UL RLC sequence number may comprise a sequence number of at least one of a RLC PDU and/or a RLC SDU may be associated. For example, the RLC PDU may comprise at least a portion of the PDCP PDU, the PDCP SDU, the RLC SDU, the RLC PDU, and/or the PDU. For example, the RLC PDU may be not received by the gNB-DU, may be discarded by the sender (e.g., UE), and/or may not be transmitted by the sender (e.g., the UE) anymore. For example, the discard information may indicate that the second RLC PDU (or the second RLC SDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) may not be transmitted, and/or may be discarded.
- [0275] In an example, the gNB-CU may receive the fifth GTP-U message at $t=t10$. Based on the received fifth GTP-U message and/or based on the discard information, the gNB-CU may determine that the transmission of the second RLC SDU (or the second RLC PDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) is aborted (terminated, stopped), that the second RLC SDU (or the second RLC PDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) is discarded, and/or that the second RLC SDU (or the second RLC PDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) may not be received from the UE (or via the gNB-DU). Based on the determination, the gNB-CU may not wait for the delivery of the second RLC SDU (or the second RLC PDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) anymore, may stop a time period for a buffered PDCP PDU (e.g., the third PDCP PDU), and/or may forward one or more buffered PDUs (e.g., PDCP SDUs, SDAP SDUs) to the UPF. For example, the gNB-CU may forward the third PDCP SDU (or the third PDU, the third SDAP SDU) and/or the fourth PDCP SDU (or the fourth PDU, the fourth SDAP SDU) to the UPF.
- [0276] FIG. 21 may depict one example embodiment of the present disclosure. By informing information of discarded units (e.g., PDCP PDUs, PDCP SDUs, SDAP SDUs,

SDAP PDUs, RLC PDUs, RLC SDUs, PDUs), gNB-CU may reduce unnecessary delay in data delivery. Similar to the previous figure (e.g., FIG. 19, 20), the sender (e.g., a UE) may transmit the one or more PDUs (e.g., PDU 1, 2, 3, 4) to the receiver (e.g., gNB-DU and/or gNB-CU). For brevity, redundant details will be omitted.

[0277] In an example, the sender may transmit the one or more RLC PDUs (e.g., RLC PDU 1, 2, 3, 4) to the gNB-DU. The gNB-DU may receive the first RLC PDU (e.g., the first RLC SDU) at $t=t_1$, the third RLC PDU (e.g., the third RLC SDU) at $t=t_3$, and the fourth RLC PDU (e.g., the fourth RLC SDU) at $t=t_4$. From the one or more received RLC PDUs, the gNB-DU may recover the one or more RLC SDUs. For example, the gNB-DU may recover the first RLC SDU (e.g., PDCP PDU 1) from the first RLC PDU, the third RLC SDU (e.g., PDCP PDU 3) from the third RLC PDU, the fourth RLC SDU (e.g., PDCP PDU 4) from the fourth RLC PDU.

[0278] In an example, the gNB-DU may forward the one or more recovered RLC SDUs (e.g., RLC SDU 1, 3, 4) to the gNB-CU. For the delivery of the one or more recovered RLC SDUs from the gNB-DU to the gNB-CU, the gNB-DU may send one or more GTP-U messages.

[0279] In an example, the gNB-DU may send the one or more GTP-U messages. The one or more GTP-U messages may comprise a first GTP-U message (GTP-U 1), a third GTP-U message (GTP-U 3), a fourth GTP-U message (GTP-U 4). The first GTP-U message may comprise a first GTP-U header and/or the first RLC SDU (the first PDCP PDU). The third GTP-U message may comprise a third GTP-U header and/or the third RLC SDU (the third PDCP PDU). The fourth GTP-U message may comprise a fourth GTP-U header and/or the fourth RLC SDU (the fourth PDCP PDU). For example, the gNB-DU may send the first GTP-U message at $t=t_5$, the third GTP-U message at $t=t_6$, the fourth GTP-U message at $t=t_7$.

[0280] In an example, the gNB-DU may determine that the delivery of the second RLC PDU (or the second RLC SDU) fails. Based on the determination, the gNB-DU may send a sixth GTP-U message to the gNB-CU. The sixth GTP-U message may comprise a sixth GTP-U header. For example, when the gNB-DU determines to send the sixth GTP-U message, the gNB-DU may not have a RLC SDU (e.g., a PDCP PDU) which is available for the delivery from the gNB-DU to gNB-CU. In other words, because the gNB-DU forwarded RLC SDU 1, 3, and 4 to gNB-CU and/or because the gNB-DU does not receive other RLC SDU from the UE afterword, there is no RLC SDU for the gNB-DU to send to the gNB-CU. In this case, the gNB-DU may determine not to include a RLC SDU into the sixth GTP-U message. The sixth GTP-U message may be a dummy GTP-U message because the sixth GTP-U message does not comprise data from the application. In an example, the sixth GTP-U header may comprise a sixth NR-RAN container. In an example, the sixth NR-RAN container may comprise at least one of:

- [0281] the PDU type.
- [0282] the NR-U sequence number.
- [0283] the DL discard NR PDCP PDU SN.
- [0284] the Lost Packet Report.
- [0285] the Number of lost NR-U Sequence Number ranges reported.
- [0286] the Highest successfully delivered NR PDCP Sequence Number.

[0287] the Highest transmitted NR PDCP Sequence Number.

[0288] the Assistance Information Type.

[0289] the discard information.

[0290] In an example, the gNB-CU may receive the sixth GTP-U message at $t=t_{10}$. Based on the received sixth GTP-U message, the gNB-CU may determine that the transmission of the second RLC SDU (or the second RLC PDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) is aborted (terminated, stopped), that the second RLC SDU (or the second RLC PDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) may be discarded, and/or that the second RLC SDU (or the second RLC PDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) may not be received from the UE (or from the gNB-DU). Based on the determination, the gNB-CU may not wait for the delivery of the second RLC SDU anymore, may stop a time period for a buffered PDCP PDU (e.g., the third PDCP PDU), and/or may forward one or more buffered PDUs (e.g., PDCP SDUs, SDAP SDUs) to the UPF. For example, the gNB-CU may forward the third PDCP SDU (or the third PDU, the third SDAP SDU) and/or the fourth PDCP SDU (or the fourth PDU, the fourth SDAP SDU) to the UPF.

[0291] FIG. 22 may depict one example embodiment of the present disclosure. By informing information of discarded units (e.g., PDCP PDUs, PDCP SDUs, SDAP SDUs, SDAP PDUs, RLC PDUs, RLC SDUs, PDUs), gNB-CU may reduce unnecessary delay in data delivery. Similar to the previous figure (e.g., FIG. 19, 20, 21), the sender (e.g., a UE) may transmit the one or more PDUs (e.g., PDU 1, 2) to the receiver (e.g., gNB-DU and/or gNB-CU). For brevity, redundant details will be omitted. In FIG. 22, not only RLC entity information (e.g., RLC sequence number) but also PDCP entity information (e.g., PDCP sequence number) may be utilized to assist the gNB-CU for handling of a PDCP entity.

[0292] In an example, an application of a sender (e.g., a UE) may generate one or more PDUs, and/or may deliver the one or more PDUs (e.g., a first PDU, a second PDU) to a first entity (e.g., a PDCP entity, a SDAP entity, and/or both). The first entity may receive the one or more PDUs. For example, the SDAP entity may receive the first PDU as a first SDAP SDU (e.g., SDAP SDU 1), generate a first SDAP PDU (e.g., SDAP PDU 1), and/or deliver the first SDAP PDU to the PDCP entity. The PDCP entity may receive the first SDAP PDU as a first PDCP SDU (e.g., PDCP SDU 1), generate a first PDCP PDU (e.g., PDCP PDU 1), and/or deliver the first PDCP PDU to a second entity (e.g., a RLC entity). Similar handling may apply to the second PDU (e.g., PDU 2).

[0293] In an example, the PDCP entity may assign one or more PDCP sequence numbers to the one or more PDCP PDUs. For example, the PDCP entity may assign a first PDCP sequence number to the first PDU (e.g., SDAP PDU 1, PDCP PDU 1, SDAP SDU 1, PDCP SDU 1). For example, the first PDCP sequence number may be N. A PDCP PDU header of the first PDCP PDU may comprise the first PDCP sequence number. For example, the PDCP entity may assign a second PDCP sequence number to the second PDU (e.g., SDAP PDU 2, PDCP PDU 2, SDAP SDU 2, PDCP SDU 2).

For example, the second PDCP sequence number may be $N+1$. For example, to assist for a receiver to do in-order delivery, the PDCP entity may assign the PDCP sequence number sequentially (e.g., N , $N+1$, $N+2$, $N+3$ and so on). A PDCP PDU header of the second PDCP PDU may comprise the second PDCP sequence number. The PDCP entity may deliver the one or more PDCP PDUs (e.g., first PDCP PDU, the second PDCP PDU) to the RLC entity. When the PDCP entity delivers the one or more PDCP PDUs to the RLC entity, the PDCP entity may indicate to the RLC entity, one or more PDCP sequence numbers associated with the one or more PDCP PDUs.

[0294] In an example, the RLC entity of the sender may receive the one or more PDCP PDUs as one or more RLC SDUs (e.g., first RLC SDU, second RLC SDU) from the PDCP entity of the sender. The RLC entity may process the one or more RLC SDUs, and/or may generate one or more RLC PDUs. For example, the RLC entity may generate the first RLC PDU (e.g., RLC PDU 1) from the first RLC SDU (e.g., RLC SDU 1, PDCP PDU 1), the second RLC PDU (e.g., RLC PDU 2) from the second RLC SDU (e.g., RLC SDU 2, PDCP PDU 2). The RLC entity may assign one or more RLC sequence number to the one or more RLC SDUs (or RLC PDUs, PDCP PDUs). For example, the RLC entity may assign a first RLC sequence number (e.g., K) to the first RLC SDU (or the first RLC PDU). For example, the RLC entity may assign a second RLC sequence number (e.g., $K+1$) to the second RLC SDU (or the second RLC PDU). Based on the one or more assigned sequence numbers, the RLC entity may construct the one or more RLC PDUs. For example, the first RLC PDU may comprise a first RLC PDU header. The first RLC PDU header may comprise the first RLC sequence number. For example, the second RLC PDU may comprise a second RLC PDU header. The second RLC PDU header may comprise the second RLC sequence number.

[0295] In an example, the RLC entity of the sender may transmit the one or more RLC PDUs to the receiver. For example, the RLC entity may transmit the first RLC PDU (the first RLC SDU) at $t=t_1$, the second RLC PDU (the second RLC SDU) at $t=t_2$. In an example, the RLC entity of the receiver (e.g., gNB-DU) may receive the first RLC PDU (the first RLC SDU).

[0296] In an example, the RLC entity of the sender may perform retransmission of the second RLC SDU (the second RLC PDU). When the number of transmission for the second RLC SDU (the second RLC PDU) reaches the configured maximum number of transmission, the RLC entity of the sender may not attempt transmission of the second RLC SDU (the second RLC PDU) anymore, and/or may discard the second RLC SDU (the second RLC PDU).

[0297] In an example, the RLC entity of the sender may determine delivery failure for the second RLC PDU (equivalently, the second RLC SDU, the second PDCP PDU, the second SDAP SDU, the second SDAP PDU, the second SDAP SDU, the second PDU). Based on the determination, the RLC entity of the sender may send to the RLC entity of the receiver, a discard assistance information. The discard assistance information may comprise the indication for the second RLC PDU. For example, the indication of the second RLC PDU may indicate that the transmission of the second RLC PDU (or the second RLC SDU) is aborted/stopped, that the second RLC PDU (or the second RLC SDU) is discarded, that a RLC receiving window needs to be updated,

and/or that the second RLC PDU (or the second RLC SDU) may not be transmitted. For example, the discard assistance information may be delivered via a RLC control PDU. For example, the discard assistance information may comprise the RLC sequence number of the second RLC PDU, and/or the PDCP sequence number associated with the second RLC PDU (e.g., the second RLC SDU). In one example, the RLC control PDU may comprise information of one or more discarded RLC SDUs (or RLC PDUs) that the sender may discard and/or that the sender may not transmit any more. The RLC control PDU may comprise one or more RLC sequence numbers of the one or more discarded RLC SDUs and/or one or more PDCP sequence numbers of the one or more discarded RLC SDUs.

[0298] In an example, the RLC entity of the receiver may receive the discard assistance information at $t=t_3$. Based on the received discard assistance information, the RLC entity of the receiver may construct the discard information. For example, to construct the discard information as shown in the example of FIG. 19, FIG. 20 and FIG. 21, the gNB-DU may use the discard assistance information. Reverting back to FIG. 22, for example, the discard information of the GTP-U message (not shown in the figure) may comprise the RLC sequence number (e.g., $K+1$) of the second RLC PDU and/or the PDCP sequence number (e.g., $N+1$) of the second PDCP PDU associated with the second RLC PDU. The gNB-DU may send the discard information to the gNB-CU.

[0299] In an example, the gNB-CU may receive the discard information sent by the gNB-DU. Based on the discard information, the gNB-DU may determine which PDCP PDU (RLC SDU) may not be received. For example, because the discard information indicates that PDCP sequence number (e.g., $N+1$) is discard, the gNB-CU may not wait for the second PDCP PDU (associated with the PDCP sequence number $N+1$) anymore, and/or may forward buffered PDCP SDUs (e.g., PDCP SDU associated with a PDCP sequence number (e.g., $N+2$, $N+3$, and so on) larger than $N+1$) to UPF.

[0300] FIG. 23 may depict one example embodiment of the present disclosure. By informing information of one or more discarded units (e.g., PDCP PDUs, PDCP SDUs, SDAP SDUs, SDAP PDUs, RLC PDUs, RLC SDUs, PDUs), gNB-CU may reduce unnecessary delay in data delivery. Similar to the previous figure (e.g., FIG. 20, 21, 22), the sender (e.g., a UE) may transmit the discard assistance information. For brevity, redundant details will be omitted. In FIG. 23, PDCP entity information may be utilized to assist the gNB-DU handling of RLC entity.

[0301] In an example, the PDCP entity of the sender (e.g., a UE) may generate a discard assistance information. For example, based on indication from a RLC entity of the sender that one or more RLC SDUs are not delivered to the receiver, the PDCP entity of the sender may determine that one or more PDCP SDUs (or PDCP PDUs) associated with the one or more RLC SDUs are discarded and/or may discard the one or more PDCP SDUs (or PDCP PDUs). For example, based on the indication from the RLC entity of the sender that the RLC SDU 2 is not delivered to the receiver, the PDCP entity of the sender may determine to send the discard assistance information to the PDCP entity of the receiver. Based on the determination, the PDCP entity of the receiver may send a discard assistance information to the PDCP entity of the receiver. The discard assistance information may comprise information of the discarded PDCP PDU (or PDCP SDU). For example, the discard assistance

information may comprise information that the second PDCP PDU (the second PDCP SDU) associated with the RLC SDU 2 is discarded and/or that the second PDCP PDU (the second PDCP SDU) is discarded. For example, the discard assistance information may comprise a sequence number (e.g., N+1) of the second PDCP PDU.

[0302] In an example, the PDCP entity of the receiver may receive the discard assistance information. Based on the received discard assistance information, the receiver (e.g., gNB-CU) may send to the gNB-DU, a seventh GTP-U (e.g., GTP-U 7) message. The seventh GTP-U message may comprise a seventh GTP-U header. The seventh GTP-U header may comprise at least one of:

- [0303]** the PDU type.
- [0304]** the NR-U sequence number.
- [0305]** the DL discard NR PDCP PDU SN.
- [0306]** the Lost Packet Report.
- [0307]** the Number of lost NR-U Sequence Number ranges reported.
- [0308]** the Highest successfully delivered NR PDCP Sequence Number.
- [0309]** the Highest transmitted NR PDCP Sequence Number.
- [0310]** the Assistance Information Type.
- [0311]** the discard information.
- [0312]** the discard information. For example, the discard information may indicate one or more UL (uplink, from the UE to a NG-RAN) sequence number. For example, each of the one or more UL sequence number may comprise at least one of a UL PDCP sequence number, a UL RLC sequence number, UL SDAP sequence number, and/or a sequence number of a PDU. The UL PDCP sequence number may comprise a sequence number of at least one of a PDCP PDU and/or a PDCP SDU. The UL RLC sequence number may comprise a sequence number of at least one of a RLC PDU and/or a RLC SDU. For example, the RLC PDU may comprise at least a portion of the PDCP PDU, the PDCP SDU, the RLC SDU, and/or the PDU. For example, the RLC PDU may be not received by the gNB-DU, may be discard by the sender (e.g., UE), and/or may not be transmitted by the sender anymore. For example, the discard information may indicate that the second PDCP PDU (or the second RLC SDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) may not be transmitted, and/or may be discarded.

[0313] In an example, the gNB-DU may receive the seventh GTP-U message. Based on the received seventh GTP-U message, the gNB-DU may determine that the transmission of the second RLC SDU (or the second RLC PDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) associated with the second PDCP PDU (the second PDCP SDU) is aborted (terminated, stopped), that the second RLC SDU (or the second RLC PDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) may be discarded, and/or that the second RLC SDU (or the second RLC PDU, the second PDCP PDU, the second PDCP SDU, the second SDAP SDU, the second SDAP PDU, the second PDU, the first PDU set) may not be received from the UE (or from the gNB-DU). Based on the determination, the

RLC entity of the gNB-DU may not wait for the delivery of the second RLC SDU (or associated RLC PDU) from the UE anymore, may process other received RLC SDUs (or associated RLC PDUs), may adjust one or more RLC variables (e.g., a RLC receiving window, one or more sequence numbers associated with management of the RLC receiving window), and/or may update one or more RLC timers.

[0314] FIG. 24 may depict one example embodiment of the present disclosure. By informing information of a PDU set, a gNB-CU and/or a gNB-DU may reduce unnecessary delay in data delivery.

[0315] In an example, an application of a UE may generate one or more PDU sets. The one or more PDU sets may comprise a seventh PDU set and/or an eighth PDU set. The seventh PDU set may comprise a seventh PDU (e.g., PDU 7), and/or an eighth PDU (e.g., PDU 8). The eighth PDU set may comprise a ninth PDU (e.g., PDU 9). The UE may deliver the one or more PDUs (e.g., PDU 7, 8, 9) of the one or more PDU sets (e.g., PDU set 7, 8) to a first network node (e.g., gNB-DU) via one or more RLC PDUs (or RLC SDUs). In an example, the one or more RLC PDUs may comprise information of one or more PDU sets associated with the one or more RLC PDUs.

[0316] In an example, the first network node may receive the one or more RLC PDUs and/or may recover one or more RLC SDUs (PDCP PDUs) from the one or more RLC PDUs. The first network node may deliver the one or more RLC SDUs to a second network node (e.g., gNB-CU) via one or more GTP-U messages. For example, the one or more GTP-U messages may comprise a 17th GTP-U message, 18th GTP-U message, and/or 19th GTP-U message. Each of the one or more GTP-U messages may comprise at least one of a GTP-U header and/or a RLC SDU (RLC PDU, PDCP PDU, PDCP SDU, SDAP PDU, SDAP SDU, PDU). For example, the GTP-U header may comprise information of a PDU set associated with the RLC SDU (RLC PDU, PDCP PDU, PDCP SDU, SDAP PDU, SDAP SDU, PDU). For example, the 17th GTP-U message may comprise a seventh RLC SDU and/or a 17th GTP-U header. For example, the 17th GTP-U header may comprise information of the PDU set (e.g., PDU set 7) associated with the seventh RLC SDU. For example, the information of the PDU set may be a sequence number of the PDU set. For example, the seventh RLC SDU may comprise the seventh PDU. Alternatively, the seventh RLC SDU may comprise information of the PDU set (e.g., PDU set 7) associated with the seventh RLC SDU. For example, the 18th GTP-U message may comprise an eighth RLC SDU and/or a 18th GTP-U header. For example, the 18th GTP-U header may comprise information of the PDU set (e.g., PDU set 7) associated with the eighth RLC SDU. For example, the eighth RLC SDU may comprise the eighth PDU. Alternatively, the eighth RLC SDU may comprise information of the PDU set (e.g., PDU set 7) associated with the eighth RLC SDU. For example, a 19th GTP-U message may comprise a ninth RLC SDU and/or the 19th GTP-U header. For example, the 19th GTP-U header may comprise information of the PDU set (e.g., PDU set 8) associated with the ninth RLC SDU. For example, the ninth RLC SDU may comprise the ninth PDU. Alternatively, the ninth RLC SDU may comprise information of the PDU set (e.g., PDU set 8) associated with the ninth RLC SDU. For example, the information of the PDU set associated with the ninth RLC SDU may be a sequence number of the PDU set (e.g., PDU set 8)

[0317] In an example, the gNB-CU may receive the one or more GTP-U messages. Based on the one or more PDU set information of the one or more GTP-U messages, the gNB-CU may be able to determine when to forward the one or more PDUs of the one or more GTP-U messages to a UPF.

[0318] In an example, an application server may generate one or more PDU sets. The one or more PDU sets may comprise a fifth PDU set and/or a sixth PDU set. The fifth PDU set may comprise a first PDU (e.g., PDU 1). The sixth PDU set may comprise a second PDU (e.g., PDU 2), and/or a third PDU (e.g., PDU 3). The application server may deliver the one or more PDUs (e.g., PDU 1, 2, 3) of the one or more PDU sets (e.g., PDU set 5, 6) to a UPF. The UPF may forward the one or more PDUs to the second network node (e.g., gNB-CU) via N3 interface.

[0319] The second network node may receive the one or more PDUs. The second network node may deliver the one or more PDUs to the first network node (e.g., gNB-DU) via one or more GTP-U messages. For example, the one or more GTP-U messages may comprise a 21st GTP-U message, 22nd GTP-U message, and/or 23rd GTP-U message. Each of the one or more GTP-U messages may comprise at least one of a GTP-U header and/or a PDU (RLC SDU, RLC PDU, PDCP PDU, PDCP SDU, SDAP PDU, SDAP SDU). For example, the GTP-U header may comprise information of a PDU set associated with the PDU (RLC SDU, RLC PDU, PDCP PDU, PDCP SDU, SDAP PDU, SDAP SDU). For example, the information of a PDU set associated with PDU may be a sequence number of the PDU set. For example, the 21st GTP-U message may comprise the first PDU and/or the 21st GTP-U header. For example, the 21st GTP-U header may comprise information of the PDU set (e.g., PDU set 5, PDU set SN=5) associated with the 21st PDU. For example, the 22nd GTP-U message may comprise the second PDU and/or the 22nd GTP-U header. For example, the 22nd GTP-U header may comprise information of the PDU set (e.g., PDU set 6, PDU set SN=6) associated with the second PDU. For example, the 23rd GTP-U message may comprise the third PDU and/or the 23rd GTP-U header. For example, the 23rd GTP-U header may comprise information of the PDU set (e.g., PDU set 6, PDU set SN=6) associated with the third PDU.

[0320] In an example, the gNB-DU may receive the one or more GTP-U messages (e.g., 21st, 22nd, 23rd GTP-U messages). Based on the one or more PDU set information of the one or more GTP-U messages, the gNB-DU may be able to determine when to forward the one or more PDUs of the one or more GTP-U messages to the UE and/or when to discard the one or more PDUs of the one or more PDU sets.

[0321] FIG. 25 may depict one example embodiment of the present disclosure. Similar to the previous examples (e.g., FIG. 20, 21, 22, 23, 24), the GTP-U message may comprise one or more information that assist a receiver to efficiently perform data delivery. For brevity, redundant details will be omitted.

[0322] Each of the GTP-U messages may be at least one of:

[0323] GTP-U signaling message. This message may be used for user plane path management, or for user plane tunnel management.

[0324] G-PDU. This may carry an original packet (T-PDU) and/or a GTP-U header. For example, the original packet may be an IP datagram, Ethernet frame or unstructured PDU Data, from a UE, or from a

network node in an external packet data network. For example, the original packet may be a PDCP PDU, a PDCP PDU, a RLC SDU, a SDAP SDU, a SDAP PDU, a PDU of a PDU set and/or the like.

[0325] In an example, the G-PDU may be a NR user plane protocol frame and/or a NR RAN container. For example, the NR user plane protocol frame and/or the NR RAN container may be at least one of DL User Data, DL Data Delivery Status, Assistance Information Data, UL User data, UL Data Delivery Status, and/or the like.

[0326] In an example, the GTP-U header (or a header of the NR user plane protocol frame) may further comprise at least one of:

[0327] Version field.

[0328] Protocol Type (PT).

[0329] Message Type.

[0330] One or more extension headers.

[0331] In an example, the one or more extension headers may comprise at least one of:

[0332] PDU type.

[0333] NR-U sequence number.

[0334] DL discard NR PDCP PDU SN.

[0335] Lost Packet Report.

[0336] Number of lost NR-U Sequence Number ranges reported.

[0337] Highest successfully delivered NR PDCP Sequence Number.

[0338] Highest transmitted NR PDCP Sequence Number.

[0339] Assistance Information Type.

[0340] Discard information. For example, the discard information may indicate one or more UL (uplink, from the UE to a NG-RAN) sequence numbers of a one or more discarded PDUs (or RLC PDUs, PDCP PDUs, SDAP PDUs, RLC SDUs, PDCP SDUs, and/or the like). The discard information may indicate that one or more RLC PDU (or RLC PDUs, PDCP PDUs, SDAP PDUs, RLC SDUs, PDCP SDUs, and/or the like) are discarded. For example, each of the one or more UL sequence number may comprise at least one of a UL PDCP sequence number (e.g., UL discard/lost NR PDCP PDU SN and/or the like), a UL RLC sequence number (e.g., UL discard/lost NR RLC PDU SN and/or the like), UL SDAP sequence number (e.g., UL discard/lost NR SDU PDU SN and/or the like), and/or a sequence number of a PDU (e.g., UL discard/lost PDU SN and/or the like). In other example, the discard information may comprise one or more PDU set sequence numbers of one or more discarded PDU sets. For example, each of the one or more PDU set sequence numbers may indicate a sequence number of a PDU set which is discarded by a sender, and/or which may not be further transmitted by the sender. The one or more PDU set sequence numbers may comprise at least one of a UL PDU set sequence number and/or a DL PDU set sequence number. For example, if a PDU 2 is discarded by a sender and if the PDU 2 belongs to a PDU set 1, then the PDU set sequence number (of the discard information) may be set to the sequence number of the PDU set 1 (e.g., PDU set SN 1). In other example, the discard information may indicate one or more DL (downlink, from a NG-RAN to a UE) sequence number. For example, the gNB-DU may receive one or more DL data (e.g., DL PDCP PDUs, DL

RLC SDUs) from the gNB-CU. Due to congestion or radio condition, the gNB-DU may discard one or more DL data from the one or more DL data (received from the gNB-CU), and/or the gNB-DU may indicate one or more sequence numbers of the one or more discarded DL data, to the gNB-CU. The gNB-CU may configure the gNB-DU with information on whether the gNB-DU is allowed to discard the one or more DL data. If the gNB-DU is configured to be allowed to discard the one or more DL data, the gNB-DU may discard the one or more DL data.

[0341] A PDU set sequence number. This may indicate the sequence number of the PDU set to which a PDU (which is included in the GTP-U message) is associated with.

[0342] Assistance Information Type.

[0343] Last PDU in a PDU set indication. This may indicate whether the PDU (which is included in the GTP-U message) is the last PDU in the PDU set to which the PDU is associated with. For example, in the example of FIG. 15, PDU 2 is the last PDU for the PDU set 1 and PDU 4 is the last PDU for the PDU set 2. For the PDU 2 and the PDU 4, the last PDU in a PDU set indication may indicate that the PDU 2 and the PDU 4 are the last PDU in the PDU set. For example, in the example of FIG. 15, PDU 1 is not the last PDU for the PDU set 1 and PDU 3 is the last PDU for the PDU set 2. For the PDU 1 and the PDU 3, the last PDU in a PDU set indication may indicate that the PDU 1 and the PDU 3 are not the last PDU in the PDU set. This may be end of data burst indication, the last PDU of the data burst, the last PDU of the current PDU set, a last PDU indication, and/or the like.

[0344] FIG. 26 may depict one example embodiment of the present disclosure. Similar to the previous examples (e.g., FIG. 20, 21, 22, 23, 24, 25), the gNB-DU may receive the one or more RLC PDUs from the UE. The one or more GTP-U messages sent by the gNB-DU to the gNB-CU may comprise the last PDU in a PDU set indication. This may assist the gNB-DU to determine whether to forward the one or more buffered PDUs to the UPF, may reduce data delivery delay. For brevity, redundant details will be omitted.

[0345] In an example, the gNB-DU receives the RLC PDU 3 at $t=3$, the RLC PDU 4 at $t=4$. From the one or more received RLC PDUs (e.g., RLC PDU 1, RLC PDU 3, RLC PDU 4), the gNB-DU may recover the one or more RLC SDUs (e.g., RLC SDU 1, RLC SDU 3, RLC SDU 4).

[0346] In an example, the gNB-DU may send to the gNB-CU, the one or more RLC SDUs, via the one or more GTP-U messages. The one or more GTP-U messages may comprise a first GTP-U message (GTP-U 1), a third GTP-U message (GTP-U 3), a fourth GTP-U message (GTP-U 4). The first GTP-U message may comprise a first GTP-U header and/or the first RLC SDU (the first PDCP PDU). The third GTP-U message may comprise a third GTP-U header and/or the third RLC SDU (the third PDCP PDU). The fourth GTP-U message may comprise a fourth GTP-U header and/or the fourth RLC SDU (the fourth PDCP PDU). For example, the gNB-DU may send the first GTP-U message at $t=5$, the third GTP-U message at $t=6$, the fourth GTP-U message at $t=7$.

[0347] In an example, each of the one or more GTP-U messages may comprise information of whether each of the one or more PDUs (SDAP SDUs, PDCP SDUs, RLC SDUs)

is the last PDU of a PDU set. For example, based on that the RLC PDU 1 does not comprise the last PDU (e.g., PDU 2) of the PDU set 1, the first GTP-U message may not comprise the indication of last PDU in a PDU set and/or may comprise an indication that the PDU is not the last PDU in a PDU set. For example, based on that the RLC PDU 3 does not comprise the last PDU (e.g., PDU 4) of the PDU set 2, the third GTP-U message may not comprise the indication of last PDU in a PDU set and/or may comprise an indication that the PDU is not the last PDU in a PDU set. For example, based on that the RLC PDU 4 comprises the last PDU (e.g., PDU 4) of the PDU set 2, the fourth GTP-U message may comprise the indication of last PDU in a PDU set and/or may comprise an indication that the PDU is the last PDU in a PDU set. In other example, the gNB-CU may determine whether a PDU is the last PDU in a PDU set or not, based on the one or more headers of the one or more RLC PDUs, one or more PDCP PDUs, and/or one or more SDAP PDUs. For example, the header of the first RLC PDU, the header of the first PDCP PDU, and/or the header of the first SDAP PDU may not comprise an indication of the last PDU in a PDU set and/or may comprise an indication that the PDU is not the last PDU in a PDU set. For example, the header of the fourth RLC PDU, the header of the fourth PDCP PDU, and/or the header of the fourth SDAP PDU may comprise an indication of the last PDU in a PDU set and/or may comprise an indication that the PDU is the last PDU in a PDU set.

[0348] In an example, the gNB-CU may receive the one or more GTP-U messages. Based on the one or more GTP-U messages, the gNB-CU may determine whether each of the one or more PDUs of the one or more GTP-U messages are the last PDU in a PDU set. For example, based on that the first GTP-U message does not comprise the indication of the last PDU in a PDU set, based on that the first GTP-U message comprise the indication that the PDU is not the last PDU in a PDU set, and/or based on the header of first RLC PDU (or first PDCP PDU, first SDAP PDU), the gNB-CU may determine that the first RLC SDU (or the first PDCP PDU, the first PDCP SDU, the first SDAP SDU, the first SDAP PDU, the first PDU) is not the last PDU of a PDU set (e.g., the first PDU set). For example, based on that the fourth GTP-U message comprises the indication of the last PDU in a PDU set, based on that the fourth GTP-U message comprise the indication that the PDU is the last PDU in a PDU set, and/or based on the header of the fourth RLC PDU (or the fourth PDCP PDU, the fourth SDAP PDU), the gNB-CU may determine that the fourth RLC SDU (or the fourth PDCP PDU, the fourth PDCP SDU, the fourth SDAP SDU, the fourth SDAP PDU, the fourth PDU) is the last PDU of a PDU set (e.g., the second PDU set). Based on the determination that the PDCP PDU 4 (the fourth PDU, the fourth PDCP SDU) is the last PDU of a PDU set, the gNB-CU may reduce the time period for waiting the second PDCP PDU and/or may deliver the PDU 3 and/or the PDU 4 to UPF.

[0349] In an example, the delivery of indication of the last PDU in a PDU set described above may be used in one or more GTP-U messages sent from the gNB-CU to the gNB-DU. For example, not only the GTP-U 17, 18 and 19, the GTP-U 21, 22 and 23 (of FIG. 24) may comprise indication of whether the PDU is the last PDU of a PDU set or not.

[0350] FIG. 27 may depict one example embodiment of the present disclosure. The gNB-CU may configure a UE and/or a gNB-DU with information on data delivery. By

configuration, the gNB-CU may be able to control whether a gNB-DU needs to send the discard information or not, may be able to indicate whether the gNB-CU sends to the gNB-DU the discard information or not.

[0351] In an example, gNB-CU may send to a UE, a RRC message. For example, the RRC message may be a RRC Reconfiguration message and/or the like. For example, the RRC message may comprise information of PDU processing and/or the like. For example, the information of PDU processing may indicate at least one of:

[0352] whether the UE is allowed to (or needs) to send a discard assistance information to a NG-RAN (e.g., gNB-CU, gNB-DU).

[0353] whether the UE needs to process the discard assistance information received from the NG-RAN.

[0354] whether the NG-RAN sends the discard assistance information or not.

[0355] whether the UE needs to send the indication of the last PDU in a PDU set to the NG-RAN.

[0356] whether the NG-RAN sends the indication of the last PDU in a PDU set to the UE.

[0357] whether the UE needs to send a PDU set information

[0358] whether the NG-RAN sends the PDU set information.

[0359] whether a radio bearer (or a data radio bearer, a logical channel, a RLC entity, a PDCP entity, a SDAP entity) is configured to send a PDU set information, the discard assistance information, the indication of the last PDU in a PDU set.

[0360] In an example, the UE may receive the RRC message sent by the gNB-CU. Based on the information of PDU processing, the UE may send to the NG-RAN, the discard assistance information when a PDU (or RLC SDU, PDCP PDU, PDCP SDU and/or the like) is discarded. For example, if the information of PDU processing indicates that the UE needs to send the discard assistance information, and if the PDU is discarded, the UE may send the discard assistance information indicating the discard of the PDU.

[0361] In an example, in response to the RRC message, the UE may send a RRC response message (e.g., RRC Reconfiguration complete) to the NG-RAN.

[0362] In an example, the gNB-CU may send a F1 message to the gNB-DU. For example, the F1 message may be a UE context Setup message, a UE context modification message and/or the like. For example, the F1 message may comprise at least one of:

[0363] DRB to be setup list. This may indicate one or more parameters of one or more DRBs to be setup for the UE. This may comprise an information of PDU delivery and/or the like.

[0364] DRX cycle. This may indicate DRX parameter to be applied for the UE

[0365] gNB-CU UE F1AP ID. This may indicate an identifier allocated for the UE by the gNB-CU.

[0366] gNB-DU UE F1AP ID. This may indicate an identifier allocated for the UE by the gNB-DU.

[0367] In an example, the information of PDU delivery and/or the like may comprise:

[0368] whether the gNB-DU is allowed to (or needs) to send a discard assistance information to the UE.

[0369] whether the gNB-DU needs to process the discard assistance information received from the UE and/or the gNB-DU needs to send the discard information to gNB-CU.

[0370] whether the gNB-DU needs to send the indication of the last PDU in a PDU set to the UE and/or to gNB-CU.

[0371] whether the gNB-DU needs to send the PDU set information to the UE and/or to gNB-CU.

[0372] whether a radio bearer (or a data radio bearer, a logical channel, a RLC entity, a PDCP entity, a SDAP entity) is configured to send a PDU set information, the discard assistance information, the indication of the last PDU in a PDU set.

[0373] In an example, the gNB-DU may receive the F1 message sent by the gNB-CU. Based on the information of PDU delivery and/or the like, the gNB-DU may send to the UE, the discard assistance information when a PDU (or RLC SDU, PDCP PDU, PDCP SDU and/or the like) is discarded. For example, if the information of PDU delivery indicates that the gNB-DU needs to send the discard assistance information, and if the PDU is discarded, the gNB-DU may send to the UE, the discard assistance information indicating the discard of the PDU. For example, if the information of PDU delivery indicates that the gNB-DU needs to send the discard information, and if a PDU is determined to be discarded, the gNB-DU may send to the gNB-CU, the discard information indicating the discard of the PDU. For example, if the information of PDU delivery indicates that the gNB-DU needs to send the indication of the last PDU in a PDU set, and if a PDU is determined to be the last PDU in a PDU set, the gNB-DU may send to the gNB-CU, the indication of the last PDU in a PDU set.

[0374] In an example, in response to the F1 message, the gNB-DU may send a F1 response message to the gNB-CU. For example, the F1 message may be a UE context Setup response message, a UE context modification response message and/or the like.

[0375] FIG. 29 may depict one example embodiment of the present disclosure.

[0376] In an example, a gNB-DU may receive from a UE, one or more first RLC SDUs. The UE may send to the gNB-DU, the one or more first RLC SDUs using one or more first RLC PDUs. The gNB-DU may send one or more first GTP-U messages to the gNB-CU. For example, the one or more first GTP-U messages may comprise the one or more first RLC SDUs. The one or more first RLC SDUs may be the one or more first PDCP PDUs. The gNB-CU may receive the one or more first PDCP PDUs sent by the gNB-DU.

[0377] In an example, the gNB-DU may receive from the UE, an indication that the second RLC SDU is discarded. For example, after one or more transmission of the second RLC SDU via one or more RLC PDUs, if the UE fails to receive acknowledgement from the gNB-DU, the UE may send to the gNB-DU, the indication that the second RLC SDU (or, the second RLC PDU, a second PDCP PDU associated with the second RLC SDU) is discarded. The indication may be that the second RLC SDU is not transmitted any more.

[0378] In an example, based on the indication that the second RLC SDU is discarded, the gNB-DU may send a

second GTP-U message comprising information that the second RLC SDU (e.g., the second PDCP PDU) is discarded.

[0379] FIG. 30 may depict one example embodiment of the present disclosure.

[0380] In an example, a gNB-CU may receive from a gNB-DU, one or more first GTP-U messages. The one or more first GTP-U messages may comprise one or more PDCP PDUs (e.g., one or more RLC SDUs).

[0381] In an example, the gNB-CU may recover one or more data units (DUs, e.g., SDAP SDUs, PDCP SDUs) from the one or more PDCP PDUs.

[0382] In an example, based on that a second RLC SDU (or a second PDCP PDU) is not received, the gNB-CU may store the one or more DUs

[0383] In an example, the gNB-CU may receive from the gNB-DU, a second GTP-U message. The second GTP-U message may comprise an information that the second RLC SDU (or the second PDCP PDU) is discarded.

[0384] In an example, based on the information that the second RLC SDU is discarded, the gNB-CU may deliver the one or more buffered DUs to an upper layer. For example, the upper layer may comprise at least one of a SDAP entity, a UPF, and/or an application.

[0385] In an example, a UE may send one or more RLC SDUs to the gNB-DU via one or more RLC protocol data units (PDUs). The one or more RLC SDUs may comprise the one or more first RLC SDUs and/or a second RLC SDU. The one or more RLC PDUs may comprise one or more first RLC PDUs and/or a second RLC PDU. Each RLC PDU of the one or more RLC PDUs may comprise at least a portion of a RLC SDU associated with the RLC PDU, a RLC PDU header, and/or a RLC control PDU. For example, each of the one or more first RLC PDUs may comprise at least one of a RLC PDU header, and/or at least a portion of one of the one or more first RLC SDUs. For example, the second RLC PDU may comprise at least one of a portion of the second RLC SDU, a second RLC PDU header, and/or the RLC control PDU. For example, the one or more RLC SDUs may belong to a same data radio bearer, a same QoS flow, a same PDU session, a same application and/or the like. The one or more RLC SDUs may be one or more PDCP PDUs. For example, a PDCP entity of the UE may send the one or more PDCP PDUs to a RLC entity of the UE. For each PDCP PDU (or associated PDCP SDU) of the one or more PDCP PDUs, the PDCP entity may assign a PDCP sequence number. The RLC entity of the UE may receive the one or more PDCP PDUs as the one or more RLC SDUs. For example, the RLC entity of the UE may receive the second PDCP PDU of the one or more PDCP PDUs as the second RLC SDU. The RLC entity may receive the one or more RLC SDUs, and the RLC entity of the UE may assign one or more RLC sequence numbers to the one or more RLC SDUs.

[0386] In an example, the UE may send the one or more RLC SDUs via the one or more RLC PDUs. In an example, the gNB-DU may receive from the UE, the one or more first RLC SDUs via the one or more first RLC PDUs. For example, a RLC entity of the gNB-DU may receive, the one or more first RLC SDUs sent by the RLC entity of the UE.

[0387] In an example, the gNB-DU may fail to receive the second RLC SDU.

[0388] In an example, based on not receiving the second RLC SDU, the gNB-DU may send a negative acknowledg-

ment of the second RLC SDU to the UE. For example, based on not receiving an acknowledgement of reception of the second RLC SDU and/or based on receiving the negative acknowledgement of the second RLC SDU, the UE may perform retransmission of the second RLC SDU. For example, after performing a number of retransmission of the second RLC SDU, the UE may stop further transmission of the second RLC SDU and/or may discard the second RLC SDU. Because the UE may not transmit the second RLC SDU anymore, the gNB-DU may fail to receive the second RLC PDU. Based on not transmitting the second RLC SDU anymore and/or discarding the second RLC SDU, the UE may send an indication that the second RLC SDU is discarded.

[0389] In an example, the gNB-DU may receive from the UE, the indication that the second RLC SDU (or the second RLC PDU associated with the RLC SDU) is discarded. For example, the gNB-DU may receive the indication, after receiving the one or more first RLC SDUs. For example, a RLC control PDU may comprise the indication.

[0390] In an example, the indication that the second RLC SDU is discarded may comprise at least one of:

[0391] a sequence number of the second RLC SDU. For example, the RLC entity of the UE may assign the sequence number of the second RLC SDU.

[0392] a sequence number of the second RLC PDU associated with the second RLC SDU. For example, the second RLC PDU may comprise at least a portion of the second RLC SDU. For example, the RLC entity of the UE may assign the sequence number of the second RLC PDU.

[0393] a sequence number of the second PDCP PDU associated with the second RLC SDU. For example, the second PDCP PDU may be the second RLC SDU. For example, a PDCP entity of the UE may assign the sequence number of the second PDCP PDU.

[0394] a sequence number of the second PDCP SDU associated with the second RLC SDU. For example, the second RLC SDU may comprise at least a portion of the second PDCP SDU. For example, the PDCP entity of the UE may assign the sequence number of the second PDCP SDU.

[0395] In an example, the indication that the second RLC SDU is discarded may indicate at least one of:

[0396] that the UE may not send at least one of the second RLC SDU, a portion of the second RLC SDU, the second RLC PDU associated with the second RLC SDU, the second PDCP PDU associated with the second RLC SDU, the second PDCP SDU.

[0397] that the second RLC SDU is transmitted a configured number of times by the UE.

[0398] that a time period associated with the second RLC SDU expires;

[0399] that a transmitting RLC window is adjusted.

[0400] that a receiving RLC window of the receiver (e.g., of gNB-DU) needs to be adjusted.

[0401] that the second RLC SDU is discarded by the UE.

[0402] In an example, the gNB-DU may send to a gNB-central unit (CU) one or more general packet radio service tunneling protocol user plane (GTP-U) messages. The one or more GTP-U messages comprise at least one of:

[0403] the one or more first RLC SDUs.

[0404] information that the second RLC SDU is discarded.

[0405] For example, the information that the second RLC SDU is discarded may comprise at least one of the sequence number of the second RLC SDU, the sequence number of the second PDCP PDU.

[0406] For example, based on the indication received from the UE, the gNB-CU may include the information into the one or more GTP-U messages. For example, a header of the one or more GTP-U messages may comprise the information that the second RLC SDU is discarded.

[0407] For example, if a RLC SDU (of the one or more first RLC SDU to be delivered to the gNB-CU) to deliver to the gNB-CU is not available, the gNB-DU may send the one or more GTP-U messages not comprising the one or more first RLC SDUs.

[0408] For example, based on the information that the second RLC SDU is discarded, the RLC entity of the gNB-DU may update a RLC receiving window. For example, the gNB-DU may move the lower edge (bound) of the RLC receiving window, may move the higher edge (bound) of the RLC receiving window, may update one or more sequence numbers (e.g., RX_Next (e.g., sequence number following the last in-sequence completely received RLC SDU), RX_next_status_trigger (sequence number following the sequence number triggering t-reassembly timer) managed by the RLC entity of the gNB-DU.

[0409] For example, based on the information that the second RLC SDU is discarded, the gNB-DU may discard the one or more RLC PDU segments of the second RLC SDU. For example, the one or more RLC PDU segments may comprise at least a portion of the second RLC SDU.

[0410] In an example, alternatively and/or additionally, the gNB-DU may use expiration of a time period. For example, when the time period expires for the second RLC SDU, the gNB-DU may determine that the second RLC SDU is discarded. In this case, the gNB-DU may act (e.g., delivering the discard information to the gNB-CU) as if the gNB-DU receives from the UE, the information that the second RLC SDU is discarded.

[0411] In an example, if the gNB-DU receives from the gNB-CU, a configuration requesting the gNB-DU to send an information of RLC SDU discard, the gNB-DU sends to the gNB-CU, the information that the second RLC SDU is discarded.

[0412] In an example, a gNB-CU may receive from a gNB-DU, one or more first general packet radio service tunneling protocol user plane (GTP-U) messages. The one or more first GTP-U messages may comprise one or more first packet data convergence protocol (PDCP) protocol data units (PDUs).

[0413] In an example, the gNB-CU may store one or more first data units (DUs) associated with the one or more first PDCP PDUs. For example, from the received one or more first PDCP PDUs, the gNB-CU may recover the one or more first DUs (e.g., PDCP SDUs, SDAP SDUs, PDUs). The gNB-CU may determine that a second PDCP PDU (e.g., the second RLC SDU) is not received from the gNB-DU. For example, when the gNB-CU receives the one or more first PDCP PDUs, the gNB-CU may determine one or more first PDCP sequence numbers of the one or more first PDCP PDUs. For example, the one or more first PDCP sequence numbers may be 3, 4, 5 and so on. The gNB-CU may not receive a second PDCP PDU (of a sequence number 2).

Because the second PDCP PDU with the sequence number (e.g., 2) preceding the sequence number (e.g., 3, 4 and 5) is not received, the gNB-CU may determine that the second PDCP PDU (with the sequence number 2) is not received, and/or may determine to buffer the one or more first DUs.

[0414] In an example, the gNB-CU may receive from the gNB-DU, a second GTP-U message comprising information that the second PDCP PDU (e.g., the second RLC SDU) is discarded. For example, the information that the second PDCP PDU is discarded may be that the second RLC SDU is discarded.

[0415] In an example, based on the information that the second PDCP PDU is discarded, the gNB-CU may deliver the one or more stored DUs. For example, the gNB-CU may deliver the one or more stored DUs, to an application server, an UPF, a SDAP entity, and/or the like.

[0416] In an example, a gNB-DU may send to a gNB-CU, an information that a RLC SDU (or a PDCP SDU, a PDCP PDU) is discarded.

[0417] In an example, a gNB-CU may receive from a gNB-DU, an information that LC SDU (or a PDCP SDU, a PDCP PDU) is discarded.

[0418] In an example, a second network device (e.g., a gNB-CU) may receive from a UE, an indication that a SDU (e.g., PDCP SDU, PDCP PDU, RLC SDU) is discarded by the UE. Based on receiving the indication, the gNB-CU may send to a first network device (e.g., a gNB-DU), a GTP-U message. The GTP-U message may indicate that the SDU is discarded.

[0419] In an example, a second network device (e.g., a gNB-CU) may send to a first network device (e.g., a gNB-DU), one or more GTP-U messages. Each GTP-U of the one or more GTP-U messages may comprises at least one of

[0420] a packet data units (PDUs) received from a user plane function (UPF);

[0421] an information of a PDU set associated with the PDU; and

[0422] an information of a PDU set discarded by the second network device.

[0423] In an example, a first network device (e.g., a gNB-DU) may send to a second network device (e.g., a gNB-CU), one or more GTP-U messages. Each GTP-U message of the one or more GTP-U messages may comprises at least one of

[0424] a packet data units (PDUs) received from a UE;

[0425] an information of a first PDU set (e.g., sequence number of the first PDU set) associated with the PDU; and

[0426] an information of a second PDU set (e.g., sequence number of the second PDU set) discarded by the UE.

[0427] In an example, a first network device (e.g., a gNB-DU) may receive from a second network device (e.g., a gNB-CU), one or more first PDUs. The first network device may send to the second network device, a GTP-U message. The GTP-U message may comprise information (e.g., one or more sequence numbers) of one or more second PDUs from the one or more first PDUs. The one or more second PDUs may be discarded by the first network device.

[0428] In an example, a first network device (e.g., a gNB-DU) may receive, a PDU of a PDU set. For example, the PDU set may comprise one or more PDUs. The one or more PDUs may comprise the PDU. The first network

device may send to a second network device (e.g., a gNB-CU), a GTP-U message. The GTP-U message may comprise at least one of the PDU and/or the information of the PDU set to which the PDU belongs.

[0429] In an example, a first network device (e.g., a gNB-DU, a gNB-CU) may receive, one or more PDUs of a PDU set. For example, the PDU set may comprise the one or more PDUs. The first network device may send to a second network device (e.g., a gNB-DU, a gNB-CU), a GTP-U message. The GTP-U message may comprise at least one of a PDU of the one and/or the indication of whether the PDU is the last PDU of the PDU set.

[0430] In an example, a UE may send one or more RLC PDUs. For example, the one or more RLC PDUs may comprise one or more PDCP PDUs. The UE may send a RLC control PDU. The RLC control PDU may comprise at least one of one or more RLC sequence numbers of the one or more discarded RLC PDUs and/or one or more PDCP sequence numbers of the one or more PDCP PDUs associated with the one or more discard RLC PDUs.

1. A method comprising:
sending, by a first base station (BS) to a second BS, a message comprising an information of a service data unit (SDU) of a protocol data unit (PDU) set.
2. The method of claim 1, wherein at least one of the first BS and the second BS is a BS-distributed unit (DU) and at least one of the first BS and the second BS is a BS-central unit (CU).
3. The method of claim 1, wherein the SDU is at least one of a radio link control (RLC) SDU or a packet data convergence protocol (PDCP) SDU.
4. The method of claim 1, wherein the first BS send the message, based on at least one of:
receiving from a wireless device, a first indication; or
failing to receive, from the wireless device, the SDU.
5. The method of claim 4, wherein at least one of a RLC control PDU or a PDCP control PDU comprises the first indication.
6. The method of claim 4, wherein the first indication indicates at least one of:
that the SDU of the PDU set is discarded;
that the PDU set is discarded;
that the SDU is transmitted a configured number of times by the wireless device; or
that a time period associated with the SDU expires.
7. The method of claim 4, further comprising adjusting, by the first BS and based on the first indication, at least one of:
an RLC receiving window;
an RLC transmitting window;
a PDCP receiving window; or
a PDCP transmitting window.
8. A first base station (BS) comprising one or more processors and memory storing instructions that, when executed by the one or more processors, cause the first base station to:
send, to a second BS, a message comprising an information of a service data unit (SDU) of a protocol data unit (PDU) set.

9. The first BS of claim 8, wherein at least one of the first BS and the second BS is a BS-distributed unit (DU) and at least one of the first BS and the second BS is a BS-central unit (CU).

10. The first BS of claim 8, wherein the SDU is at least one of a radio link control (RLC) SDU or a packet data convergence protocol (PDCP) SDU.

11. The first BS of claim 8, wherein the first BS send the message, based on at least one of:

receiving from a wireless device, a first indication; or
failing to receive, from the wireless device, the SDU.

12. The first BS of claim 11, wherein at least one of a RLC control PDU or a PDCP control PDU comprises the first indication.

13. The first BS of claim 11, wherein the first indication indicates at least one of:

that the SDU of the PDU set is discarded;
that the PDU set is discarded;
that the SDU is transmitted a configured number of times by the wireless device; or
that a time period associated with the SDU expires.

14. The first BS of claim 11, wherein the instructions further cause the first BS to adjust, based on the first indication, at least one of:

an RLC receiving window;
an RLC transmitting window;
a PDCP receiving window; or
a PDCP transmitting window.

15. A non-transitory computer-readable medium comprising instructions that, when executed by one or more processors of a first base station (BS), cause the first BS to:

send, to a second BS, a message comprising an information of a service data unit (SDU) of a protocol data unit (PDU) set.

16. The non-transitory computer-readable medium of claim 15, wherein at least one of the first BS and the second BS is a BS-distributed unit (DU) and at least one of the first BS and the second BS is a BS-central unit (CU).

17. The non-transitory computer-readable medium of claim 15, wherein the SDU is at least one of a radio link control (RLC) SDU or a packet data convergence protocol (PDCP) SDU.

18. The non-transitory computer-readable medium of claim 15, wherein the first BS send the message, based on at least one of:

receiving from a wireless device, a first indication; or
failing to receive, from the wireless device, the SDU.

19. The non-transitory computer-readable medium of claim 18, wherein at least one of a RLC control PDU or a PDCP control PDU comprises the first indication.

20. The non-transitory computer-readable medium of claim 18, wherein the first indication indicates at least one of:

that the SDU of the PDU set is discarded;
that the PDU set is discarded;
that the SDU is transmitted a configured number of times by the wireless device; or
that a time period associated with the SDU expires.

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